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Flame-Jet assisted drilling technology

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Flame-jet assisted drilling technology

A dissertation submitted to

ETH ZÜRICH

for the degree of

Doctor of Sciences

presented by

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Abstract

The current thesis concerns the experimental and theoretical investigation of flame-jet drilling technology in order to define the most influencing drilling parameters and the best way for further technology development. Flame-jet drilling is an alternative drilling technology where a flame-jet impinges a rock surface, causes a high temperature gradient in the surface layer and generates thermal stresses, localized fracturing (thermal spallation) and ultimately ejection of the created flakes (spalls). The final goal of flame-jet drilling is reaching geothermal energy which presents an alternative energy source stored deep underground.

A first flame-jet drilling tool, built at ETH and operating at high temperature and pressure, did not successfully drill rock samples in the past. It simulates deep underground conditions, with no optical access and complex experimental procedure. With this system, basic knowledge about flame-jet drilling could not be obtained and technology development could not proceed. Therefore, a new experimental flame-jet drilling setup has been built. It operates at ambient pressure with firing rates between 23 and 54 kW, dimensionless nozzle-to-target distances from 5 to 14 and exit nozzle diameter of the either 7.9 or 12.1 mm.

In order to explain and characterize the influence of process parameters on flame-jet drilling, an extensive heat transfer study was conducted. A new heat flux sensor with a calorimetric working principle was constructed generating surface temperatures comparable to the temperature of the rock surface at the onset of spallation. The spallation temperature was previously obtained with infrared thermometry being between 430 and 600 °C.

For that purpose, the emissivity of the treated rock (Grimsel granite) was determined with a technique developed in this work. The technique resulted in 25 °C of measuring error, which is acceptable for the purpose of thermal spallation. I. a., the results of the heat transfer study showed that the surface temperature of the target (up to 600 °C) did not remarkably influence the heat transfer from an impinging flame jet.

The flame-jet drilling experiments were performed for varying firing rates of the impinging flame jet (between 23 and 38 kW) and varying uniaxial load applied on the rock samples (0 to 47 MPa). A larger amount of transferred heat to the rock surface and larger uniaxial load were beneficial for the flame-jet drilling, i.e. more rock volume was removed. An existing mathematical model was applied to explain the experimental results. It was concluded that in case of changing the heat transfer properties, mostly the compression in the rock surface layers is influenced. The more heat flux is transferred to the rock surface, the larger the compressive stresses get. Also, increasing the uniaxial load resulted in increase of the tension below the free surface layers. This disintegrates the connections between the grains, which are then easier to separate and spall when exposed to compression at the surface caused by heating. Consequently, the study also indicates that flame-jet drilling is improved applying larger uniaxial load. This indicates that drilling efficiency should increase when drilling at larger depths.

Contrary to previous flame-jet drilling studies, a detailed heat transfer study has been performed that explains the influence of heat transfer to the rock surface in order to improve the flame-jet drilling technology. The tested rock type (Grimsel granite) was successfully drilled with a heat flux of 4 MW/m^2, and improvement of the heat transfer increased the removal rate. High temperatures (larger than 1000 °C) of the impinging fluid are not necessary, but excellent heat transfer to the rock surface is of great importance. Also, the mechanism of thermal spallation is better understood. The importance of tensile stresses below the impinged surface is proven for the first time.

Zusammenfassung

Die vorliegende Doktorarbeit behandelt die experimentelle und theoretische Untersuchung des sogenannten Flammenstrahlbohrens mit dem Ziel, die wesentlichen Bohrparameter zu definieren und die Weiterentwicklung der Technologie voranzutreiben. Flammenstrahlbohren ist eine alternative Bohrmethode, bei dem ein Flammenstrahl gegen eine Gesteinsoberfläche gerichtet wird und hohe Temperaturgradienten in den Oberflächenschichten verursacht. Die entstehenden thermischen Spannungen resultieren in lokalem Materialversagen und schlisslich in dem Abplatzen von Bohrsplittern (die sogenannte Spallation). Das Fernziel der Entwicklung des Flammenstrahlbohrens ist die Erschliessung von tiefen unterirdischen Geothermie-Reservoirs, welche als alternative Energiequelle an Bedeutung gewinnen.

Ein erster, an der ETH entwickelter Flammenstrahlbohrer konnte noch nicht erfolgreich auf Gesteinsproben angewendet werden. Dieser Bohrer arbeitet bei hohen Temperaturen und Drücken und simuliert damit die Bedingungen im tiefen Untergrund. Er hat aber keinen optischen Zugang und erfordert komplexe experimentelle Abläufe. Mit diesem System war es nicht möglich, grundlegende Kenntnisse zu den Mechanismen des Flammenstrahlbohrens zu erarbeiten, und die Weiterentwicklung der Technologie stockte. Daher wurde für die vorliegende Arbeit ein neuer Versuchsaufbau konstruiert und errichtet. Er arbeitet bei Umgebungsdruck mit Verbrennungsleistungen zwischen 23 und 54 kW, dimensionslosen Abständen von 5 bis 14 zwischen Düse und Ziel, sowie Düsendurchmessern von 7.9 oder 12.1 mm.

Es wurde eine umfangreiche Studie zum Wärmeübergang zwischen dem

vom neuen Versuchsaufbau produzierten, runden, turbulenten Flammenstrahl und einer flachen Oberfläche durchgeführt, um den Einfluss der Prozessparameter auf das Flammenstrahlbohren zu charakterisieren. Für die Studie wurde ein neuartiger Wärmeflusssensor konstruiert, welcher Oberflächentemperaturen entwickelt, die im Bereich der Temperaturen von Gesteinsoberflächen während des Spallationsprozesses liegt. Diese Spallationstemperaturen (zwischen 430 und 600 °C) wurden im Vorlauf mittels Pyrometrie gemessen. Für die Anwendung der Pyrometrie ist die Bekanntheit der Emissivität des untersuchten Gesteins (Grimsel Granit) Voraussetzung, welche mit einer eigens entwickelten Methode gemessen wurde. Die gesamte experimentelle Prozedur resultiert in einem für die Untersuchungen akzeptablen Messfehler von 25 °C für die Spallationstemperatur. Die Studie zeigte u.a., dass die Oberflächentemperatur des Sensors den Wärmeübergang von einem aufprallenden Flammenstrahl nicht merkbar beeinflusst.

Die Spallationsversuche wurden für eine Feuerungsleistung des aufprallenden Flammenstrahls durchgeführt (zwischen 23 und 38 kW). Ausserdem wurden die Gesteinsproben mit variierender einachsiger Last beaufschlagt (zwischen 0 und 47 MPa). Sowohl eine Erhöhung der Feuerungsleistung als auch eine Erhöhung der Last führte zu einer Erhöhung des entfernten Gesteinsvolumens pro Zeit. Ein existierendes mathematisches Modell wurde für die Erklärung der Ergebnisse verwendet. Es wurde geschlussfolgert, dass eine Änderung des Wärmeübergangs hauptsächlich die Druckbeanspruchung der Oberflächenschichten beeinflusst. Bei höherem Wärmeübergang steigt die Druckbeanspruchung. Durch eine Erhöhung der einachsigen Last der Gesteinsprobe wird hauptsächlich die Zugbeanspruchung unterhalb der Oberflächenschichten intensiviert. Dadurch werden die Verbindungen zwischen den Gesteinskörnern geschwächt, welche bei Eintritt in die Oberflächenschichten unter Druckbelastung einfacher getrennt und von der Gesteinsprobe gelöst werden können. Die Studienergebnisse weisen folglich darauf hin, dass die grössere einachsige Last das Bohren verbessert. Das könnte beweisen dass sich die Effizienz des Flammenstrahlbohrens mit grösserer Bohrtiefe erhöht.

Im Gegensatz zu früheren Arbeiten wurde eine detaillierte Studie durchgeführt, welche den Einfluss des Wärmeübergangs zur Gesteinsoberfläche mit dem Ziel erklärt, die Technologie des Flammenstrahlbohrens weiterzuentwickeln. Der verwendete Gesteinstyp (Grimsel Granit) konnte erfolgreich mit einem Wärmefluss von $4\,\mathrm{MW/m^2}$ gebohrt werden. Eine Verbesserung des Wärmeübergangs erhöhte das entfernte Gestein pro Zeit. Fluidtemperaturen von über $1000\,^{\circ}\mathrm{C}$ des aufprallenden Flammenstrahls sind nicht notwendig, aber exzellenter Warmeübergang zum Ziel des Prallstrahls. Ausserdem trägt die vorliegende Arbeit zum besseren Verständnis der Mechanismen des Spallationsprozesses bei: Die Wichtigkeit der Zugbelastung unterhalb der Gesteinsoberfläche ist zum ersten Mal nachgewiesen worden.

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Nomenclature

Abbreviations

$2D$	two-dimensional
AC	Alternating Current
CFD	Computational Fluid Dynamics
CO_2	Carbon dioxide
CU	Copper
CH_4	Methane
DAQ	Data AQuisition
EGS	Enhanced Geothermal Systems
FIC	Flow rate Indicator and Control
$FTIR$	Fourier Transform Infrared
$FJDS$	Flame-jet drilling setup
H_2O	Water
HHV	Higher heating value
IR	Infrared
N_2	Nitrogen
$NIST$	National Institute of Standards and Technology
O_2	Oxygen
PI	Pressure indicator
PID	Proportional-Integral-Derivative
PSI	Paul-Scherrer-Institut
ROP	Rate of penetration
SP	Setpoint
STP	Standard temperature and pressure

TI	Temperature indicator
TR	Temperature-resistance
$V-i$	Valve number i

Subscripts

0	At the nozzle
a	Average
ad	Adiabatic
air	Air
amb	Ambient
$bolt$	Bolt
$burner$	Burner
ch	Chamber
$cal.water$	Calorimetric water
cam	Camera
cav	Cavity
$conf$	Confined
d	Device
f	Firing/Flame
$fuel$	Fuel
h	Ceramic heater connected to AC power supply
ho	Ceramic heater heated in oven
in	In
$inlet$	Inlet
IR	Infrared
jet	Jet
max	Maximum
$multi$	Multimeter
n	Index
$nozzle$	Nozzle

out	Out
outlet	Outlet
p	Constant pressure
paint	Paint
Pyr	Pyro
ref	Reference
real	Real
removed	Removed
rock	Rock
s	Sonic
sample	Sample
sen	Sensor
spall	Spall
st	Static
stoich	Stoichiometric
surf	Surface
thermo	Thermocouple
v	Constant volume

Latin letters

a	Coefficient in Gaussian distribution	-
A	Area/Cross section	m^2
b	Regression coefficient	-
C	Constant of infrared camera	-
C_p	Heat capacity	$Jkg^{-1}K^{-1}$
D	Diameter	mm
E	Radiated power	Wm^{-2}
E	Young's modulus	Pa
f	External body force per volume	Nm^{-3}
F	Force	N

g	Gravitational constant	ms^{-2}
h	Convective heat transfer coefficient	$\mathrm{Wm}^{-2}\mathrm{K}^{-1}$
k	Coefficient in Gaussian distribution	-
i	Index variable	-
I	Electric current	A
j	Index variable	-
L	Dimensionless nozzle-to-target distance	-
L	Rock sample length	-
m	Mass	kg
$\dot{m}$	Mass flow rate	kgh^{-1}
M	Molar mass	kgmol^{-1}
n	Wavelength dependent parameter of IR camera	
n	Index variable	-
N	Number of measur. points for least sq. fit	-
p	Pressure	bar
P	Load	bar
$\dot{q}$	Heat flux	Wm^{-2}
$\dot{Q}$	Heat duty	W
r	Radius	m
R	Electrical resistance	Ω
R	Surface roughness	m
R	Ideal gas constant	$\mathrm{JK}^{-1}\mathrm{mol}^{-1}$
R^2	Coefficient of determination	-
S	Nozzle-to-target distance	m
T	Temperature	$^{\circ}\mathrm{C}, \mathrm{K}$
$\dot{T}$	Temperature change with time	$^{\circ}\mathrm{Cs}^{-1}$
U	Voltage	V
$\dot{V}$	Volume flow rate	$\mathrm{m}^3\mathrm{h}^{-1}$
w	Velocity	ms^{-1}
Y	Variable in uncertainty calculation	

Greek letters

α	Linear thermal expansion coefficient	K^{-1}
δ	Uncertainty/Delta	-
ϵ	Emissivity	-
ϵ	Strain tensor	-
$\dot{\epsilon}$	Strain rate tensor	s^{-1}
η	Dynamic viscosity	Pas
γ	Isentropic expansion factor	-
κ	Thermal conductivity	$\mathrm{Wm}^{-1}\mathrm{K}^{-1}$
λ	Air-fuel equivalence ratio	-
ν	Poisson's ratio	-
ρ	Density	kgm^{-3}
σ	Standard deviation	-
σ	Stefan-Boltzmann constant	JK^{-1}
σ	Stress	Pa
$\overline{\sigma}$	Average stress for one surface	Pa

Dimensionless numbers

Ma	Mach number	-
Re	Reynolds number	-

Chapter 1

Introduction and motivation

From the overall electricity supply in Switzerland 37 % is related to nuclear power [1]. After the nuclear Fukushima disaster in Japan in 2011, the Swiss government got more aware of the risks dedicated to nuclear power plants and decided to stop building new ones and if possible shut down the already existing ones [2]. With this motivation, between 2012 and 2013, the Swiss Federal Council came up with the new "Energy Strategy 2050" that has the goal to securely supply energy at affordable prices, while emitting less CO_2 than today [1]. To achieve this goal, a larger amount of consumed electricity should also be generated from renewable sources, one of them being geothermal energy. Geothermal energy is the Earths natural heat originating from radioactive decay, primarily of naturally radioactive isotopes of uranium, thorium and potassium [3]. In Switzerland, it is assumed that geothermal energy has high potential because Switzerland has the highest density of ground source heat pump systems in the world (with drilling depths of up to 70 m) [4]. However, there is still not enough knowledge about the deep underground (more than 2 km of depth). Therefore, the government supports many projects that are oriented to develop a geothermal map in Switzerland and exploit the geothermal energy for electricity production [5, 6]. It is planned to produce around 4 400 GWh_{el} of electrical power per year by deep geothermal power plants until 2050 [1], which is 7 % of the current overall energy consumption in Switzerland [7]. Considering that no electricity is being produced from geothermal sources in

Switzerland at the moment, this goal is very ambitious.

The temperature in the Earth is higher towards its center and changes in average 25−30 °C/km, which is called geothermal gradient [8]. At regions with high gradient large temperatures can be reached at shallow depths, thus it is less costly to reach the geothermal energy reservoirs than in case of low gradient. The maximum target temperature is usually 300 °C [8]. Resources with temperatures lower than 150 °C are exploited for heating purposes (in industries, spas and housings) and those with temperatures above 150 °C can be used for electricity production. In that case, some geothermal wells are mostly deeper than 2 km, but rarely more than 3 km [8]. After drilling a deep well with the goal to produce electricity the remaining engineering issue is to extract the heat from the deep underground. Utilisation of geothermal energy is conducted with a carrier (e.g. water in liquid or vapour phase) which transfers the heat from deep hot zones to, or near, the surface. The hot water can directly be extracted from aqueous reservoirs (deep aquifers) which are naturally present underground, or artificially created ones. After the heat exchange at the surface this water is sent back to the underground [9]. Another approach to extract the heat is to create a closed water circuit, or so-called Enhanced Geothermal System (EGS): The water is passing through an injection and production well which are connected through a network of fractures in rock layers existing between the bottoms of the wells. The installations at the surface close the water circuit. Before building such a system a certain technical and economic feasibility of geothermal power production at a certain location is required. In Switzerland this has not been evaluated yet in extended manner [4]. Still, there were and are ongoing projects dedicated to the development of geothermal systems (Deep Heat Mining Basel 2010, St. Gallen 2012, Haute-Sorne, Avenches, Etzwilen, Pfaffnau and Triengen planned from Geo-Energie Suisse Ltd., etc.).

Exploitation of deep geothermal energy requires drilling of wells through hard crystalline rock where high temperatures are present. The first step toward drilling a well is drilling a conductor hole (up to 12 m) with a

usually rented drill rig. Afterwards, the drilled hole is secured with a casing and concrete. This step is repeated for a couple of times, resulting in an narrower and deeper hole. The hole is continuously stabilized and cleaned from cuttings during drilling using drilling mud, which is filtered and cleaned from unwanted drilling products at the surface in order to reuse it. The narrowest and deepest hole is the last drilled hole named production line and it is used to extract the heat at the end. Before each cementing step, the drill bit has to be taken out of the hole, which costs drilling time (the so-called tripping time). Additionally, a conventional drill bit has a life time of around 20−40 m of drilling, thus its replacement is sometimes required before completion of a drilling step. This increases the tripping time and also the drilling costs. Other common problems during drilling are loss of circulation of the drilling fluid, wellbore instability, incomplete cementing, pipe stuck [10, 11, 12, 13]. Loss of circulation is related to losing the drilling fluid used to remove the created cuttings in the already existing rock fractures when circulating through a wellbore. Wellbore instability results in a wall break (falling in) due to an unbalanced pressure in the well [11]. This can be caused by drilling that destabilise the rock deep underground and cause breaking of the well after a certain period of the heat extracting. Altogether, it was concluded that the drilling can account for up to 50 % of the overall investment of a geothermal power plant [14].

For accomplishing the drilling task in an energy-efficient and cost-effective way, existing drilling methods should be improved, or new ones developed. Flame-jet drilling presents one of the developing technologies that has a goal to solve some of above mentioned problems (wellbore stability, necessity for tripping, faster drilling) and to reduce the drilling costs of geothermal wells. It is a contact-less drilling technology where a flame jet impinges the rock surface, generates a high temperature gradient in the rock surface layer and causes localized fracturing due to the created thermal stresses. As a result, small flakes (so-called spalls) are ejected from the surface. The mechanism of spalls ejection is called thermal spallation. If the thermal stresses cannot be generated no thermal spallation will occur. The technology is already known for more than 50 years [15], but it was mostly used for

shallow drilling and mining purposes. The deepest hole ever drilled with this technology was 331 m deep with a maximum velocity of 30 m/h [16].

There are couple of aspects how the flame-jet drilling technology can reduce the drilling cost. Pressurized rock deep underground is naturally under stress, which could be beneficial for the flame-jet drilling, but usually not for conventional technologies [17] due to increase of rock strength [18]. The pre-existing stresses could positively influence generation of thermal stresses cause by a flame jet. Also, the well diameter can be adjusted with drill bit moving velocity, without the requirement for adjusting the drilling tool. It is also possible to adjust the shape of the hole according to the present local stresses, which could maybe improve the stability of the well. Potter et al. [19] mentioned that drill bit stability is an advantage of flame-jet drilling due to no direct contact with the rock and no abrasion, thus no drill bit replacement is required while drilling and the tripping time can be reduced. Also, there is no need for drill head rotation, which simplifies the drilling procedure [19, 20].

During previous development of the drilling technologies it was noticed that flame-jet drilling has a drawback. Not all rock types are spallable [21]. For example, sandstone belongs to the soft rocks which are easy to treat mechanically, but the thermal stresses required for thermal spallation cannot be generated because of the soft mineral structure. Also, certain rock types melt before the generated stresses become strong enough to start the spallation. The molten rock cannot further be spalled and the flame-jet drilling cannot continue, even after cooling it down. But on the contrary rocks like granite have high hardness which is a rock property that makes the drilling with conventional techniques more difficult [22], but at the same time they are the most suitable for flame-jet drilling technology due to generation of larger thermal stresses.

The final goal of flame-jet drilling is to reach large depths (larger than 2 km), thus a first flame-jet drilling test was performed at the conditions of deep underground (pressure up to 400 bar and temperature up to 500 °C). For that reason, a high pressure setup for drilling in aqueous environment

had previously been built at the Institute of Process Engineering (IPE), ETH Zrich. Unfortunately, the test resulted with no cavity formation in a rock sample. Also, the experiments in the high pressure setup are time and energy demanding, the transferred heat from the applied flame jet can not be easily measured, and there is no optical access to follow the flame-jet drilling progress. Thus, the reason for the unsuccessful drill test could not easily be found. Therefore, the lack of basic knowledge about flame-jet drilling and necessity to further develop the technology led to the presented thesis.

1.1 Objectives

It was already shown that flame-jet drilling is a successful drilling technology when applied on rock formations [16, 23, 24]. However, there are many process parameters that can be adjusted about an impinging flame in order to make the system applicable. Browning et al. adjusted fuel and oxidant flow rate in order to influence the jet velocity and combustion power of the flame [16]. But any effects on the heat transfer and thermal stress development in the rock have not been studied in detail. Therefore, the main goal of the thesis was to find the most influencing parameters on the flame-jet drilling process and understand their impact. As a result, a path toward improvement of rock drilling with the mentioned technology can be defined and be useful for further research. The experiments should be fast, flexible in terms of combustion power, distance between the rock and the exit nozzle, diameter of the exit nozzle and size of rock samples.

The drilling results can be analyzed and explained only if a proper performance indicator for the drilling progress is defined. Usually, drilling is characterized with the rate of penetration (ROP) that defines the drilling speed in m/h when penetrating through a rock with a drill bit. However, this indicator is not fully capturing the flame-jet drilling performance in the laboratory. As example, the resulting hole diameter is also of interest

while drilling and it is dependent on nozzle diameter and penetration time, thus a proper performance indicator has to be defined which, if possible, can gather more drilling effects and also be related to conventional drilling.

It is important that the tested rock samples are representative for geothermal projects in Switzerland. Also they should be spallable, not to melt at applied conditions and well described rock type with known physical properties. Also the size of the rock samples plays an important role. While performing drilling experiments they should be easy to handle, which means to be carried and quickly replaced under the flame jet. However, the smaller the rock probes, the easier the thermal stresses are released by breaking the whole rock sample, which stops further thermal spallation. This has to be avoided when drilling the rock probes. Therefore, there is a conflict between last two requirements and a compromise has to be found.

The influence of heat transferred between the impinging flame jet and rock surface on the developed thermal stresses and flame-jet drilling has not been quantitatively described before. It is dependent on firing rate, dimensionless distance and nozzle diameter, thus a heat transfer study is required that can describe the influence of the process parameters on the heat transfer.

Knowing that thermal stresses are the main cause of thermal spallation, it is assumed that pre-existing pressures on the rock sample would influence their development. Uniaxial pressure is a parameter that increases with depth and causes the increase of stress tensor within the rock, but its influence on flame-jet drilling is unknown. Therefore, an additional objective of the thesis is to predict the behaviour of flame-jet drilling process when drilling in large depths (up to 2 - 3 km).

Finally, after experimental and theoretical analysis of the flame-jet drilling process the work should be concluded with a recommendation how to optimize it and proceed with the drilling technology development.

1.2 Outline

The thesis starts with a chapter on basics and state of the art of flame-jet drilling that helps to understand the performed steps and obtained results (chapter 2). In order to perform fast and flexible experiments in terms of combustion power, distance between the rock and the exit nozzle, diameter of the exit nozzle and size of rock samples, a new setup has been constructed. It operated at ambient pressure and it was an open system, thus the treated rock sample could be visually observed while drilling. More details about the setup are presented in chapter 3.

The heat transfer study is more significant for the flame-jet drilling process if the surface temperature of the target (applied heat flux sensor) has a similar value as the temperature of the rock sample when the first spall is ejected (spallation temperature). The spallation temperatures were obtained by a nonintrusive measuring technique, known as infrared thermometry. For that purpose the emissivity of the rock sample had to be previously determined, which is described in section 4.1. The surface temperature measurement is explained in section 4.2.

For the heat transfer study, a heat flux sensor was designed and constructed which withstands high temperatures and high heat fluxes. The heat flux sensor and the heat transfer study are presented in chapter 4.2.3

Afterward, flame-jet drilling experiments were performed and presented in chapter 5. The drilling performance depending on firing rate, nozzle diameter, distance between the exit nozzle and target and uniaxial load applied on rock samples was analyzed. The obtained experimental results were explained with thermal stresses calculated from a mathematical model, which helped to extend the already existing theories of thermal spallation.

At the end, the technology was applied for drilling a larger rock probe in order to test the durability of a drilling device that could be applied in field. The constructed burner and the experiment are described in chapter 6.

The thesis is concluded with its main discoveries (chapter 7). Further potential steps to proceed with flame-jet drilling technology research are proposed in the outlook of the thesis (chapter 8).

Chapter 2

Basics and state of the art

2.1 Flame-jet drilling technology

Flame-jet drilling presents a drilling technology of hard crystalline rocks by an impinging flame-jet. It is a process of continuous rock disintegration based on the physical phenomenon called thermal spallation. As a final result a hole is created. In this chapter, the current theory on the thermal spallation mechanism is presented. Furthermore, the development of flame-jet drilling technology in the past years is summarized.

2.1.1 Current theory on thermal spallation mechanism

If a brittle solid body of low thermal conductivity is suddenly exposed to high temperature (hot thermal shock) the increase of the temperature inside the body is much smaller than at the surface. This results into a steep temperature gradient beneath the surface. As a result, thermal stresses are generated due to a difference in expansion of body parts. Here, thermal stress is a common name for tensile and compressive stress. They cause fracturing, enlargement of the fracture and finally body breakage. The elements at the surface expand in all directions but the cold elements prevent the expansion causing the compression of the heated ones. As a

result, a very thin surface zone is under compression, with a steep stress change along the axis normal to the surface. The expanding hot elements pull the deeper cold ones creating a tension zone with less steep stress gradient [25]. As an example, if a sphere is exposed to a thermal shock the surface layer would be under tangential compression and the core in radial tension [26]. Finally, when the compressive stress is larger than the tensile strength of the material, a fracture will occur [27].

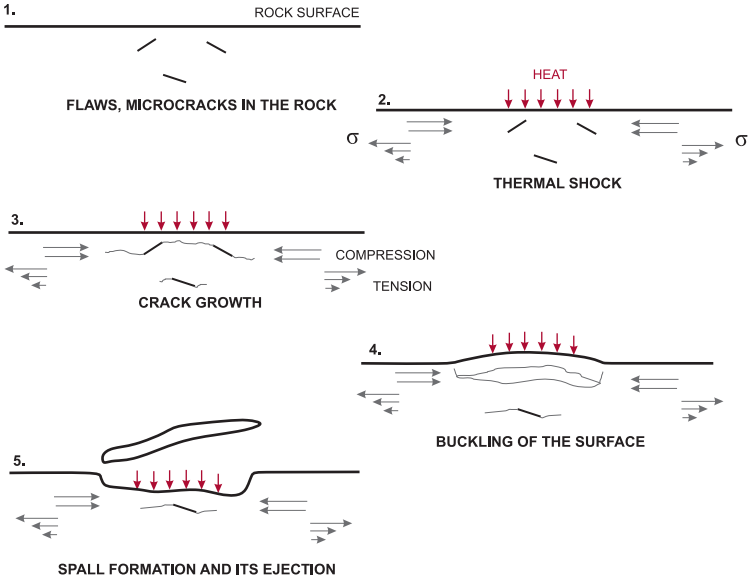


Figure 2.1: Schematic overview of thermal spallation mechanism presented with steps from heating the rock surface toward spall ejection.

In thermal spallation, the same phenomena occur but on microscopic scale and very locally, as presented graphically in Fig. 2.1. After exposure of a hard crystalline rock to an impinging flame jet thermal stresses arise due to difference in expansion of grains and crystals. Also, rock is full of randomly oriented microcracks and flaws [24] that initiate crack propagation in the compressive region as explained by Preston and White [26] (see

Fig. 2.1, step 1 and 2). Compressive stress does not break the material directly. Under compression, tensile stresses arise that actually propagate rock fracturing and its failure [26]. The weaknesses can interact with each other and grow (Fig. 2.1, step 3) as experimentally shown on thin plates by Nemat-Nasser and Hori [28]. They proved that a pre-existing crack in a brittle material extends in the direction of the applied compressive stress. In case of a flaw being not parallel with the surface, shear stress at both ends of the crack indirectly causes its propagation in alignment to the stress field. According to Rauenzahn the same phenomenon is present in thermal spallation forming a near surface crack parallel to the surface plane (see Fig. 2.1, step 3) [24]. When the crack reaches a critical diameter, buckling of the incipient spall takes place (Fig. 2.1, step 4). Fairhurst and Cook [29] concluded that under compressive load, the presence of many interacting microcracks all extending in the same direction will initiate buckling parallel to the nearest free surface. Keer et al. [30] concluded that buckling will occur only if the distance between the crack and the free surface is less than 10 % of the crack length. At the moment the spall disconnects from the core material it is violently expelled from the surface due to rapid release of the elastic energy present in the spall. The spall is consequently taken away with the flow of the exhaust gases (Fig. 2.1, step 5).

The above is an overall explanation of thermal spallation mechanism as summary from literature work. For the first time, thermal spallation was observed by Norton et al. 1925 [31]. Their theory is based on shearing mechanism between layers which was experimentally observed by heating a brick. Later, Preston and White were the first ones who proposed a contra-theory about spalling stating that rock fractures grow because of the high compressive stresses [26], as already explained. In addition, they described also the relationship between heating rate, spallation time and spall thickness. It was stated that both the spallation time and the spall thickness were disproportional to the applied heat flux. Thirumalai [32] experimentally studied well spallable rocks measuring a temperature change beneath the impinged surface. He concluded that considerable temperature change occurs only up to short distance (approximately 1 mm). From

experimental results, Preston et al. [33, 26] concluded that the size of the heated area may only be large up to 10 % of the sample surface in order to guarantee zero far-field displacement. Far-field displacement means expansion of elements at the boundaries of the rock sample, which essentially results in release of stress. This also implies that the required thermal stresses for onset of spallation must be reached, before more than 10 % of the sample volume is affected by the heating.

The rock mineral composition and structure are important for thermal spallation, but until today, their influence on the mechanism has not completely been understood. The stresses due to grain-grain interactions arise primarily from different coefficients of thermal expansion between the grains [24]. For example, it was noticed that quartz and nepheline promote spallation, while mafic, micaceous, and other soft, pliable minerals have an adverse effect on thermal spallation [21]. Freeman et al. [34] explained that quartz, due to large expansion coefficient, induces spallation in a polycrystalline rock such as granite, but biotite (mica) and feldspar inhibit spallation due to fusion at lower temperature. In sedimentary rocks, which are consisting mostly of rock parts and minerals bounded (cemented) with soft clay, localized thermal stresses can not be generated like in igneous rocks, which are formed from molten magma and which are much harder than sedimentary rocks. Therefore, they are not appropriate for flame-jet drilling [24]. Calaman and Rolseth [15] reported that thermal spallation is influenced by numerous rock properties whose interaction is very complex. Therefore, they defined a property called spallability, which is directly proportional to thermal diffusivity, expansion coefficient (Young's modulus) at critical temperature, grain size and reciprocal compressive strength at critical temperature (temperature at which the rock becomes elastic or sufficiently plastic to prevent further spallation). The larger the spallability of a rock type, the more efficient flame-jet drilling can be assumed for that rock type. On the contrary, Soles and Geller [21] concluded that there is no single rock property that can be related with a good or bad thermal spallation.

Based on the presented theory about thermal spallation, the first simulation

of the complete flame-jet drilling process was taken by Rauenzahn. He developed a model which was supposed to predict flame-jet drilling results [24]. He applied a Weibull failure theory to explain the crack initiation and spall formation, as suggested by Morvelli et al. [35]. It is a statistically based method that considers the number, orientation and size distribution of cracks. At the end the distribution of failure under compression was predicted semi-empirically. A prediction of the hole diameter was very successful, but the predicted heat transfer rates were not aligned with the experimental data.

The thermal spallation mechanism was further investigated and simulated by Germanowich et al. [25]. Their work was based on development of a $2-$D micromechanical model of thermal spallation in the compressive region below the surface in order to explain the growth of one crack. According to Germanowich et al., the crack growth is occurring in the compression zone beneath the heated surface. They also concluded that the magnitude of generated stresses and their distribution describes the spallation properties of rock. The model is very simplified and did not consider many phenomena present while spalling (spalls removal, grained structure, and presence of multiple cracks). However, it is useful to understand the stress development in brittle media when exposed to heat and to describe the influence of generated stresses on crack propagation. Germanowich et al. are the first who reported the existence of the tension zone when spalling beneath the compression zone, but they did not give much attention to it and neither to a possible relation with the thermal spallation mechanism.

Even though many researchers tried to find parameters that influence flame-jet drilling, there are still phenomena that have not been explained yet to full extent. Therefore, an improved modelling of the flame-jet drilling process and prediction of drilling results is not possible yet. Also, none of the previous works was committed to examine the influence of the heat flux on the thermal spallation. Only Rauenzahn conducted a heat transfer study at few conditions and used it to interpret the experimental flame-jet drilling results with the help of a model.

2.1.2 Development of the flame-jet drilling technology in the past - dry flame-jet drilling

Flame-jet drilling was for the first time practically applied in 1937 by the Soudan Mine of the Oliver Iron Mining Company near Tower, Minnesota [36]. The results were not much successful because an oxy-acetylene torch was used which melted the rock. The development of the technology continued and it was greatly improved with application of rocket burners. They enabled to combine high temperatures and high momentum in order to achieve spallation of the rock and not its melting. This improved drilling velocity and also made the technology more efficient. The technology was applied mostly for mining purposes, tunnelling, blasthole drilling or quarry channelling in that time [15]. Around 1950 it became commercially applicable [15] and it was applied all around the world for around 30 years. Other names for the technology are jet piercing, jet channelling, rocket jet mining and quarrying, flame cutting, or thermal rock fragmentation. Countries like Russia, Canada, Germany, England, South Africa, and France were all working on development and application, but the most progress was made in the USA [37]. The most successful companies in the USA were The Linde Air Products Division of Union Carbide Corporation (Tonawanda, New York, from 1930), Browning Engineering Corporation (Hanover, New Hampshire, from 1961) and Flame Jet Partners Limited (Encino, California, year 1976) [38, 37]. A maximum depth of 331 m and a drilling velocity of 30 m/h were reached by Browning Engineering Corporation [37]. In 1977, James Browning also managed to apply the drilling technology for nine straight hours through 426 m of polar ice [39]. After the crash of oil prices in 1980, flame-jet drilling companies were not financially supported anymore by the government, thus the flame-jet drilling development and the research were stopped for a while. Later in 2004, Potter Drilling Inc. from California was founded, which applied thermal spallation again for drilling purposes. Their goal was to reduce drilling costs of a geothermal well by up to 50 % compared to conventional drilling.

Flame-jet drilling equipment by Browning Engineering

The drilling system developed by Browning Engineering, which was commercially available, is presented in Fig. 2.2. For convenient transportation the system was mounted on a truck. The core of the system was the burner where the combustion took place. It was a simple construction made of steel tubing that was hard-coated at the bottom to reduce its erosion. As a fuel, No. 2 fuel oil was used together with air. According to the reported work of James Browning, drilling of granite was very successful when the burner exit nozzle was designed in a way to generate supersonic velocities [16]. The burner was mounted on the lower end of a hose. Also water was used for quenching of the hole walls, for noise and dust spreading reduction and cooling of the exhaust gases in order to protect the hoses made of a rubber. The water jets were located above the burner in order not to interact with the impinging flame jet [37]. With this system a hole of 331 m was drilled [37].

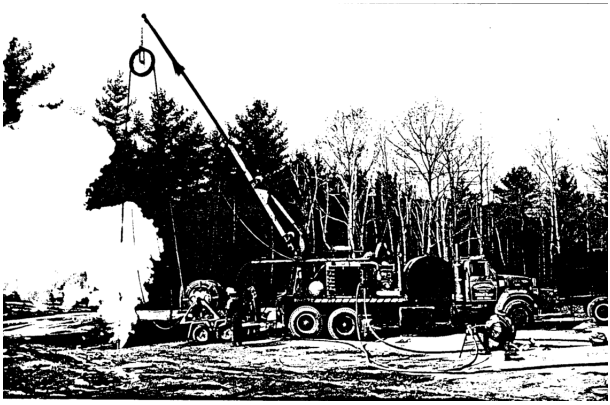


Figure 2.2: Flame-jet drilling rig constructed by Browning Engineering [37].

In general, Browning did many tests varying the oxidizer (pure oxygen, nitrogen diluted oxygen or air), nozzle design, burner design and fuel. He also

inserted abrasive particles in the exhaust flow to enable drilling also of non-spallable rocks. According to Rinaldi et al. [37] the relative drilling costs were much lower compared to the systems constructed by Linde Division and Flame Jet Partners.

Flame-jet drilling equipment from Potter Drilling, Inc.

Potter Drilling, Inc. had the goal to drill geothermal wells of a couple of kilometres of depth with a hot impinging jet. In order to stabilise the borehole they decided to fill it with a water circuit, which also removed the created cuttings while drilling. Based on these ideas a new drilling system was developed, which was called hydrothermal spallation tool (see Fig. 2.3) [40]. Simply explained, it is a hot jet that exists in aqueous environment. A mixture of 38 mass % peroxide, 12 mass % methanol and 50 % deionised water was passing through Instant Steam catalyst in order to start a flameless catalytic oxidation producing a hot jet that reached 800 °C [23]. The heat flux from the hot fluid (mostly steam) to the rock surface was transferred in a more efficient way (up to 20 MW/m^2) than in case of the dry flame jet used by Browning Engineering, due to increased density, volumetric heat capacity and thermal conductivity of the impinging phase. The reacting system could be applied at pressures up to 240 bar which represents conditions deep downhole (equivalent to up to 2.2 km depth). To cool down the reaction products mixed with the rock particles, additional cold water was injected in upward direction. The spalls were separated from the water at the surface and the water was sent back to the drill head. The bottom hole assembly (the reactor) was connected to a coiled tubing unit placed on a truck. The reactants were prepared at the surface and pumped to the catalytic bed with a flow rate of 3.8 L/min. The reaction products were passing through a 4.8 mm diameter exit nozzle creating a cavity of 102 mm in diameter.

Figure 2.3: Hydrothermal spallation drilling system constructed by Potter Drilling, Inc. [41].

2.1.3 Flame-jet drilling in aqueous environment

In a deep borehole (deeper than 2.5 km), pressure can reach 260 bar and temperatures of 400 °C. In order to simulate those conditions in laboratory and to test the flame-jet drilling technology under such a conditions a high pressure plant was constructed at IPE by Stathopoulos [42]. He developed a setup where a flame as a product of hydrothermal oxidation exits the reaction zone through a nozzle entering an aqueous environment. The reactants are water-diluted ethanol (up to 32.5 %) and gaseous oxygen. A first experiment with an unconfined rock sample resulted in detection of a few spalls. Later, the experiment was repeated with a confined rock sample to prevent the release of stresses required for spallation. Much more rock mass was removed. Any further optimisation of hydrothermal flame-jet drilling process has not been performed yet due to the complex experimental procedure (preparation for the experiment, sealing test, eventual equipment damages) and deficient number of measuring devices that can operate at high pressure and temperatures.

2.1.4 Particle size distribution

While drilling a borehole, there is not much information that can be obtained in order to indicate the drilling success. It is difficult to estimate the direction and the velocity of penetration, which is however important for the movement of the drill head. The only indication that the thermal spallation takes place are the resulting spalls. According to literature [16], the spalls can describe the quality of heat transfer between the hot impinging fluid and the rock surface. Browning et al. [16] reported that the heat transfer influences temperature gradient in the rock and the spall size, too. He noticed that particles of dust size are the result of high heat transfer from a supersonic flame jet and a steeper temperature gradient in the rock surface layers. Unfortunately, no further investigation about the resulting spalls was conducted. Preston et al. concluded that the spall size is related to stress gradient in the rock surface. In case of low stress gradient the spalls are thicker and broader compared to steep stress gradient. Afterward, Dey and Kranz [43] observed spalls (which they called flakes) with a microscope and with a high speed camera while spalling. They were also comparing the ratio between width and length of the spalls, which was around 0.6 in average. This experimentally obtained result was in agreement with theoretical predicted result (0.577) when simulating the mechanism proposed by Preston et al. [26]. Combining the presented conclusions, a high temperature gradient is supposed to induce high stress gradient, which results in smaller spall size.

The latest work that reports characterisation of spalls was done by Potter et al. [40]. They reported particle sizes from 20 µm up to 1 mm, but the largest fraction was in the range between 100 and 500 µm. They concluded that the particles were small enough to be lifted and transported by water, which was the most important issue for them. The resulting particle sizes were also applicable for the model development of Walsh et al. [44] that predicts spalls production.

2.2 Combustion and circular jet flow dynamics

Combustion is an exothermic reaction between a fuel and an oxidant. The energy required for initiation of the reaction is provided by temperature increase in form of a spark, heater, or a supporting flame. As a result the hot gaseous products exit the reaction zone (usually recognized as a flame). Some properties of the flame are temperature profile, concentration profile of present species, flame speed, released heat, stability, size, shape, and colour [45, 46]. The properties can be influenced by type of the fuel, type of the oxidant, flow rates of reactants and their ratio, flow dynamics, mixing quality of reactants, temperature of reactants, etc.

2.2.1 Combustion reaction

In this research the reactants of combustion were methane (fuel) and air (oxidant). The chemical equation for the complete oxidation of methane can be written as follows:

$$CH_4 + 2(O_2 + \frac{79}{21}N_2) \rightarrow CO_2 + 2H_2O + 2\frac{79}{21}N_2 \tag{2.1}$$

In case the oxidation is incomplete (e.g. because of insufficient mixing of fuel and oxidant or insufficient amount of oxidant) products of intermediate or side reactions that can remain are amongst others: CH_3^+, CH_2O, HCO^+, CO, NO and NO_2.

Gross heat release rate, or firing rate ($\dot{Q}_f$ in kW), can be calculated according to Eq. 2.2 as suggested by Baukal [47]:

$$\dot{Q}_f = \dot{m}_{CH_4} \cdot HHV \tag{2.2}$$

where $\dot{m}_{CH_4}$ presents the mass flow rate of methane in kg/s and HHV is the higher heating value of methane (55.5 MJ/kg).

The combustion can be influenced by the amount of injected air. The flame temperature, composition of exhaust gases and also physical properties of a flame (length, shape, colour) are usually all depending on the amount of reacting air. Therefore, there is a physical property called air-fuel equivalence ratio (λ) which is calculated as in Eq. 2.3:

$$\lambda = \frac{\frac{m_{air}}{m_{fuel}}}{(\frac{m_{air}}{m_{fuel}})_{stoich}} = \frac{m_{air}}{(m_{air})_{stoich}} \tag{2.3}$$

where the subscript *stoich* is denoting the stoichiometric amount of air required for complete combustion of methane.

The way reactants are mixed influences the combustion considerably. Consequently, two distinct types of flames are premixed and diffusion flames. In case of a premixed flame, reactants are mixed before the ignition. Better mixing of reactants improves the combustion efficiency. The main disadvantage of premixed flames is the possibility of a flashback (described and studied in literature [48]). This can be caused by an imbalance in the local flame and flow speeds, thus the flame propagates in upstream direction through the hose of reactants, potentially causing an unwanted damage of the equipment (explosion) and hazard for operators [49]. If a fuel and an oxidant diffuse into each other and combust at the same time the resulting flame is called a diffusion flame. Those flames are quite stable and flashback is less possible, but the combustion is heavily dependent on the mass transfer between fuel and oxidant phase, thus it can be less efficient.

2.2.2 Fluid dynamics of a round free turbulent jet

The high temperature products generated in a combustion chamber can be released through an exit nozzle. In that case the resulting flow field

downstream of the nozzle depends on initial and boundary conditions [50]. One way to describe the flow of compressible fluids is as subsonic (being turbulent or laminar), sonic or supersonic, indicating the range of developed velocities. The subsonic flows reach velocities below the speed of sound and the supersonic flows higher than the speed of sound (c in m/s) at temperature and pressure at the nozzle. This can better be characterised with the Mach number as calculated in Eq. 2.4:

$$Ma = \frac{w_0}{c} \tag{2.4}$$

where w_0 represents the velocity at the nozzle in m/s. For subsonic flow $Ma < 1$, sonic flow $Ma = 1$, otherwise it is supersonic.

When a subsonic jet exits the nozzle it spreads radially entering a receiving fluid. Three regions are defined in an axisymmetric round jet: potential core, transition and fully developed region [51] as presented in Fig. 2.4. The potential core exists directly after the exit nozzle and it is characterised by the fluid velocity being not less than 95 % of the jet velocity at the nozzle exit [52]. The length of the potential core is up to 6.2 x D_{nozzle} [53] independent on Reynolds number, and its width reduces with increasing distance from the nozzle because of entrainment of the surrounding air into the shear layer. The shear layer is highly turbulent because of velocity differences between the ambient air and the jet. The entrainment of air increases the jet mass flow downstream of the nozzle, but flattens the jet velocity profile [54]. Subsequently, the jet enters the transition region, where the centreline velocity starts to decay. It is followed by a fully developed region with the maximal velocity at the centreline of the jet, which decreases with increasing distance from the exit nozzle. Important to notice is that the static pressure remains constant along the flow.

Free jets can additionally be described with another dimensionless number called Reynolds number (Re_0) at the exit nozzle, which is defined as in Eq. 2.5:

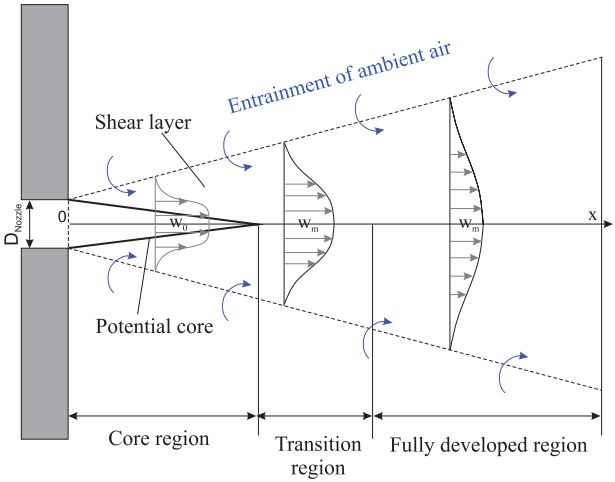


Figure 2.4: Turbulent round free jet [55].

$$Re_0 = \frac{w_0 D_{nozzle} \rho_0}{\eta_0} \tag{2.5}$$

where w_0 is jet velocity at the exit nozzle in m/s, D_{nozzle} is the exit nozzle diameter in m, ρ_0 is the fluid density in kg/m^3 and μ_0 is the dynamic viscosity in m^2/s at the exit nozzle conditions.

If the jet flow is observed as steady, one dimensional, adiabatic and as an ideal gas, the velocity at the exit nozzle can be calculated as in Eq. 2.6:

$$w_0 = \sqrt{2 C_p T_0 (1 - \frac{T}{T_0})} \tag{2.6}$$

where C_p is the heat capacity in J/(kg K), T_0 is the temperature at the exit nozzle and T is the temperature in the reservoir (in this case the combustion chamber) in K. In case the temperature at the nozzle and in the reservoir are not available, the temperature ratio can be calculated as in Eq. 2.7

$$\frac{T}{T_0} = \frac{p}{p_0}^{\frac{\gamma-1}{\gamma}} \tag{2.7}$$

where p is the pressure at the exit nozzle and is p_0 is the pressure in the reservoir. The isentropic expansion factor γ is also known as the heat capacity ratio (C_p/C_v).

If the pressure ratio between the chamber pressure and the pressure of atmosphere, where the exhaust gases are released to, is higher than certain critical value, underexpansion of the flow can occur. In case of air, the critical pressure ratio is 1.893 [56]. In the chamber, along the centreline of the flow in downstream direction the pressure decreases until the exit nozzle, but in case of underexpansion the pressure at the nozzle is still higher than the ambient one. As a result, instead of having a smooth pressure drop after the nozzle exit, in the underexpanded flow the pressure oscillates along the jet axis [56] and it can reach supersonic velocities ($Ma > 1$). The sudden expansion of the jet occurs first and it is followed by periodical compression and expansion until the static pressure in the centerline becomes equal to the ambient pressure. The flow is also called shocked flow due to resulting and visible shock cells. An illustration of an underexpanded jet is presented in Fig. 2.5 according to the description given by Franquet et al. [55]. When the expansion acoustic waves (expansion fan) reach the constant pressure streamline, they are reflected into compression waves. They converge towards the inner jet and coalesce forming an oblique shock. Afterward a new expansion fan is initiated.

When the flow is underexpanded, the pressure at the exit nozzle is equal to the Laval pressure p_s and the velocity at the nozzle is equal to sonic velocity. For underexpanded flow, pressure p_s in Pa and sonic velocity w_s in m/s at the exit nozzle can be calculated as given in Eq. 2.8 and Eq. 2.9, respectively.

$$p_s = p_0 \frac{2}{\gamma+1}^{\frac{\gamma}{\gamma-1}} \tag{2.8}$$

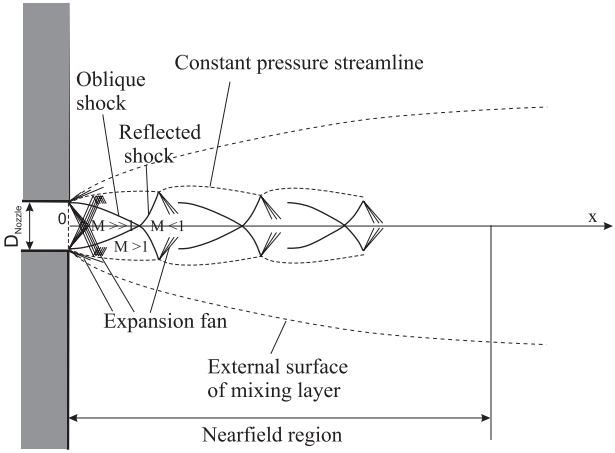


Figure 2.5: Structure of a moderately underexpanded free jet [55].

$$w_s = \sqrt{\frac{\gamma R T_0}{M}} \tag{2.9}$$

where R is the ideal gas constant ($8.314\,\mathrm{J/(mol\,K)}$) and M is molar mass in mol/K of the gas passing through the nozzle.

2.2.3 Fluid dynamics of an impinging jet

During flame-jet drilling, the flame jet impinges a flat rock surface and causes thermal spallation. In order to understand the occurring phenomena and the heat transfer to the rock surface the knowledge reported in literature about impinging flame jets is be presented in this chapter. Impinging flame jets and the resulting heat transfer phenomena were widely observed in order to describe existing heat transfer mechanisms [57], influencing parameters on heat transfer [58], flow structure of an impinging flame jet [59, 60] and theoretical heat transfer to an impinged surface [61, 62]. All those works are summarized in reviewing publications [63, 64, 65, 66]. Over-

all, impingement of isothermal jets (e.g. air) was a more frequent study subject [67, 68, 69, 70, 71] because of lower medium temperatures and thus, more convenient measurement. However, it was confirmed that independent on the impinging medium (air, or hot gaseous mixture in case of combustion) the flow is divided in the following regions: free jet region, stagnation region and wall jet region [66].

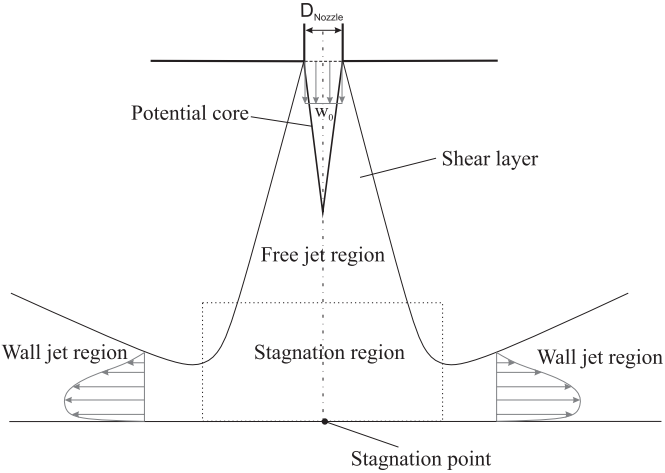


Figure 2.6: Structure of impinging jet.

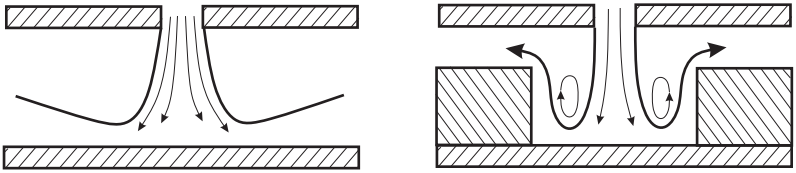
The free jet zone is the region that is largely unaffected by the presence of the impingement surface [68]. The potential core exists within the free jet region and the turbulence level is very low. Between the potential core and the ambient air there is a shear layer where the turbulence is more intense and the mean velocity is lower than at the nozzle exit (w_0). After the potential core the velocity along the centreline decreases downstream and the turbulence intensity increases. The stagnation region is characterized by a pressure gradient and axial velocity deceleration of the flow which stops at the stagnation point and turns radially outward. In the stagnation region the flow is deflected from the axial direction to the side along the target surface entering the wall jet region.

A characteristic of the impinging jet is the distance between the exit nozzle and the target surface is dimensionless distance (L), which is calculated as in Eq. 2.10:

$$L = \frac{S}{D_{nozzle}} \tag{2.10}$$

where S presents the distance between the nozzle exit and the target surface.

When drilling a rock sample a cavity is created and the jet becomes confined. Studies of confined impinging jets that impinge a cavity or curved surface are very limited. More attention was directed to confined air jets with a design as illustrated in Fig. 2.7a [72, 73]. In this case, the impinging jet contains the same regions as an impinging free jet.



a) b)

Figure 2.7: Example of confined impinging jets. a) Common example in literature, b) observed in this study.

Confined jet construction that is of interest in this study is presented in Fig. 2.7b and it has never been observed before. A similarly confined impinging system was investigated by Rothenfluh et al. and Schuler et al. [74, 75]. They performed an experimental and numerical heat transfer study of near-critical water jets impinging a flat surface at the end of an annular wall, which confined the impinging jet. The resulting trends were mostly in agreement with literature. Any influence of the confinement on the heat transfer was not reported.

Chapter 3

Flame-jet drilling setup

A flame-jet drilling setup (FJDS) was built to perform experimental research of flame-jet drilling technology at ambient pressure. Convenient and fast drilling experiments are possible due to simple replacement of rock samples, simple ignition, fast startup and shutdown procedure. Free optical access to the rock samples enables observation of the thermal spallation, thus temperature measurement of the rock sample with an infrared camera, or recording of the spallation process with a high speed camera is possible. The setup is also flexible regarding the size of the rock sample. The overall time consumption of drilling experiments is estimated to be around 20 times lower than for comparable experiments in the hydrothermal drilling setup (chapter 2.1.3) constructed for flame-jet drilling experiments at high pressures.

Setup is designed according to the dry flame-jet drilling tools described in literature. The driving parameters were those temperatures and velocities of the impinging medium that would rapidly heat up the rock surface up to 600 °C and a jet force of exhaust gases that can quickly remove the created spalls. Successful drilling in the past at ambient pressure was performed with firing rates up to 100 kW and flame temperatures above 2700 °C (propane and oxygen) [24]. Browning Engineering successfully drilled with a flame of more than 1 MW and velocities at the nozzle higher than 1000 m/s when combusting 151 L of No. 2 fuel oil [37]. Pumps

at the surface had an output pressure for air up to 48 bar and for the fuel 82 bar. However, Wideman et al. reported successful drilling also with a fluid temperature of 800 °C [23] using steam as impinging medium and generating velocity at the nozzle of 850 m/s. In this case, the heat transfer was very efficient due to more dense impinging fluid, thus high fluid temperatures as for dray flame-jet drilling were not required.

A schematic overview of the experimental system with a rock sample as a target is presented in Fig. 3.2. The combustion reaction takes place in the water-cooled combustion chamber (i.e. burner). The flow of the fuel (i.e. methane) and the air are adjusted with control valves. After each experiment the methane line was purged with nitrogen. The rock samples are positioned below the burner with a two-axial positioning system. The largest part of the equipment is placed in an insulation box in order to reduce the generated noise and to limit the area exposed to particles and dust produced while drilling. A view on the insulation box and the working desk with electronic box is shown in Fig. 3.1.

Figure 3.1: Flame-jet drilling setup inside of the insulation box (lefthand side) with the working desk (righthand side).

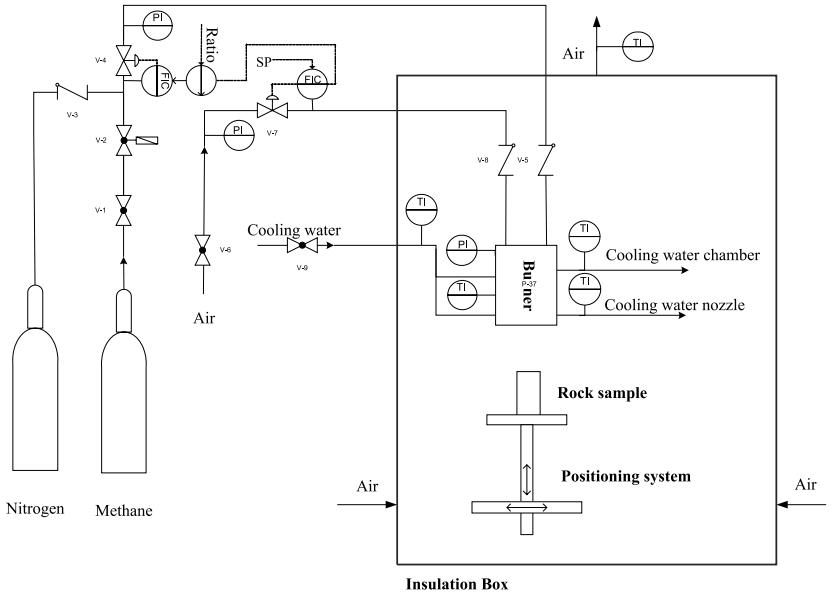


Figure 3.2: Schematic overview of the flame-jet drilling setup. Methane is used as a fuel and air as oxidizer to produce a flame that impinges a target from the top. The water line is required to cool down the burner.

3.1 Burner

Before entering the combustion chamber (see Fig. 3.3), methane and air passes through the mixing pipe placed at the top of the burner. Turbulence and mixing of the reactants is provided with a perforated 5 mm thick plate installed perpendicular to the gas flow. After the mixing pipe the reactant mixture enters the combustion chamber where ignition takes place. The installed igniter is a ceramic heater (Type EF, Bach RC, Werneuchen, Germany) made of silicon nitride (Si_3N_4) doped yttrium oxide (Y_2O_3) having a resistance of 40 Ω at 20 °C. It is connected to alternating current (AC) power supply and its temperature is adjusted by means of an AC transformer. The minimal electrical power required for ignition of the gas mixture is 205 W. The dimensions of the combustion chamber (inner diameter 80 mm and 140 mm in length) are chosen in a way to enable installation of a pressure sensor and igniter at the top of the chamber. The igniter is placed in an independent channel, thus its replacement can be performed without demounting the whole burner. When calculating the length of the burner it is considered how the released heat would be influenced by the cooling. In case of a sudden explosion the combustion chamber has to withstand high pressures. Therefore, a pressure test was performed up to 40 bar, which the burner successfully withstood.

The main role of the cooling water is to protect the material from overheating and to extend the life time of the burner. The flow rate of the cooling water was fixed being larger than the minimum required mass flow, which was calculated for maximal expected firing rate (70 kW), maximal possible flame temperature (1957 °C) and desired maximal inner wall temperature of stainless steel (300 °C). Finally, the mass flow was set to 0.221 kg/s.

The maximum firing rate is limited by the supply pressure in the air line. The air-fuel equivalence ratio (λ) could be varied from 0.9 up to 1.3 (the range of stable flame, no shut-off). For the performed experiments a value of 1.1 has been chosen because it provided the maximum heat transfer to the heat flux sensor at process conditions required for drilling (see chapter

4.3). For the selected λ, the firing rate ($\dot{Q}_f$) was varied between 23 and 54 kW.

The combustion products are directed to the bottom of the burner where an exit nozzle was positioned. Two convergent nozzles were used with a diameter (D_{nozzle}) of 7.9 mm (D_1) and 12.1 mm (D_2). Depending on the nozzle diameter and the firing rate the pressure in the combustion chamber (p_{ch}) varied between 1.54 and 3.25 bar. The diameters of the nozzles are chosen such that the resulting flame-jet has subsonic and sonic flow. The following equations are applied to approximate the velocity at the nozzle (w in m/s) and to calculate the resulting nozzle cross section area (A_{nozzle} in m^2):

$$w = Ma \cdot w_s \qquad A_{nozzle} = \frac{\dot{V}}{w} \tag{3.1}$$

where Ma is Mach number, which is equal to 1 when the flow is sonic at the nozzle, w_s is the sonic velocity obtained as described in chapter 2.2.2 in m/s, $\dot{V}$ is approximated volume flow rate of exhaust gas when the fuel is completely combusted in m^3/s. The properties of the exhaust gas components were taken from NIST Standard Reference Database. During design of the burner, the exact pressure in the combustion chamber was unknown, thus the volume flow rate was calculated with roughly approximated temperatures and pressures in the system. Afterward, according to the system properties measured during the combustion experiment (chamber pressure, chamber temperature, flame temperature at the exit nozzle) the resulting exit nozzle velocities are approximately up to maximum 850 m/s and Reynolds numbers up to 35 000 indicating that the flow is in turbulent regime.

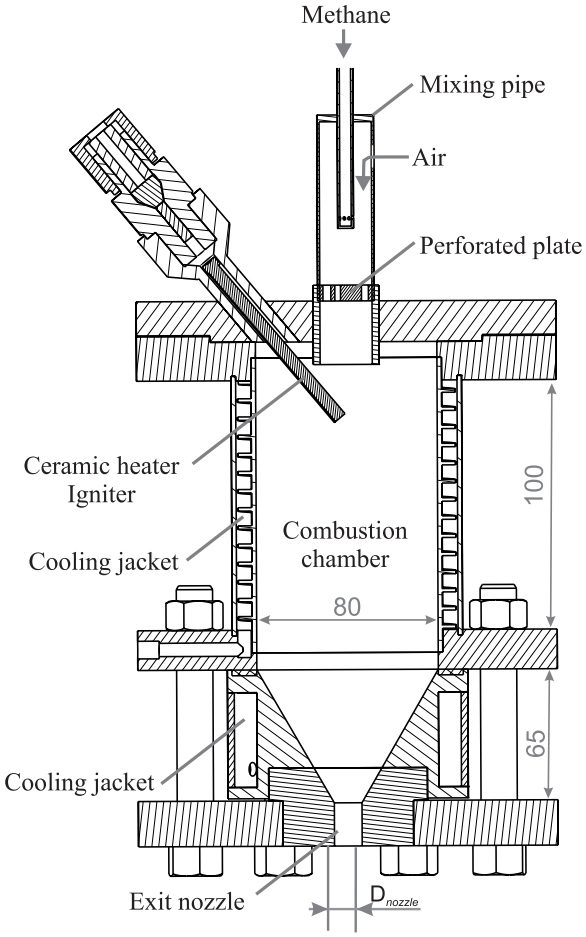


Figure 3.3: Schematic overview of the burner. Methane and air are mixed in the mixing pipe. The gas mixture enteres the combustion chamber where the ignition is initiated by a ceramic heater. The combustion products exit the burner at the bottom through a replaceable exit nozzle. The presented dimensions are in mm.

3.2 Piping

The fluids used in FJDS are methane as fuel, air as oxidizer, nitrogen to purge the fuel line and cooling water. The adiabatic flame temperature (T_{ad}) of methane when mixed with air at stoichiometric conditions is 1957 °C, which is not much different than other fuels used in flame-jet drilling experiments ($T_{ad}(propane)$ = 1980 °C [24]). The advantage of methane is lower risk of soot formation due to less present carbon atoms. Therefor it is decided to used methane as a fuel. Two methane bottles of 50 L are connected in series and placed in a separate cabinet for explosive gases. As oxidizer the more convenient solution is to use compressed air from the supply lines of the laboratory. The supply pressure of air is 7 bar, which limited the firing rate to a maximum of 54 kW at an air-fuel equivalence ratio of 1.1. Check valves are mounted on both gas lines (fuel and oxidizer) close to the combustion chamber. They prevent back flow in case of undesired pressure change in the burner or supply lines. The methane line is additionally secured with a fail-closed magnet valve which activates in case of power loss (or emergency stop). In addition, the same valve can only be opened by starting the ventilation system, which is a safety precaution to perform the experiments only when the ventilation works. After each experiment, the pipes were purged with nitrogen provided from a 50 L bottle. The cooling water line is directed in two ways: one to the cooling jacket of the combustion chamber and another to the conical part of the burner.

3.3 Insulation box

The FJDS produced a noise of 85 dB, which can cause hearing problems at extended exposure. Therefore it is placed in an insulation box which reduced the noise by 20 dB. In addition, area exposed to the dust and particles produced during flame-jet drilling experiments was confined. The nitrogen bottle, methane closet, electronics and flow controllers are placed

outside of the insulation box. The rest of the FJDS is placed inside of it. Also, 5000 m^3/h of air circulate through the insulation box, which prevents overheating of the equipment. All exhaust gases of the combustion are directed to the exit channel at the top of the box, which is connected to the ventilation system.

3.4 Control system

An electronic schema of FJDS is presented in Fig. 3.4. The electronic control unit (presented in Fig. 3.5) consisted of electronic devices that are required to provide safe and proper operation of the FJDS and its elements. It was supplied with 32 A current at 400 V. The power supply can be mechanically blocked by pressing an emergency stop button. The main part is the group of EtherCAT Terminals that send required signals to flow controllers, alarm lamps, switches and linear actuators; and received signals from flow controllers, thermocouples, pressure sensor and igniter. All installed terminals are listed in Fig. 3.4 with the marked connection to a corresponding device. The ceramic igniter is supplied with alternating current by an AC transformer. The applied voltage and current are measured with the EtherCAT terminals. In case some temperatures (cooling water temperature, temperature in the combustion chamber and the insulation box) exceeds their maximal defined values the Nanodac controller stops the electric supply of the fail-closed magnet valve, which blocks the flow of methane to the burner and shut down the flame. The maximal temperatures are defined in the Nanodac controller.

The mass flow rate of methane, air and nitrogen (for purging the pipeline) is measured and controlled with PID controllers (Bronkhorst High-Tech B.V.). The pressure in the combustion chamber (p_{ch}) is measured with a pressure transducer (STS Sensor Technik Sirnach AG, ATM.1ST). The temperature of cooling water before and after the burner ($T_{burner,in}$, $T_{burner,out}$) are recorded together with the temperature in the combustion chamber

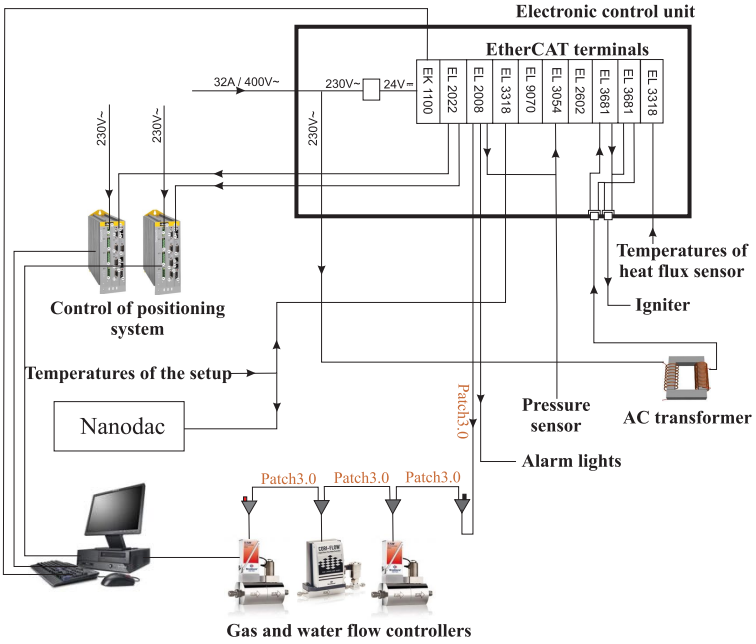


Figure 3.4: Electronic schema of the installed sensors and electronic devices of the flame-jet drilling setup.

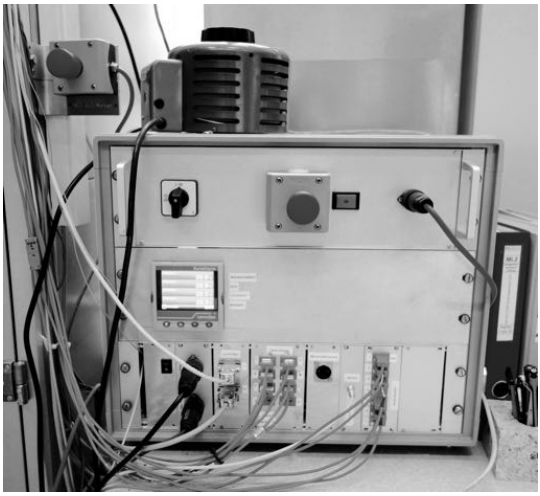


Figure 3.5: Electronic control unit with AC transformer on the top, connected sensors and Nanodac unit (with the screen).

(T_{ch}). The thermocouple that measures T_{ch} is placed at the beginning of the conical part with the tip 10 mm of lateral distance from the chamber wall. The temperature in the insulation box, at the exit of the air-exhaust gases mixture, is also monitored because it is not allowed to exceed 85 °C when entering the exit channel.

As a laboratory control system software, LabView (National Instruments) is used for measurement data acquisition, instrument control and system monitoring.

3.5 Positioning system

The positioning system consisted of two linear actuators (Schaeffler Group, INA) which are combined to reposition an object (e.g. a rock sample) in two directions: vertically and horizontally. The maximal vertical and horizontal

repositioning was 450 mm and 240 mm, respectively. The accuracy of the positioning system is 0.1 mm. The maximal moving speed is 0.2 m/s.

3.6 Safety considerations

The following safety precautions were considered when building the FJDS:

- The methane bottles are placed in an ex-protected gas bottle closet.
- The methane closet is connected to a ventilation system that is continuously operating and ex-protected.
- Concentration of the methane is measured in the methane closet and in the insulation box. In case of exceeding the low explosive limit for methane (5.0 vol.%) the methane flow would be blocked by the magnet valve and the alarm would be activated.
- In case the critical temperature of the outlet cooling water (80 °C), burning chamber (1100 °C) and insulation box (85 °C) are reached the combustion would be stopped by blocking the methane flow with the magnet valve.
- Before reaching any of the critical temperatures signalization lights will be activated in order to react, if possible, before the automatic shut down.
- The FJDS could operate only if the insulation box is ventilated. Otherwise, the magnet valve is closed and the combustion could not be started.
- To prevent a flash back or flame propagation in backward direction a perforated plate (described in chapter 3.1) is installed in the mixing pipe.
- Upward flow of gases is prevented with check valves in both gas lines (methane and air).

3.7 Setup performance analysis

After the construction and initial commissioning of FJDS, its performance was analyzed. The generated flow field was observed, flame temperature at the exit nozzle and pressure in the chamber were measured. Also, the preliminary drilling experiments are performed in order to select a rock type the most suitable for the study.

3.7.1 Flame description with resulting process conditions

The flame-jet generated after the smaller nozzle ($D_1 = 7.9\,\text{mm}$) was subsonic at 23 and 28 kW and underexpanded (shocked flow) at 33 and 38 kW. The underexpansion of the exhaust gases was clearly registered with an infrared camera as presented in Fig. 3.6.

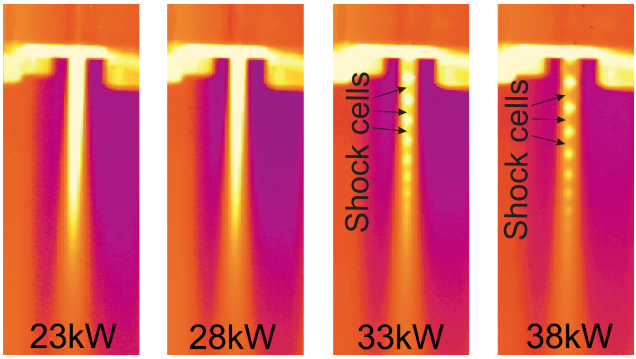


Figure 3.6: Infrared view of free jet at different firing rates. Exit nozzle was $D_1 = 7.9\,\text{mm}$. Underexpansion of the jet (shocked flow structure) occured for the firing rate above 31 kW.

The transition to underexpanded jet occurred at firing rate 31 kW when the pressure in the chamber (p_{ch}) reached 2.25 bar. At this point, instead

of a constant pressure decrease after the nozzle exit, the pressure oscillates along the jet axis, which results into a shock-cell structure of the flow. This periodical expansion and compression happened until the pressure in the exhaust gases became equal to the ambient pressure. Further increase of firing rate results in the more visible shocks. The camera registered more and less dense regions (light spots are more dense) of the exhaust gases that are related to the pressure oscillations present in the underexpanded jet [76]. With the larger nozzle $D_2 = 12.1\,\text{mm}$ the flow was only subsonic for all firing rates.

The operating process conditions with the resulting pressures in the chamber (p_{ch}) are summarized in Fig. 3.7. The pressure increased with the mass flow of reactants due to increase of the pressure drop at the exit nozzle until the maximum of 3.25 bar for the smaller nozzle (with diameter D_1). Larger pressure could not be reached due to limiting supply pressure of the air line. The pressure drop at the larger nozzle (with the diameter D_2) is lower than at the D_1 nozzle, thus the resulting chamber pressure is also lower at the same operating conditions.

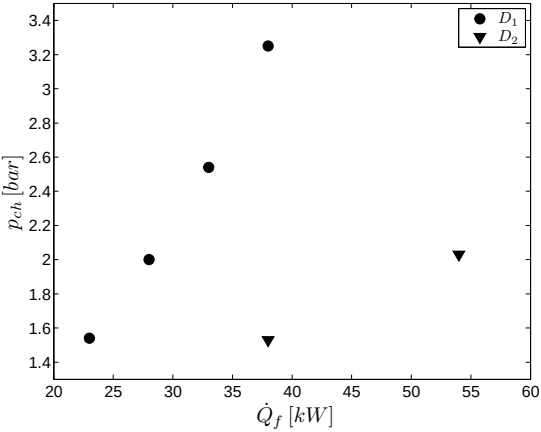


Figure 3.7: Pressure in the combustion chamber related to the firing rate and nozzle diameter.

The flame temperature at the nozzle (T_f) was measured with two methods described by Baukal [47]: 1) by means of K-type thermocouples (thickness of 1, 2 and 3 mm) and 2) bare S-type thermocouple coated with Ceramabond 569 as described by Burton et al. [77]. The correction of the temperature obtained with S-type thermocouple was done as suggested by Siewert [78]. The method 1) resulted in averadge 130 °C higher temperatures than method 2). The mean value of the results obtained with both methods was used as flame temperature in this study. Obtained results are presented in Fig. 3.8 depending on the firing rate. The increase of the flame temperature is a result of increase of firing rate and more released energy. In case of a larger nozzle diameter and for 38 kW flame power larger flame temperature was obtained probably due to increase of the nozzle cross section and more energy release per area.

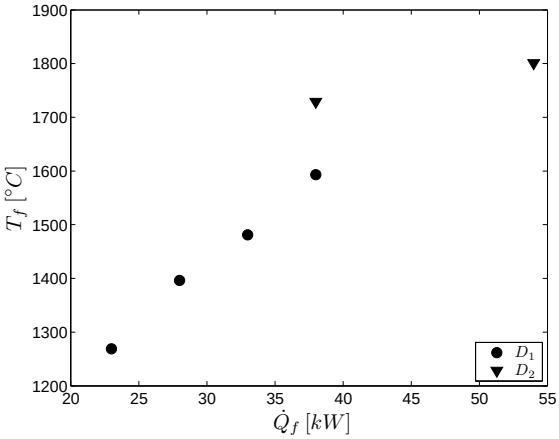
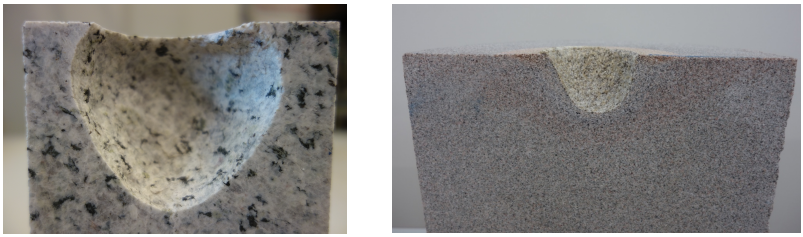


Figure 3.8: Flame temperatures measured at the exit nozzle depending on firing rate and nozzle diameter.

3.7.2 Rock type selection

Different rock types were tested in order to find the one that is the most suitable for flame-jet drilling study. Suitable means that the rock is spallable, relevant to geothermal projects in Switzerland and also can be easily delivered. The tested rock types were granite (Grimsel CHE, Bethel USA, Gotthard CHE, Paradiso IND), gabbro (Gabbro CHN) and sandstone (Rorschach CHE). Concerning the rock type granite was spalling the most. The least successful drilling was performed with sandstone due to its melting. This is probably because it contains a lot of soft clay that binds the grains, thus required thermal stresses could not be generated in the soft material and no thermal spallation occurred. Achieved results depending on the rock type are presented in Fig. 3.9).



a) b)

Figure 3.9: a) Grimsel granite (cylinder with 100 mm in diameter) and b) Rorschach sandstone (cylinder with 100 mm in diameter) after drilling at 38 kW of firing rate for a) 90 s and b) 6 min.

As a conclusion, Grimsel granite was selected for flame-jet drilling experiments. The main reason is that this rock type is a subject of many geothermal projects in Switzerland and it is well described in literature. In addition, it could be successfully spalled for the most tested process conditions as evaluated by measuring the removal rate (removed rock volume per time, in m^3/min). Rock samples were obtained from a quarry in Chuenzentennlen close to Grimselpass in Switzerland. The rock consists mainly of quartz and K-feldspar, but other present minerals are plagioclase

and biotite (Fig. 3.10). The physical properties from literature of the rock are listed in Table 3.1. The rock samples used in the experiments had a cylindrical shape with 100 mm in diameter and 150 mm in length.

Table 3.1: Considered physical properties of Grimsel granite [79, 80]

Physical property	Symbol	Value	Unit
Thermal conductivity	k	3.22	$\mathrm{W/(m\,K)}$
Poisson's ratio	ν	0.33	–
Young's modulus	E	47.3	GPa
Density	ρ	2706	$\mathrm{kg/m^3}$
Heat capacity	C_p	1025	$\mathrm{J/(kg\,K)}$
Thermal expansion coefficient	α	$8.5 \cdot 10^{-6}$	1/K

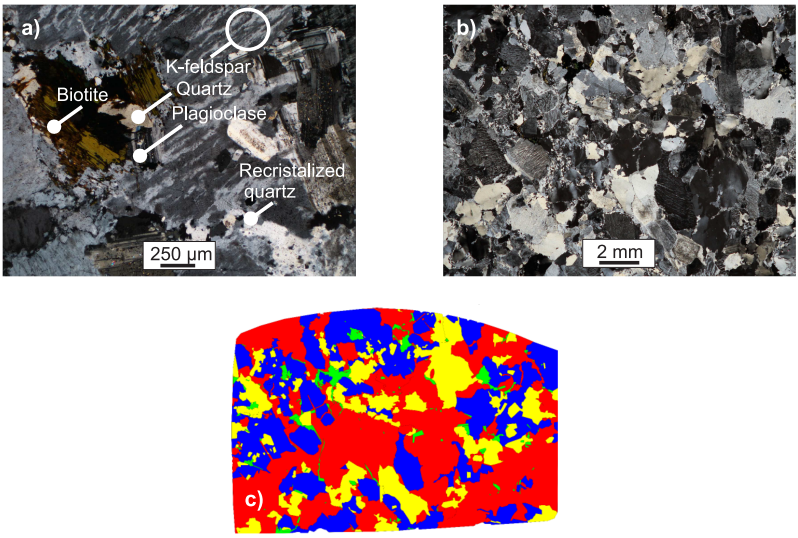


Figure 3.10: Grimsel granite; (a) and (b) present macroscopic photographs of a a section, (c) mineral composition of one section: quartz in red, K-feldspar in blue, plagioclase in yellow and biotite in green [81].

Chapter 4

Surface temperatures at flame-jet drilling onset

The required heat transfer study should describe the heat transfer between the impinging flame jet and the rock surface. This is possible if the surface temperatures of the rock sample, at the moment when the thermal spallation starts, and the heat flux sensor, applied in the heat transfer study, have similar values. The measurement of the rock surface temperature at which the spallation starts (spallation temperature) is presented in section 4.2. It was determined applying non-intrusive radiation thermometry, a method that does not interfere with the thermal spallation process. The radiation thermometry however requires an emissivity of observed object (rock surface), which experimental determination is described in section 4.1.

4.1 Emissivity measurement of a rock sample

4.1.1 Introduction and literature review

Radiation thermometry is based on measuring radiated power (E) from an observed surface. According to the Stefan - Boltzmann law (Eq. 4.1), E

(in W/m^2) can be used to calculate the temperature of the surface (in K) if its emissivity (ϵ_{surf}) is known:

$$(T_{surf})^4 = \frac{E}{\epsilon_{surf}\sigma} \tag{4.1}$$

where σ is the Stefan Boltzmann constant ($5.670 \times 10^{-8}\,W/m^2K^4$). A device that measures power radiated by a surface and converts it into the surface temperature based on Eq. 4.1 is an infrared (IR) camera. It offers an important advantage for investigating thermal spallation as it allows observation of the entire rock surface and monitoring its temperature profile over time, which is essential for identification of both the location and the moment of the first spall.

The emissivity of the rock surface (ϵ_{rock}), however, is not a common property of the material. It depends on both, surface properties of the rock (color, roughness, origin, grain size, etc.) and process conditions (wavelength of electromagnetic radiation, surface temperature, observation angle and environmental conditions) [82, 83]. The emissivity of rocks reported in the literature [84, 85] are thus specific to the investigated samples, thermal radiation measuring protocol, and/or environmental conditions for which the measurements were performed. As effective investigation of the spallation mechanism imposes testing a number of different rock types, a reliable technique for convenient determination of rock sample emissivity is required. The emissivity has to be valid for the temperature range between 230 °C and 650 °C [24, 86], different observation angles, and environmental conditions existing in thermal spallation experiments (drilling in atmosphere).

Emissivity is defined as a ratio between the energies radiated from a sample and a reference radiator at the same temperature [82]. Thus, the principle of measuring emissivity commonly used in literature is presented with Eq. 4.2 [87] where U_{sample} and U_{ref} (both in V) are the induced voltages in a device that measures the radiated energies from the sample and the

reference radiator. The reference radiator is commonly represented by a blackbody having an emissivity close to one such as a cavity [88, 89, 90] or blackbody furnace [91].

$$\epsilon_{sample} = \frac{U_{sample}}{U_{ref}} \tag{4.2}$$

In order to measure emissivity at elevated temperature, some possibilities are reported in literature and listed in Tab. 4.1. Typically, the temperature of the sample is measured by a thermocouple mounted (or welded) inside the sample, or on its surface [91, 88, 90], or by contactless bichromatic pyrometry [89]. The sample can be heated up by an electric resistor (resistive wires) [91, 88, 90], a lamp image furnace [89], or by a high-power laser [92]. Furthermore, the radiated energies are recorded at various wavelength ranges, observation angles [91, 92] and in vacuum of inert medium [91, 88, 89] commonly obtained via Fourier transform infrared (FTIR) spectrometers [91, 89, 90] having different infrared detectors.

Table 4.1: Emissivity measurement at elevated temperature reported in literature.

Author	Max. Temp. [°C]	Heating	Measurement of sample temp.
Hameury et al. [88]	2800	electric resistor	contact thermometry
Hay et al. [89]	1500	lamp furnace	bichromatic pyrometry
Rendgrove et al. [90]	1000	heater wire	contact thermometry
Del Campo et al. [91]	777	resis. spiral wire	contact thermometry
Lopes et al. [92]	1820	laser	ratio thermometry

In this work the applied radiation measuring device is the same IR camera that was also installed to monitor temperature development on the rock surface until the first spall was ejected (Optris PI450). The temperature of an observed surface could be measured after the emissivity was entered in a software designed for visualization and recording of camera signals (IP Imager). However, the IR camera did not provide the power radiated from a surface, nor generated voltages directly (Eq. 4.2), thus, it could not be used

for emissivity measurement procedures as described in literature. Therefore, another common principle of emissivity measurement was applied: the temperature of the rock surface is obtained with the IR camera (T_{rock}) for an initially assumed emissivity which is iteratively changed until T_{rock} becomes equal to the reference temperature (actual rock temperature). The emissivity at which those two temperatures matches presented the resulting emissivity of the rock sample (ϵ_{rock}) at T_{rock}.

The reference temperature is a parameter that is equal to the temperature of the observed rock surface. It was measured also by the Optris PI450 IR camera from the observed rock surface part painted with low-reflecting black paint of known emissivity. The black paint Pyromark 2500 (Tempil, LA-CO Industries, Inc., IL) was chosen because it has high emissivity and it can withstand high temperatures without oxidation. According to the author's knowledge, Pyromark 2500 has not been used in emissivity measurement for a reference body in the temperature range of interest (between 230 °C and 1650 °C) before.

Before the rock emissivity measurement, the emissivity of Pyromark 2500 was obtained for temperatures between 230 °C and 650 °C, and at different observation angles in ambient air. The measuring principle is the same as the one applied in the measurement of the rock emissivity: an initially assumed emissivity of the paint is iteratively changed until the temperature obtained with the IR camera matches the reference temperature. The reference paint temperature was the temperature of a ceramic heater painted with Pyromark 2500, but calculated from the measured electrical resistance of the ceramic heater when being heated (similar to the temperature measurement by a resistance thermometer). Therefore, the ceramic heater had been previously calibrated as a resistance thermometer.

The described technique is flexible concerning the type of a sample, its size, position relative to the IR camera (distance, observation angle) and environmental conditions. The maximum temperature at which the emissivity of a rock can be measured is limited by properties of Pyromark 2500 (the paint can withstand temperatures up to 1093 °C) [93]. There is no example

in literature that reports emissivity of a rock over a defined temperature range. Usually emissivity of rock samples is reported for a wide range of wavelengths but valid for only one temperature [84, 85].

4.1.2 Experimental procedure

The performed experimental procedure is summarized in Tab. 4.2.

Table 4.2: Experimental steps of the rock emissivity measurement.

Experim. step	Description
Step 1	Calibration of the ceramic heater as a resistance thermometer
Step 2	Measurement of Pyromark 2500 emissivity
Step 3	Measurement of rock emissivity

Calibration of the ceramic heater - Step 1

The ceramic heater (Type EF, Bach RC, Werneuchen, Germany) is made of silicon nitride (Si_3N_4) doped yttrium oxide (Y_2O_3) with a resistance of 150 Ω at 20 °C. Its dimensions are shown in Fig. 4.1.

Electrical resistance (R) is a temperature dependent property of any electrical conductor. This relationship is the main characteristic used by resistance thermometers. Similarly, the temperature of the ceramic heater (T_h) can be determined by measuring its resistance. In order to do so, the functional dependence between T_h and R (TR *dependence*, $T_h = b_1 \cdot R + b_0$) had to be obtained.

During the calibration, the ceramic heater was stepwise heated in an oven (Nabertherm B 180). At each step, once the constant temperature of the oven was reached, electrical resistance of the heater and the oven tem-

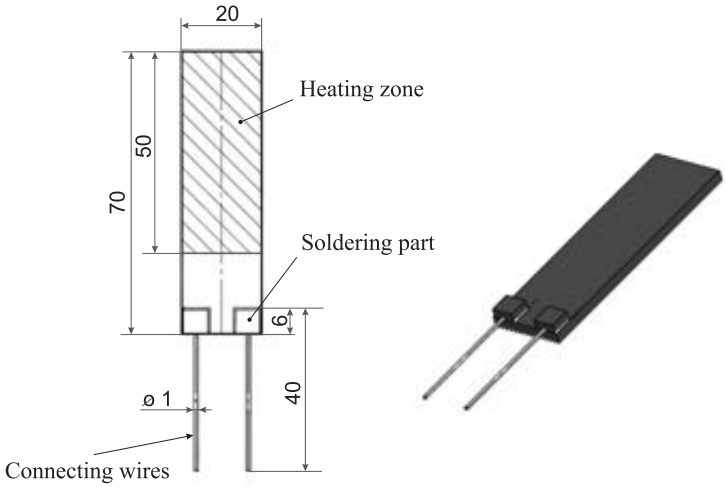


Figure 4.1: Ceramic heater applied in the black paint emissivity measurements. Dimensions are in mm. Thickness of the heater is 3 mm.

perature were measured by a multimeter (Agilent 34420A) and a K-type thermocouple respectively. The K-type thermocouple was placed next to the ceramic heater because of a non-uniform temperature distribution in the oven and its signal was recorded by an NI USB-6218 data acquisition device. It is assumed that the temperature of the heater is equal to the oven temperature (T_{ho}). The calibration of the ceramic heater is schematically presented in Fig. 4.2.

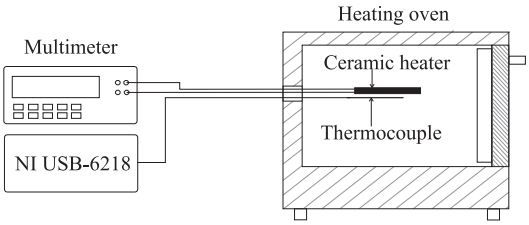


Figure 4.2: Calibration of the ceramic heater within the oven. Its electrical resistance was measured with the multimeter.

The results of the calibration are presented in Fig. 4.3. It shows the temperature of the heater (T_{ho} in °C) over the electrical resistance of the heater (R in Ω). A linear fit (Eq. 4.3, *TR dependence*) was obtained by a least - square linear regression method.

$$T_{ho} = b_1 \cdot R + b_0 \tag{4.3}$$

The coefficient of determination R^2 being equal to 0.999 gives very high confidence to use the obtained linear fit for temperature determination of the ceramic heater.

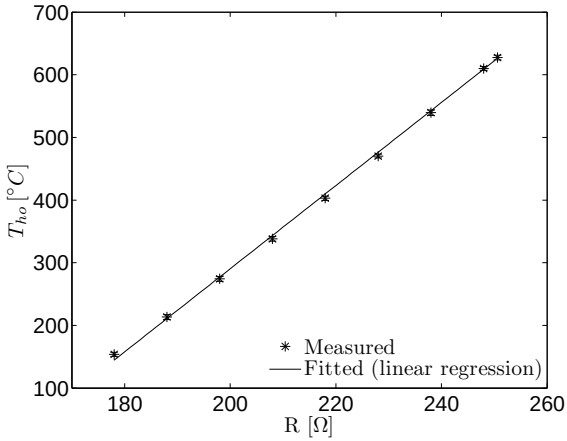


Figure 4.3: Calibration curve of the ceramic heater ($T_{ho} = b_1 \cdot R + b_0$), $R^2 = 0.999$, $b_0 = -1036.7\ °C$, $b_1 = 6.6\,°\text{C}/\Omega$.

Black paint Pyromark 2500 and its emissivity measurement - Step 2

Pyromark 2500 is a silicone-based high temperature black paint [93] that was used to color one part of the rock surface as the reference part required for determination of the rock sample emissivity. It can sustain 800 °C (the maximum heating temperature of the rock sample) without oxidation and changing its properties. Using Pyromark 2500 for the reference temperature measurement requires its emissivity to be known. The following experimental procedure was followed to obtain the black paint emissivity: The paint was applied on the ceramic heater as suggested by Ho et al. [93]. The heater temperature was adjusted by means of an AC transformer after connecting the heater to the AC power supply. It is assumed that the temperature of the applied black paint (T_{paint}) is equal to the temperature of the ceramic heater (T_h) calculated from the heater electrical resistance (R) and its TR *dependence*. The heater resistance R was calculated from the voltage (U) and current (I in A) measured at the connecting wires of the

ceramic heater with an EtherCAT terminal EL3681 (Beckhoff Automation AG) as shown in Eq. 4.4:

$$R = \frac{U}{I} \tag{4.4}$$

When the voltage was kept constant and T_h reached a steady state the radiated energy from the heater was recorded with the IR camera.

IR cameras are state of the art in the surface temperature measurement of opaque bodies (T_{surf}). The one implemented in the experimental setup of the presented work was Optris PI450 with a resolution of 382 x 288 pixel and a frequency of 80 Hz measuring the radiation in a wavelength range of 7.5 - 13 µm. The radiation detector installed in the IR camera generates a voltage (U) proportional to the surface radiated power. This signal is converted into the surface temperature T_{surf} with an algorithm that requires the surface emissivity (ϵ_{surf}) to be defined by the user. T_{surf} is calculated from the voltage (U) as shown in Eq. 4.5:

$$T_{surf} = \sqrt[n]{\frac{U - C \cdot T_{amb}^n + C \cdot \epsilon_{surf} T_{amb}^n - C \cdot T_{Pyr}^n}{C \epsilon_{surf}}} \tag{4.5}$$

where C is a specific constant of the IR camera, T_{amb} is the ambient temperature, T_{Pyr} is the temperature of the IR camera and n is a parameter dependent on the wavelength. Constants C and n, and the voltage U are unknown to the user. In the measurement of the paint emissivity ϵ_{surf} was equal to ϵ_{paint} and it was iteratively changed with the software of the IR camera until the temperature of the paint (T_{paint}) became equal to the calculated T_h.

The measuring equipment of Step 2 is schematically presented in Fig. 4.4. During the measurement of paint emissivity the heater was surrounded by air in a dark box together with the IR camera in order to be protected from the ambient radiation. The emissivity measurement of Pyromark 2500 was

repeated for three different observation angles (0° - normal to the surface, 45° and 60°) and temperatures between 230 °C and 650 °C.

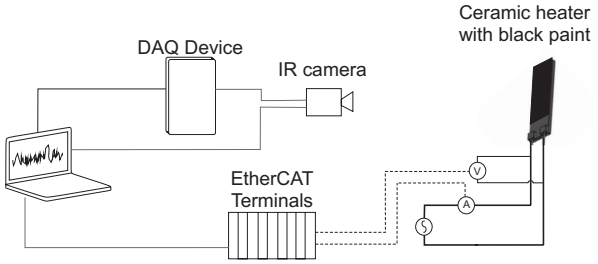


Figure 4.4: Black paint emissivity measurement. The ceramic heater was painted with Pyromark 2500 and heated up by an AC power supply. The IR camera and EtherCAT Terminals measured the surface temperature (T_h) and electrical resistance (R) of the heater, respectively, in order to obtain black paint emissivity (ϵ_{paint}).

The ceramic heater, when heated in the oven, had a uniform temperature profile along its length (in Step 1). But, during the experimental Step 2, the heater was connected to the AC power supply which induced a non-uniform temperature distribution along the heater, having the lowest values close to the wires and the highest at the top of the heater (see Fig. 4.1). Nevertheless, it is assumed that the calculated temperature of the ceramic heater connected to the AC power supply (T_h) actually represents its mean temperature and that it is equal to the temperature of the ceramic heater placed in the oven (T_{ho}) at the same electrical resistance. This assumption was made according to the definition of the mean function and the linearity of the *TR dependence*.

Rock samples and rock emissivity measurements - Step 3

Emissivity measurement was performed on the Grimsel granite rock sample used in flame-jet drilling experiments (see chapter 5). Its cross section is presented in Fig. 4.5. The sample size was 70 x 30 x 30 mm. During the

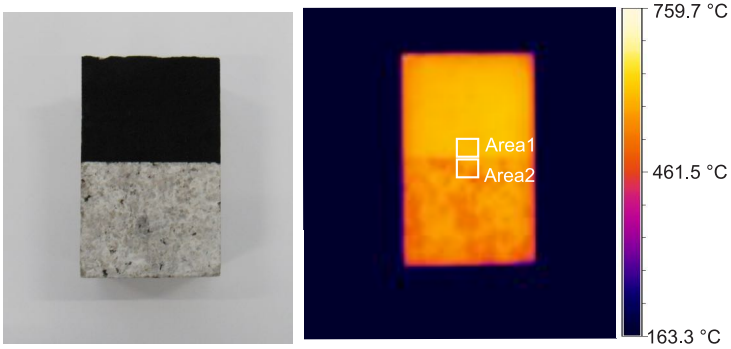
measurement of rock emissivity the rock sample was surrounded by air.

Figure 4.5: Surface of Grimsel granite which emissivity was measured

Pyromark 2500 was deposited on one half of the rock sample in order to create the reference surface for the measurement of the rock emissivity. This was done due to the assumption that the rock sample is equally heated from all sides and that the temperature profile of the painted surface is assumed to be the same as the temperature profile of the unpainted one (Fig. 4.6).

To reach high temperatures, the rock samples were heated in the oven mentioned in Step 1. The rock sample was slowly heated up to a temperature of 800 °C over a period of 3 h (heating rate of 4.4 °C/min) in order to avoid temperature shock of the material and its breakage. The emissivity measurement was performed after the rock sample was taken out of the oven and placed in a dark box together with the IR camera being protected from the radiation of surrounding. The temperature of the rock surface (painted and unpainted) was recorded with the IR camera while the rock sample was cooling down until it reached its low-end temperature of the desired range. The measurements were performed for three different observation angles (0° - normal to the surface, 45° and 60°). The recorded files were analyzed with the software IP Imager (special design from Optris GmbH) after the measurement had been performed. The software enables to select multiple observation areas of the surface and to record their mean temperatures at the same time. An example of the video analysis is presented in Fig. 4.6 where the first observation area (Area1) was located on the reference

surface while the second area (Area2) was located on the unpainted part. The emissivity valid for Area1 is ϵ_{paint} (known parameter) and for Area2 ϵ_{rock} (unknown parameter). For a selected moment from the recorded file, ϵ_{rock} was iteratively changed until the surface temperature of the unpainted part in Area2 (T_{rock}) matched the reference temperature (T_{ref}) in Area1. The resulting emissivity of the rock sample (ϵ_{rock}) is valid at the observed temperature.



(a) Tested rock sample (b) Rock sample observed with IR camera

Figure 4.6: (a)The tested rock sample, (b)rock sample recorded with IR camera. Half of the rock surface was painted with Pyromark 2500. The shown areas present the observation areas of the IR camera (Area1 - painted and Area2 - unpainted rock surface part). The emissivity of the rock (Area2) was adjusted until the IR camera showed same temperatures for both areas.

Cooling of the rock sample is a consequence of convection from the rock to the air and radiation. It was assumed that the difference in radiated heat loss from the painted and unpainted surface part during the cooling procedure of the rock samples was negligible, because the emissivity of the black paint was reported to be up to 0.92 [87, 93] which is very close to the reported rock sample emissivity of 0.9 [84]. Additionally, Area1 and Area2 (Fig. 4.6) were located next to each other, which minimizes any

temperature difference caused by uneven cooling of the rock sample through conduction. Therefore, it is assumed that the surface temperatures of the painted and unpainted observed rock surface areas are equal during the cooling process of the rock samples. It is also assumed that the layer of the black paint does not influence the temperature of the rock surface during the cooling procedure.

4.1.3 Uncertainty analysis

The equipment implemented in the emissivity measuring method and the experimental procedure introduce uncertainty in the obtained emissivity of the rock. In the following, the calculation of the uncertainty of the rock emissivity measuring procedure ($\delta\epsilon_{rock}$) is presented.

The measurement uncertainties of all devices used in the experiments are listed in Tab. 4.3.

Table 4.3: Uncertainties of the equipment in emissivity measurement.

Measuring device	Symbol	Unit	Uncertainties (of measured value)	Experim. step
Thermocouple K-typ	δT_{thermo}	°C	2.2°C or 0.75%	1
Agilent 34420A	δR_{multi}	Ω	0.0015%	1
IR camera	δT_{cam}	°C	2.00%	2, 3
EL3681 current	δI	A	0.70%	2
EL3681 voltage	δU	V	0.75%	2

Uncertainty of *TR dependence* (Step 1)

The measuring uncertainty of the installed thermocouple in the oven (δT_{thermo}) is considered in the uncertainty analysis. The uncertainty of the multimeter used to measure R (δR_{multi}) is minor compared to the other devices (Tab. 4.3) and thus, it was neglected. The resulting uncertainty (δT_{ho})

from Step 1 was calculated as in Eq. 4.6 [94] where the first term presents the uncertainty in the obtained temperature after the linearization and the second term is the uncertainty of the thermocouple (δT_{thermo}).

$$\delta T_{ho} = \sqrt{\frac{1}{N-2}\sum_{i=1}^{N}\left(T_i - b_0 - b_1 R_i\right)^2 + \left(\delta T_{thermo}\right)^2} \tag{4.6}$$

where T_i and R_i present the measured temperature (T_{ho}) and electrical resistance of the heater (R) for the each measuring point respectively, b_0 and b_1 are coefficients of the TR *dependence* and N is the number of measuring points in Step 1.

Uncertainty in paint emissivity determination (Step 2)

All sources of uncertainty in Step 2 were included in calculating the uncertainty in paint emissivity determination. First the resistance (R) of the ceramic heater was considered and it was calculated as shown in Eq. 4.4 from the voltage (U) and the current (I) obtained with EtherCAT terminals. Therefore, the uncertainties in recorded U and I were included into the resistance uncertainty of the ceramic heater (δR) as in Eq. 4.7 (general formula for error propagation). Because R is included in the calculation of T_h over the TR *dependence*, the calculation of the uncertainty in the temperature of the ceramic heater (δT_h) is presented with Eq. 4.8 where the second term was derived from the TR *dependence* (general formula for error propagation).

$$\delta R = \sqrt{\left(\frac{1}{I}\delta U\right)^2 + \left(-\frac{U}{I^2}\delta I\right)^2} \tag{4.7}$$

$$\delta T_h = \sqrt{\left(\delta T_{ho}\right)^2 + \left(b_1 \delta R\right)^2} \tag{4.8}$$

As it was explained in Step 2, in order to obtain the emissivity of Pyromark 2500, temperature of the painted heater obtained with IR camera (T_{paint}) was compared with the temperature T_h calculated from the measured resistance. Hence, the uncertainty in the temperature that was detected with the IR camera (δT_{cam}) is included in the uncertainty of T_{paint} (δT_{paint}) as in Eq. 4.9.

$$\delta T_{paint} = \sqrt{(\delta T_h)^2 + (\delta T_{cam})^2} \tag{4.9}$$

δT_{paint} was incorporated in the calculation of the uncertainty in the black paint emissivity ($\delta \epsilon_{paint}$) according to Eq. 4.10. The black paint emissivity was measured for three times and its standard deviation ($\sigma_{\epsilon_{paint}}$) was taken into account [94].

$$\delta \epsilon_{paint} = \sqrt{(\sigma_{\epsilon_{paint}})^2 + \left(\frac{\partial \epsilon_{paint}}{\partial T_{paint}} \delta T_{paint}\right)^2} \tag{4.10}$$

The derivative of ϵ_{paint} with respect to T_{paint} (the second term of Eq. 4.10) could not be analytically obtained because the coefficients in Eq. 4.5 were unknown. Therefore, it was approximated with the difference quotient of a function $\epsilon_{paint} = f(T_{paint})$ determined with the software IP Imager: at a certain temperature of the black paint (T_{paint}), ϵ_{paint} was adjusted until T_{paint} changed for δT_{paint}. The value of emissivity change was recorded and presents the uncertainty of black paint emissivity proportional to δT_{paint}.

Uncertainty in rock emissivity determination (Step 3)

As already explained in the chapter 4.1.2, the temperature of the painted rock surface (T_{ref}) was the reference temperature for the measurement of the rock emissivity. Therefore, the uncertainty in the reference temperature measurement (δT_{ref}) includes the uncertainty of the black paint emissivity ($\delta \epsilon_{paint}$) and uncertainty in temperature measurement with the IR camera

(δT_{cam}) as in Eq. 4.11

$$\delta T_{ref} = \sqrt{\left(\frac{\partial T_{ref}}{\partial \epsilon_{paint}} \delta \epsilon_{paint}\right)^2 + (\delta T_{cam})^2} \tag{4.11}$$

The derivative of T_{ref} with respect to ϵ_{paint} (first term of Eq. 4.11) is approximated with the difference quotient of a function $T_{ref} = f(\epsilon_{paint})$ which was determined with IP Imager.

The calculation of the uncertainty of rock emissivity included δT_{ref} according to Eq. 4.12. The rock sample emissivity was measured three times and its standard deviation ($\sigma_{\epsilon_{rock}}$) is also considered.

$$\delta \epsilon_{rock} = \sqrt{(\sigma_{\epsilon_{rock}})^2 + \left(\frac{\partial \epsilon_{rock}}{\partial T_{ref}} \delta T_{ref}\right)^2} \tag{4.12}$$

The derivative of ϵ_{rock} with respect to δT_{ref} (second term of Eq. 4.12) is again approximated with the difference quotient of a function $\epsilon_{rock} = f(T_{ref})$ which was determined with IP Imager.

4.1.4 Results

Step 2 - Black paint emissivity measurement

The results of Pyromark 2500 emissivity are summarized in Fig. 4.7. It can be seen, that by increasing the observation angle the emissivity decreases. This agrees with the theoretical observation for non-conductive materials [83]. The emissivity has a value between 0.8 and 0.95.

In order to compare the obtained data with data in literature, it is important to do so with the same type of emissivity. In general, emissivity can be defined as *total hemispherical emissivity* which is integrated over all wavelengths and all directions, and the *spectral directional emissivity*

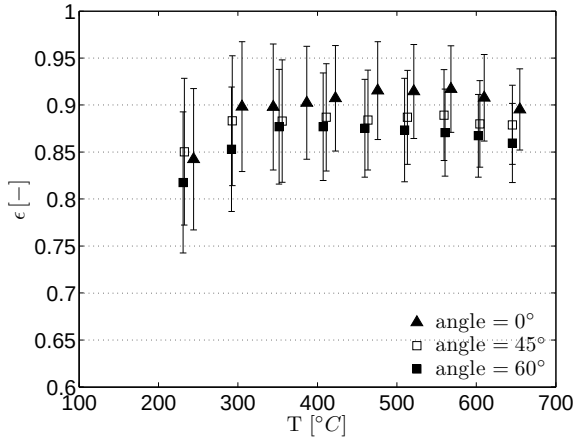


Figure 4.7: Emissivity of Pyromark 2500 measured at different temperatures and measuring angles.

which is integrated over a specific wavelength range and valid for one observation angle [82]. Because the IR camera measures radiation in a defined wavelength range, the emissivity obtained in this work could be regarded as spectral directional emissivity. For the data comparison, the work of Ho et al. [93] was found where the spectral directional emissivity of Pyromark 2500 with respect to wavelength was presented. They applied the black paint on different metallic coupons (Inconel, SS304, Cold-Rolled Steel) and obtained the emissivity for an incidence angle equal to 10 °C (near-normal) and different temperatures (26 °C, 540 °C and 600 °C). At reported temperatures the emissivity of Pyromark 2500 is around 0.9 for a wavelength between 7.5 and 13 μm. It is involved in the calculation of the total hemispherical emissivity at a hypothetical temperatures range of the surface between 100 °C and 1000 °C having a value between 0.8 and 0.88.

Still, the validation of the obtained results from this work cannot be performed completely. The resulting emissivity presents an average over the measuring wavelength range of the IR camera, thus, having higher values than total hemispherical emissivity presented in literature. This is mainly

due to the lower spectral emissivity that exists outside the wavelength range of the IR camera, which is presented in the work of Ho et al. [93]. However, the literature data is still helpful to conclude that the emissivity measurement results of Pyromark 2500 are in an agreement with expected values.

Step 3 - Rock emissivity measurement

The obtained rock emissivity is presented in Fig. 4.8. According to the results from Danov et al. [95] it is expected to see a decrease in emissivity when increasing the observation angle. Unfortunately, it is not possible to discuss this expectation because of the uncertainty of the measurement. In literature [85], the rock emissivity is reported to be high (around 0.9) which is close to the values obtained in this study.

As the emissivity of the rock sample and the emissivity of the black paint are close to each other, the assumption presented in chapter 4.1.2 about radiative heat losses of the painted and unpainted surface part having a small difference can be accepted.

The relative uncertainty of the rock sample emissivity ($\frac{\delta\epsilon_{rock}}{\epsilon_{rock}}$), (Fig. 4.9), reaches values up to 10 %. When using the resulting emissivity of rocks in spallation experiments, recorded temperatures are uncertain up to 25 °C which presents a small percentage of the expected temperature difference between the impinging flame-jet and the rock surface (around 1000 °C). It is assumed that the rock surface temperature being in the range of the obtained uncertainty negligibly influences the heat transfer to the rock surface. For the purpose of thermal spallation this relative uncertainty is small and the equipment has not been adjusted to reduce it.

Fig. 4.9 presents the relative uncertainties of paint ($\frac{\delta\epsilon_{paint}}{\epsilon_{rock}}$) and rock emissivity ($\frac{\delta\epsilon_{rock}}{\epsilon_{rock}}$). With this figure, it is possible to explain which measuring step has the largest uncertainty. The uncertainty $\delta\epsilon_{paint}$ represents 84–97 % of the uncertainty $\delta\epsilon_{rock}$ indicating the low measuring accuracy in

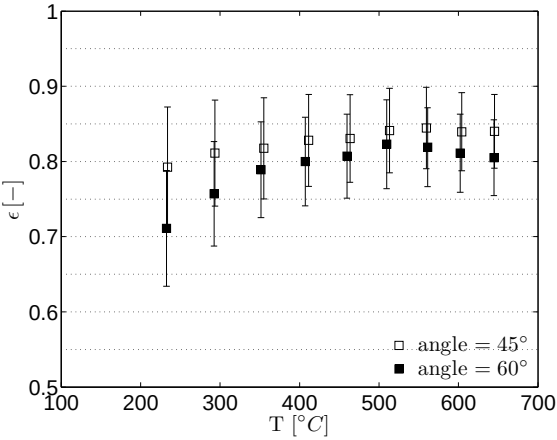


Figure 4.8: Measured emissivity of rock sample Grimsel granite at different temperatures and observation angles.

Step 1 or Step 2. Considering the absolute values of uncertainties from the equations in chapter 4.1.3, it was noticed that the second term of Eq. 4.8 has the largest impact. Therefore, by recording the voltage and the current with more accurate devices it is possible to reduce the uncertainty in the presented experimental procedure. In Fig. 4.9, it can be noticed that the gap between $\frac{\delta\epsilon_{paint}}{\epsilon_{rock}}$ and $\frac{\delta\epsilon_{rock}}{\epsilon_{rock}}$ increases with temperature, which is caused by the increasing uncertainty in temperature measurement by the IR camera (δT_{cam}).

Even though the uncertainties of the equipment increases with temperature, the decline of the relative uncertainty in Fig. 4.9 can be explained with Eq. 4.13, which is derived from the Stefan-Boltzmann's law (Eq. 4.1) (simplified Eq. 4.5):

$$\delta\epsilon_{rock} = \sqrt{\left(\frac{\partial\epsilon_{rock}}{\partial T_{rock}}\delta T_{rock}\right)^2} = \sqrt{\left(-4\frac{E}{\sigma}\frac{1}{T_{rock}^4}\frac{\delta T_{rock}}{T_{rock}}\right)^2} \tag{4.13}$$

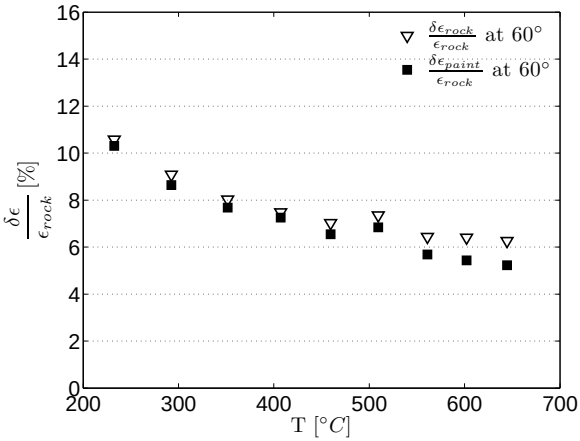


Figure 4.9: Uncertainties of emissivity of the rock sample and of black paint Pyromark 2500 relative to the rock emissivity.

Eq. 4.13 can be rewritten into Eq. 4.14

$$\left|\frac{\delta\epsilon_{rock}}{\epsilon_{rock}}\right| = \left|-4\frac{\delta T_{rock}}{T_{rock}}\right| \tag{4.14}$$

From Eq. 4.14 it can be concluded that the relative uncertainty in rock emissivity ($\frac{\delta\epsilon_{rock}}{\epsilon_{rock}}$) is proportional to the relative uncertainty in the temperature of the rock ($\frac{\delta T_{rock}}{T_{rock}}$). Additionally, it can be noticed that by increasing the temperature the electrical resistance of the ceramic heater in Step 2 increases slower than the current. Thus, due to non-proportionality between the resistance and the current existing in the second term ($-\frac{U}{I^2}\delta I = -\frac{R}{I}\delta I$) of Eq. 4.7, $\frac{\delta T_{rock}}{T_{rock}}$ decreases with temperature.

4.1.5 Conclusion

With the presented method the emissivity of the rock sample could be successfully measured. The presented experimental procedure has an un-

certainty of up to 10 %. For the purpose of thermal spallation this relative uncertainty is small due to assumption that the heat transfer to the rock remain unchanged for 25 °C of surface temperature difference and the equipment has not been adjusted to reduce it. Pyromark 2500 with high emissivity can be used for the detection of the reference temperature required for the emissivity measurement of rocks. A solution to determine the emissivity of the black paint has been presented and the obtained results were in agreement with the values from literature. The uncertainty analysis shows that the measurement of black paint emissivity (Step 2) has the largest impact on the accuracy of the presented technique. Actually, it was discovered that the accuracy can be mostly increased by using a more accurate device when measuring the electrical resistance and not other parameters (temperature of the oven, temperature of the heater and rock with IR camera). The presented experimental procedure can be applied for any opaque body. The composition of rock samples has not been investigated and thus, its influence on emissivity of rock samples can be evaluated in the future.

4.2 Rock surface temperature measurement

The heat transfer study of the impinging flame-jet was performed with a heat flux sensor, but it represents better the heat transfer between the flame-jet and the rock surface if the surface temperature of a heat flux sensor and the rock are equal. The rock surface temperature, when exposed to the flame jet, increases up to the spallation temperature (temperature when onset of spallation occurs). After spall ejection the rock surface temperature decreases for around 100 °C when the heating process starts again initiating further thermal spallation. Therefore, rock surface temperature does not reach a steady state while spalling of the rock. However, it was assumed that the heat transfer coefficient of the impinging flame jet is not influenced by the rock surface temperature in the temperature range in which it oscillates while spalling. Therefore, this oscillating temperature

range of the spallation temperature was measured first. An nonintrusive infrared thermography was applied for the spallation temperature measurement in order not to influence thermal spallation process. The obtained values were target temperatures to be reached with the heat flux sensor in the heat transfer study. This chapter describes the experimental procedure and results of the spallation temperature measurement applied on Grimsel granite.

4.2.1 Experimental setup and procedure

The rock surface temperature was recorded with infrared (IR) camera Optris PI450 with a resolution of 382 x 288 pixel and a frequency of 80 Hz measuring the radiation of observed object in a wavelength range of 7.5–13 µm. The camera was positioned next to the burner at an angle of 60° relative to the vertical axis (see Fig. 4.10). The emissivity of 0.82 (reported in chapter 4.1) was applied for the surface measurement.

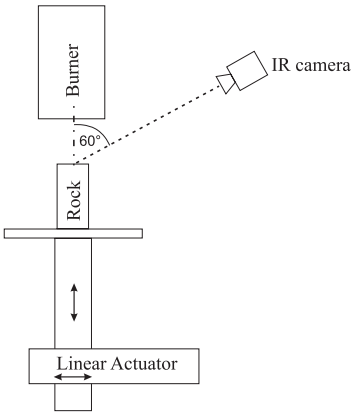


Figure 4.10: Surface temperature measurement of the rock sample at the spallation onset with an IR camera.

The rock samples were unconfined in order to provide an optical access to

impinged surface. It was moved to the desired axial position ($L = 8$) with linear actuator, but still being at certain lateral distance to the flame jet. This is the closest possible vertical position of the rock sample to the exit nozzle in flame-jet drilling experiments. Afterward, the rock samples were laterally repositioned under the flame with a moving velocity of 0.2 m/s. They were impinged for 5 s because in this time the first spall is already ejected. Firing rates of the burner were varied between 23 and 38 kW having a nozzle diameter of 7.9 mm. The dimensionless distance is kept constant. The measurement is repeated for 3 times.

The temperature of interest is the temperature at the rock surface when and where the first spall was detached (T_{spall}). Actually, the temperature of rock surface one frame before spall detachment was evaluated as the spallation temperature. The ejection of a spall is a very fast process (faster than 10 000 Hz as observed with a high speed camera). Because the IR camera could only record 80 frames/second, a possible measuring error was approximated to be 45 °C (at 38 kW), which is the temperature rise in between two frames. Still, it was assumed that the heat transfer coefficient between the flame jet and the rock surface is not influenced in the range of the measuring error.

The products of the combustion did not deposit any soot on the target and the flame varied between being invisible and blue. Thus, it is assumed that the flame does not influence the surface temperature measurement for the chosen temperature range. One reason is that the radiation of the flame jet could not be registered for the selected temperature measurement range (150 - 900 °C). This was evaluated by placing the flame jet between an object with known surface temperature (ceramic heater applied in the emissivity measurement) and the IR camera: The recorded surface temperature with the IR camera differed only negligibly (around 2 °C) from the real heater temperature.

4.2.2 Results

Mostly detachment of spalls larger than 1.5 mmcould be detected. The surface temperatures were in a range between 430 and 600 °C as presented in Fig. 4.11. The measurement was performed at different firing rates, but any trend was not recorded, probably due to uncertainty of the applied method described in chapter 4.1 and 45 °C of temperature difference between two frames. However, the resulting temperatures are in agreement with spallation temperatures of other granite rock samples reported in literature [86] being between 200 and 650 °C. The temperatures are also lower than the melting temperature of the granite (more than 900 °C [96]), thus indicating that generated thermal stresses in the rock surface could cause the ejection of spalls long before rock melting starts.

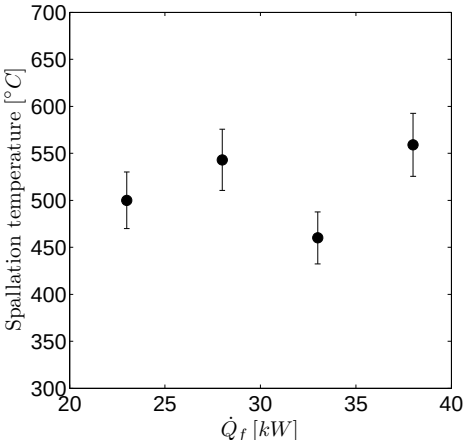


Figure 4.11: Surface temperature recorded with IR camera one frame before the first spall was ejected.

4.2.3 Conclusion

The rock surface temperature of Grimsel granite at the onset of spallation varied between 430 and 600 °C. There was not any clear influence of the firing rate in the applied range on the developed surface temperatures. The recorded values are the temperatures to be reached in the heat transfer study, thus the heat flux sensor was accordingly designed.

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4.3 Heat transfer study

The flame-jet drilling technology can be evaluated, understood and improved not only by analysing physical properties of the drilled rock, but also by analyzing the influence of process parameters on the drilling. Process parameters (firing rate, nozzle diameter, nozzle-to-target dimensionless distance) have an impact on the heat transfer to the rock surface, thus influencing the rock heating rate and development of thermal stresses in the rock. The impact of process parameters on the heat transfer has to be therefore at firts analyzed with a heat transfer study. It offers also the possibility for optimization of the flame-jet drilling process in order to reach highest heat fluxes. In addition, the confined flame jet is of interest because it simulates the physical conditions existing in a borehole.

4.3.1 Literature review

Browning et al. [16] concluded with an experimental study that the released heat from the flame has an impact on spall thickness and volumetric removal rate. One of the reasons is that the heat transferred through the rock surface influences the temperature rise in the rock, hence, the development of thermal stress and finally the drilling itself. Only the heat released through the nozzle was reported. The heat transfer to a target was not

characterized in detail.

Heat transfer from an impinging flame jet applied in flame-jet drilling experiments was further analyzed by Rauenzahn [24]. He performed a heat transfer study with a small quarry slotting tool producing supersonic flame jets. The purpose of the performed heat transfer study was to measure the boundary conditions required for a flame-jet drilling model. The heat flux was calculated according to an energy balance of a copper block impinged by the flame jet. The block had the concave impinged surface and embedded thermocouples below it. The tested process parameters, recorded temperatures of the copper block and heat fluxes are presented in Tab. 4.4.

Table 4.4: Process parameters of the heat transfer study performed by Rauenzahn et al. [24].

Property of a flame jet	Measured value	Unit
Distance to target	7.6, 10.2, 12.7, 15.2	cm
Fuel flow rate	1	L/s STP
Oxygen flow rate	5.1	L/s STP
Max. target temp.	266	°C
Max. heat flux	7.4	MW/m^2

As noticed by Rauenzahn, difficulties occurred during the performed heat transfer study while measuring temperatures within the copper block and while calculating the heat transfer to the concave surface. A prediction of the hole diameter was very successful, but the predicted heat transfer rates were not aligned with the experimental data.

Only Rauenzahn [24] studied a confined impinging flame jet, but for conditions listed in Tab. 4.4. Furthermore, there are numerous heat transfer studies dedicated to free flame jets impinging a flat surface. For the free impinging flame jets the heat transfer is dependent on the diameter of a burner exit nozzle, axial distance between the target and the nozzle, the released flame power and adiabatic flame temperature [64, 66]. In those

studies mostly laminar flame jets reaching Reynolds numbers up to 2500 were observed [97, 98, 99]. Van der Meer reached Reynolds numbers up to 4226 [100] when comparing the heat transfer from isothermal and flame jets. Higher Reynolds numbers (up to 35 300) were reported by Milson et al. [101] who observed the heat transfer on a steel plate at constant nozzle-to-target dimensionless distance. As a fuel, mostly methane [58, 97] and natural gas [100] were combusted. The largest firing rate reported in literature was applied in the studies from Baukal et al. [58] reaching 25 kW. They concluded that increasing the firing rate by 400 % the heat flux to the target increases by 470 %. Concerning the influence of the axial distance on the heat transfer the same trend was noticed by many researchers: the maximum heat transfer occurs at the end of the jet potential core [97, 99, 100, 101]. The only work investigating the influence of the nozzle diameter was presented by Chander et al. [97], who performed the study for the laminar regime. They concluded that the heat transfer was affected by the nozzle diameter only in the wall jet region.

A heat flux sensor used in heat transfer studies was usually designed depending on the applied flame properties (temperature, exhaust gas velocity, firing rate) in order to sustain the harsh environment. Examples include a water-cooled calorimeter [58, 102], a Gardon heat flux transducer [100] and a Vatell heat flux sensor [97]. The highest heat flux recorded in these heat transfer studies of impinging free flame jets was 1 MW/m^2 [58].

Both, the heat flux measuring device and the heat transfer study of the confined impinging flame jet are presented in this work. Process conditions that successfully drilled rock samples produce subsonic and underexpanded flame jets, thus, the heat transfer study was performed with both flame jet types. The influence of confinement on the heat transfer from the impinging flame jet is also discussed. The heat transfer to the target was adjusted by changing the firing rate, the nozzle diameter and axial dimensionless distance. The targets for the heat transfer study were two custom designed water-cooled calorimeters that recorded surface temperatures between 70 and 600 °C. According to Baukal et al. [64] no flame jet impingement heat

transfer study with target surface temperatures between 350 and 700 °C has been published yet. The influence of different surface temperature on the heat transfer from an impinging flame jet has also never been studied before. Heat fluxes up to 10.5 MW/m^2 were measured in the presented work, which is higher than any reported heat fluxes in literature.

4.3.2 Experimental setup

The flame-jet drilling setup applied in this heat transfer study consists of two main parts: the burner and a heat flux sensor. The flame jet exits the burner and it impinged the top of the sensor. The sensor is cooled with water (calorimetric and guard) flowing beneath its surface and absorbing the transferred heat. A schematic overview of the experimental system is presented in the Fig. 4.12.

The target assembly is designed as presented in Fig. 4.13. It consists of a water-cooled calorimeter and a confining ring. The water-cooled calorimeter measures transferred heat flux and it is placed below the exit nozzle. In order to simulate conditions which are present in drilling experiments the confining ring was placed on the top of the calorimeter. Part of the jet close to the impinged surface was confined with it. Its dimensions are 40 mm of inner diameter and 35 mm in hight.

Measurement principle of a water-cooled calorimeter

A water-cooled calorimeter is a device that measures the quantity of absorbed thermal energy from the energy balance equation of the cooling water. In general the heat flux $\dot{q}_{sen}$ through the calorimeter is calculated at steady-state condition as presented in Eq. 4.15 [103]:

$$\dot{q}_{sen} = \frac{\dot{m} C_p}{A} (T_{outlet} - T_{inlet}) \tag{4.15}$$

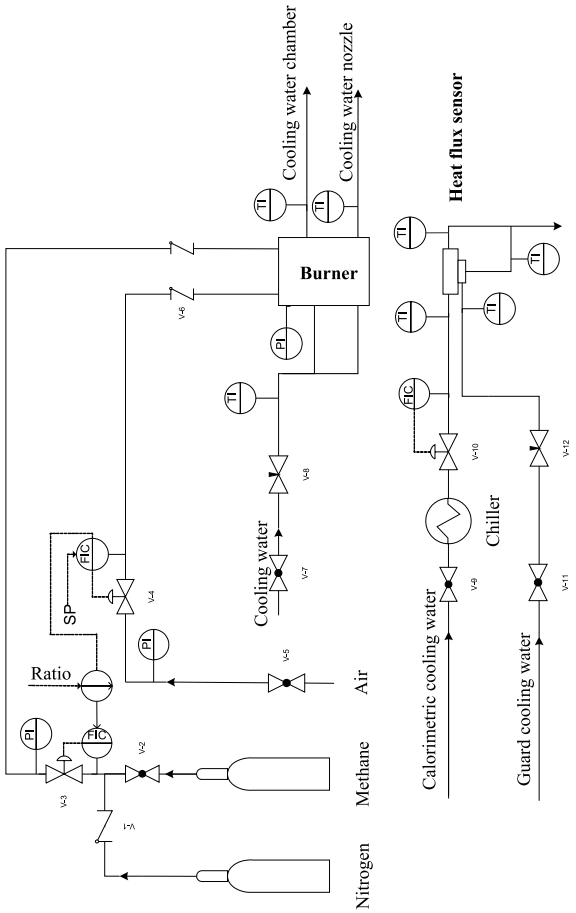

Figure 4.12: Schematic overview of the flame impinging system (flame-jet drilling setup and heat flux sensor (calorimeter)). Methane is used as a fuel, air as oxidizer and nitrogen to purge the lines at the end of experiment. The flame jet impinges the calorimeter from above. The burner and the heat flux sensor are cooled with water.

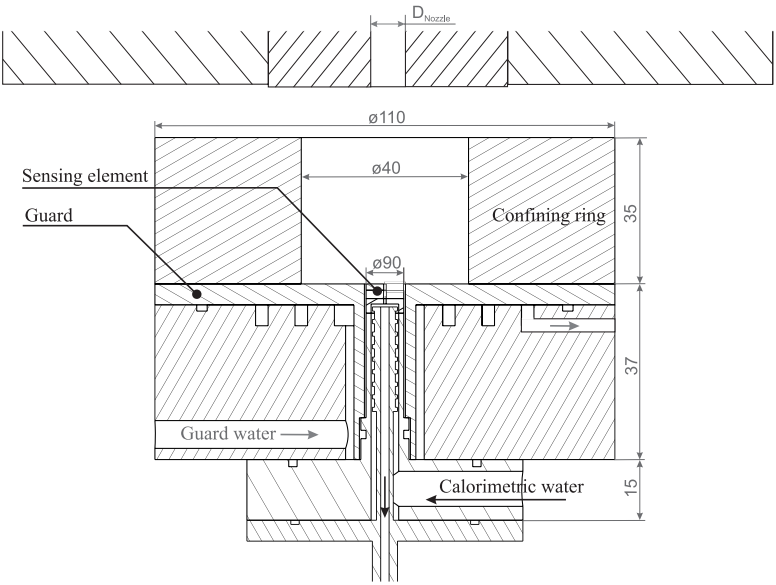

Figure 4.13: Design of the heat flux sensor (calorimeter) that was impinged by the flame jet. The measuring method is based on calorimetric measuring principle [103]. It consists of two parts: sensing element and guard. The confining ring is removable.

where $\dot{m}$ is the mass flow of the cooling water in kg/s, C_p the heat capacity of water in kJ/(kg K), A the sensing surface area in m^2 and T_{inlet}, T_{outlet} are inlet and outlet temperatures of the cooling water, respectively, in K.

The heat transfer coefficient h in W/m^2K from the flame jet to the impinged surface is defined as in Eq. 4.16:

$$h = \frac{\dot{q}_{sen}}{(T_f - T_{surf})} \tag{4.16}$$

where the difference $T_f - T_{surf}$ presents the driving force for the heat transfer from the impinging fluid jet to the impinged surface. T_{surf} is the temperature of the surface, and T_f is the temperature of the heat source, which is usually the temperature of the fluid jet at the exit nozzle [104].

Design of the water-cooled calorimeter

The calorimeter consists of two parts: 1) sensing element and 2) guard, which are presented in Fig. 4.13. The sensing element is the central part of the calorimeter. It is surrounded by the guard that isolates the sensing element from potential heat loss or gain. Both parts are cooled with water: calorimetric water and guard water, respectively.

A closer look at the cross-section of the sensing element is presented in Fig. 4.14. It has an empty inner space 5 mm beneath the sensing surface. This is the area where the calorimetric water flows. A pipe (1, Fig. 4.14) is placed along the centreline axis in order to define the direction of the cooling water flow. Thus, the cooling water enters the inner space at the bottom right corner and it flows upwards along the outer side of the pipe. At the top of pipe the water enters it flowing to the exit at the bottom (6, Fig. 4.14). The guard water also flows 5 mm beneath the guard surface. The water enters the guard inner space at the bottom left corner (2, Fig. 4.14) flowing upward nearby the sensing element. After reaching the top of the inner space, the guard water flows through a spiral channel beneath

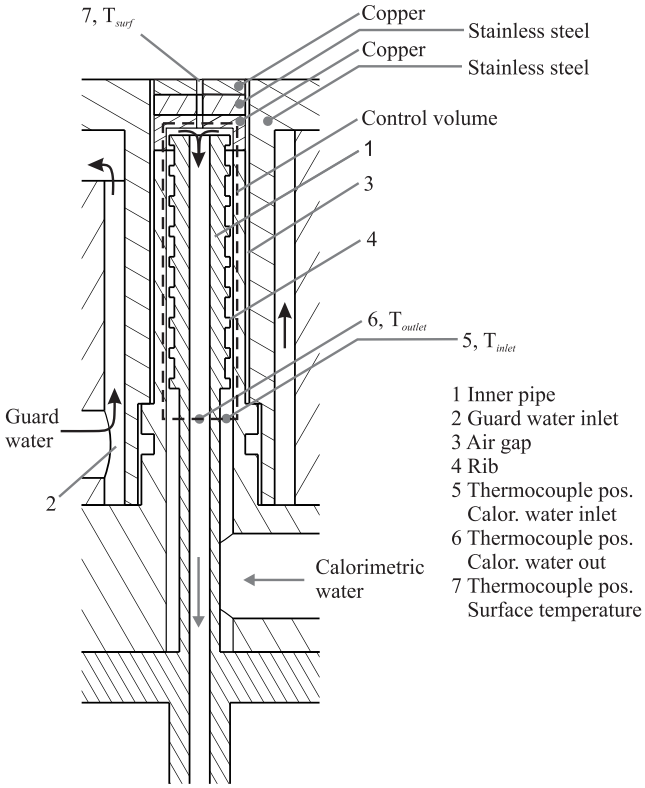

Figure 4.14: Closer look at the cross-section of the sensing element. The heat balance equation (Eq. 4.15) of the calorimetric cooling water was valid for the marked control volume. The arrows show the flow direction of cooling water (calorimetric and guard water).

the surface to the exit located on the side of the guard.

The heat flux measured with the calorimeter is the heat that enters the calorimetric cooling water ($\dot{q}_{sen}$). Ideally, this heat is equal to the heat transferred through the sensing surface ($\dot{q}_{real}$). But, in case of additional heat transfer ($\delta\dot{q}$) a) from the guard to the sensing element ($\delta\dot{q} > 0$), b) from the sensing element to the surroundings (e.g. guard, $\delta\dot{q} < 0$), or c) to the cooling water outside of the control volume, $\dot{q}_{sen}$ can be described with Eq. 4.17:

$$\dot{q}_{sen} = \dot{q}_{real} + \delta\dot{q} \tag{4.17}$$

In order to prevent heat transfer in radial direction between the sensing element and the guard, a 30 mm long and 0.5 mm thick air gap is introduced (3, Fig. 4.14). The heat transferred from the sensing element to the calorimetric water is increased with turbulence in the water flow by ribs at the inside pipe (4, Fig. 4.14). In addition, the wall thickness of the sensing element is minimized, which also increases the transferred heat to the cooling water. The operating flow rate was 120 kg/h and 129 kg/h for the calorimetric and guard water, respectively. The inlet temperature of the calorimetric water is reduced with a chiller before entering the sensing element in order to prevent eventual boiling.

The inlet temperature of the calorimetric water (T_{inlet}) is measured with three K-type thermocouples (0.5 mm diameter thickness). They are positioned shortly after the water entered the sensing element around the inner pipe in the inner space (5, Fig. 4.14). T_{inlet} from Eq. 4.15 presents the average of the recorded values with all thermocouples. The outlet temperature of the calorimetric water (T_{outlet}) is measured with one thermocouple placed in the pipe (6, Fig. 4.14). The tips of all thermocouples are positioned at the same height, thus defining a control volume (see Fig. 4.14) which Eq. 4.15 was applied for. The tips are aligned with the bottom of the air gap in order to have minimal resulting $\delta\dot{q}$. The surface temperature of the calorimeter (T_{surf}) is measured with a K-Type thermocouple soldered

in the centre of the sensing element with the tip at the surface plane (7, Fig. 4.14).

According to the measurement of the spallation temperature (chapter 4.2) the heat transfer study has be performed with a heat flux sensor that can generate temperatures between 430 and 600 °C. Also, due to measuring uncertainty of the spallation temperatures it is important to estimate the sensitivity of the heat transfer coefficient to the surface temperature of the target. The surface temperature of the target varied only about 50 °C when changing the mass flow of the calorimetric water. Larger temperature difference is required to study the heat transfer behaviour at surface temperatures of a larger temperature range. Therefore, two calorimeters were designed: the first calorimeter (CU calorimeter) completely manufactured of CuETP copper (both, the sensing element and the guard). Because of its high thermal conductivity (388 W/(m K)) the surface temperature of the sensing element (T_{surf}) did not exceed 200 °C. The second calorimeter (PSI calorimeter) is also made of CuETP copper, but a 2 mm thick layer made of stainless steel (1.4435, thermal conductivity 15 W/(m K)) was soldered 1.5 mm beneath the sensing surface (shown in Fig. 4.14). It reduces the overall thermal conductivity of the sensing element and thus increased the temperature of its surface up to a maximum of 605 °C. A thin part of the sensing element (1.5 mm) exposed to the flame jet is still made of copper to ensure a uniform surface temperature profile, as in the case of CU calorimeter. A disadvantage of the copper layer exposed to the jet is its oxidation at high temperatures. Therefore, all experiments were performed after the oxidation film had been formed in order to keep physical properties of the calorimeter unchanged. The guard is manufactured from stainless steel in order to keep the construction simple and to provide higher surface temperature than in case of CU calorimeter.

4.3.3 Experimental procedure

The flame power (firing rate $\dot{Q}_f$) was varied between 23 and 54 kW. Two nozzles were tested with a diameter (D_{nozzle}) of 7.9 mm (D_1) and 12.1 mm (D_2). The tested range of dimensionless distance L was between 5 to 14.

During the experiment, the measuring procedure was the following:

- The mass flows of the guard and the calorimetric water were chosen and the chiller was adjusted to reach a desired inlet temperature. Afterwards, the flame was ignited.
- After reaching the desired firing rate, the calorimeter was positioned at the desired dimensionless distance (L) with a linear actuator ($L = 5 - 14$). As soon as steady state had been reached the measurement was performed.
- The surface temperature of the sensing element as well as the inlet and the outlet temperature of the calorimetric and guard water were recorded for 30 s.

4.3.4 CFD modelling of the heat flux sensor

The CFD modelling is done in Comsol Multiphysics in order to estimate the heat loss or gain of the sensing element $\delta\dot{q}$ for the tested operating conditions. Only the calorimeter is subject of the numerical simulation. The geometry is observed as axisymmetric. Therefore, the spiral flow of the guard water can not be modelled and it is approximated with a straight horizontal channel. The results of interest are only those from the sensing element. Also, the thermocouples are not modelled for reasons of computation simplicity and assumed negligible influence on the results. Conservation of mass, momentum and energy in laminar regime are applied for the water regions. Conservation of energy is considered for the solid regions (including the air gap). The boundary conditions are defined as follows:

- mass flow of calorimetric water and guard water as set during the experiments
- inlet temperatures of calorimetric and guard water as measured during the experiments
- measured temperatures of sensing surface (T_{surf}), outlet temperature of calorimetric water and guard water as obtained in experiments
- open boundary on all surfaces exposed to the surrounding air with temperature of 25 °C

The heat flux profile on the impinged surface is assumed to have a Gaussian form as done in literature [74, 102]. It is defined with Eq. 4.18.

$$\dot{q} = k \cdot e^{-\frac{(r)^2}{a^2}} \tag{4.18}$$

The coefficients k and a are the resulting parameters of the model giving the estimated heat flux $\dot{q}_{real}$ through the sensing surface as a result:

$$\frac{1}{r_{sen}} \int_0^{r_{sen}} k \cdot e^{-\frac{(r)^2}{a^2}} dr = \dot{q}_{real} \tag{4.19}$$

$$\frac{1}{r_{guard} - r_{sen}} \int_{r_{sen}}^{r_{guard}} k \cdot e^{-\frac{(r)^2}{a^2}} dr = \dot{q}_{guard} \tag{4.20}$$

where r_{sen} presents the radius of the sensing element and r_{guard} the radius of the guard. It is noted, that the heat flux $\dot{q}_{real}$ could actually not be computed by Comsol Multiphysics in one run. Instead, it had to be obtained iteratively by adjusting the coefficients k and a until the calorimetric water outlet temperature (T_{outlet}) matched the experimentally measured values.

The confinement is not considered in the model because it is assumed that the defined boundary conditions as listed above contains all information about the heat entering the calorimeter. The resulting temperature profile

in PSI calorimeter is presented in Fig. 4.15). Highest temperatures were at the impinged region: sensing surface and part of the guard surface.

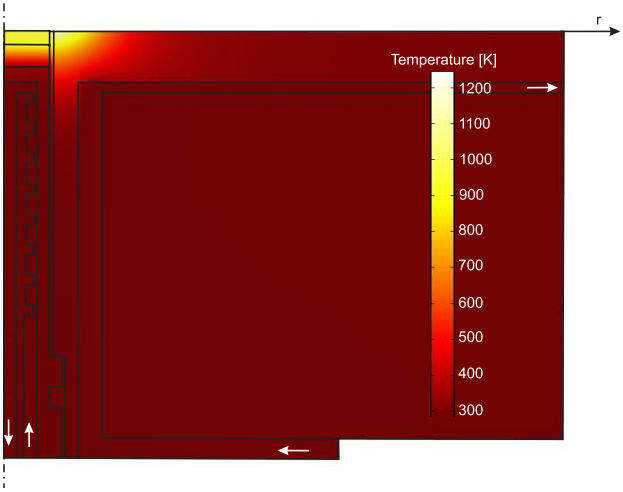

Figure 4.15: Temperature profile of the PSI calorimeter cross section obtained with the simulation. Applied boundary conditions were those obtained at 38 kW and L = 5. Arrows point the flow direction of the cooling water.

4.3.5 Uncertainty of the heat flux measurement

Devices that are included in the heat transfer study are listed in Table 4.5 with their measuring uncertainties.

The final uncertainty in measured heat flux ($\delta \dot{q}_{sen}$) and heat transfer coefficient (δh) is calculated using the method of Kline et al. [105]. It is in detail presented in Eq. 4.21 and 4.22. Eq. 4.21 presents the error calculation of heat flux. Because the measured data was recorded for a certain period of time, the standard deviation σY of each variable U ($Y \in \{\dot{m}, T_{inlet}, T_{outlet}, T_{surf}, T_f\}$) is determined and included in the un-

Table 4.5: Uncertainty of measuring devices

Measured quantity	Symbol	Instrument	Uncertainty
Calorimetric water mass flow	$\delta\dot{m}$	Flow meter	0.2% *
Inlet water temperature	$\delta T_{inlet,d}$	Calibrated K-type thermocouple	0.32 °C
Outlet water temp.	$\delta T_{outlet,d}$	Calibrated K-type thermocouple	0.32 °C
Surface temperature	$\delta T_{surf,d}$	K-type thermocouple	0.075 %
Flame temperature	$\delta T_{f,d}$	S and K-type thermocouple	0.25 % and 0.075 % *

* from measured value

certainty calculation as presented in Eq. 4.22.The K-type thermocouples, which measurs the temperature of the calorimetric water, were calibrated in order to reduce their measuring uncertainty because they had the largest influence on the final uncertainty in the heat flux. The calibration was performed with a FLUKE 9142 Metrology Well and the obtained data of the thermocouples and reference temperatures were fitted with a least squares linear regression method. Any variations in the methane and air flow can also influence the heat transfer study. The uncertainties of the methane and air flow rate measurement are ± 3.70 % and ± 0.61 %, respectively. Therefore, it is assumed that the uncertainty in the gas flow rate has a negligible influence on the heat flux measurement. The final uncertainty in the heat flux is in the best case 7 % and in the worst 45 % as shown in Fig. 4.16. The highest uncertainties results at the lowest heat fluxes because the temperature difference between the inlet and outlet cooling water ($T_{outlet} - T_{inlet}$ in Eq. 4.15) was only 1 °C. However, process conditions with heat fluxes higher than 4 MW/m^2 are required to perform flame-jet drilling. In that range, the uncertainty in obtained heat fluxes was less than 20 %.

$$\delta\dot{q} = \sqrt{\left(\frac{C_p\left(T_{outlet} - T_{inlet}\right)}{A}\delta\dot{m}\right)^2 + \left(\frac{\dot{m}C_p}{A}\delta T_{outlet}\right)^2 + \left(-\frac{\dot{m}C_p}{A}\delta T_{inlet}\right)^2} \tag{4.21}$$

$$\delta Y = \sqrt{\left(\sigma Y\right)^2 + \left(\delta Y_d\right)^2} \tag{4.22}$$

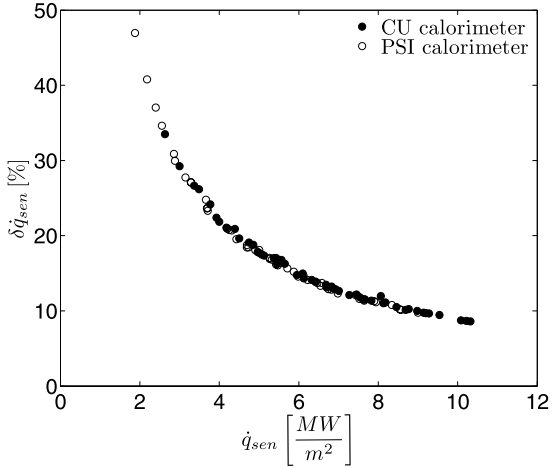

Figure 4.16: Uncertainty in heat flux measurement with both calorimeters. PSI calorimeter measured less heat flux than CU calorimeter at same operating conditions, because of its larger thermal resistance.

The mass flow of the calorimetric water influences the recorded heat flux. At the lowest tested flow rate the temperature difference of the calorimetric water ($T_{outlet} - T_{inlet}$) reached 35 °C. As a result, heat flux ($\dot{q}_{sen}$) was the lowest, surface temperature (T_{surf}) the highest and according to the CFD model more heat is lost instead of being transferred to the calorimetric water in the defined control volume. When increasing the calorimetric water flow rate, the temperature difference $T_{outlet} - T_{inlet}$ was decreasing,

$\dot{q}_{sen}$ was increasing and T_{surf} decreasing. More heat is transferred to the water in the control volume, which decreases the overall heat losses, but the calculated uncertainty $\delta\dot{q}$ increases due to a smaller ratio of temperature difference ($T_{out} - T_{in}$) and measuring uncertainty of thermocouples. As an example, $\dot{q}_{sen}$ is presented including its uncertainty range $\delta\dot{q}$ in Fig. 4.17 depending on the calorimetric water mass flow rate. The condition of the flame were unchanged (28 kW and $L = 6$) when measuring with the PSI calorimeter.

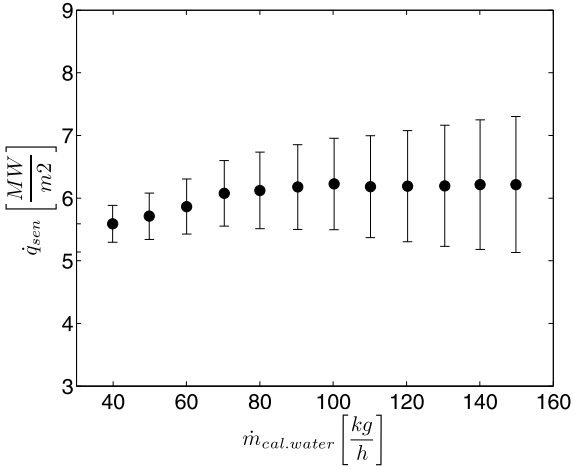

Figure 4.17: Heat flux change with the flow of the calorimetric water obtained with PSI calorimeter at 28 kW and $L = 6$.

It can be noticed, that the heat flux measurement has a considerable sensitivity to the calorimetric cooling water at low flow rates, which continuously reduces until getting negligible above 80 kg/h for the PSI calorimeter (or 120 kg/h for the CU calorimeter). Therefore, the final mass flow of the calorimetric water for the performed heat transfer study with both calorimeters was selected as 120 kg/h.

The uncertainty range of the heat transfer coefficient calculated as in Eq.

4.23 is presented together with the results in chapter 4.3.6 being between ±8 % and ±45 %, but at the process conditions relevant for flame-jet drilling the uncertainty was less than 20 %, as in case of heat flux uncertainty.

$$\delta h = \sqrt{\left(\frac{C_p\left(T_{outlet}-T_{inlet}\right)}{A\left(T_f-T_{surf}\right)}\delta\dot{m}\right)^2+\left(\frac{\dot{m}C_p}{A\left(T_f-T_{surf}\right)}\delta T_{outlet}\right)^2} + \sqrt{\left(-\frac{\dot{m}C_p}{A\left(T_f-T_{surf}\right)}\delta T_{inlet}\right)^2+\left(-\frac{\dot{m}C_p\left(T_{outlet}-T_{inlet}\right)}{A\left(T_f-T_{surf}\right)^2}\delta T_f\right)^2} + \sqrt{\left(\frac{\dot{m}C_p\left(T_{outlet}-T_{inlet}\right)}{A\left(T_f-T_{surf}\right)^2}\delta T_{surf}\right)^2} \tag{4.23}$$

4.3.6 Results

Influence of air-fuel equivalence ratio

The air-fuel equivalence ratio (λ) influences the reaction in terms of flame temperature, flame length, etc. In the presented work, a test was performed that identified value of λ at which the highest heat transfer to the heat flux sensor was obtained. Therefore, the influence of λ (from 0.9 up to 1.3) on heat flux $\dot{q}_{sen}$ was observed. The results are presented in Fig. 4.18 for a nozzle-to-target distance L between 6 and 14, the D_1 nozzle diameter and a firing rate of 38 kW. The measurement was performed with the CU calorimeter for an unconfined flame jet. It was assumed that the same trends would be obtained with a confined jet, larger nozzle, at different firing rates and with the PSI calorimeter. Tests with air-fuel equivalence ratios lower than 0.9 or higher than 1.3 resulted in flame instability.

The least efficient heat transfer (which was reflected by the lowest heat flux) was observed for $\lambda = 0.9$. For those conditions the fuel was not com-

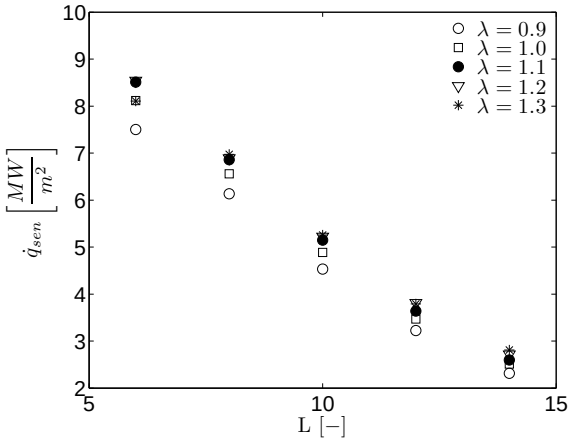

Figure 4.18: Heat flux obtained with CU calorimeter at different was dimensionless distances, different air-fuel equivalence ratios and at $\dot{Q}_f$ =38 kW. Nozzle diameter $D_1 = 7.9$ mm.

pletely combusted and a part of the combustion heat remained unreleased. In case of larger λ larger heat flux was obtained. Because the drilling experiments were conducted at $L \geqslant 8$, the most relevant results were related to those distances. The best performance was obtained with λ equal to 1.1, 1.2 and 1.3 and difference between the obtained heat fluxes was almost negligible. With $\lambda = 1.1$ less air is consumed and larger firing rates can be reached, thus this value of air-fuel equivalence ratio was applied in all other experiments.

Influence of jet confinement

Heat transfer form underexpanded jets were rarely investigated [106, 107, 108]. Yu et al. [106] studied the heat transfer from an underexpanded air jet impinging on a flat surface. It was noticed that the heat flux oscillates with the distance because of the static pressure change in the jet. The heat flux oscillations were also present in this work when the underexpanded jet

impinged the flat calorimetric surface without the confinement. As shown in Fig. 4.19, the heat flux measured with the unconfined underexpanded impinging jet was fluctuating with dimensionless distance ($\dot{Q}_f = 38$ kW and $L < 9$). At larger distances the shocks were not present, thus no oscillations in the heat flux are visible. For firing rate 23 kW the jet was subsonic without shocked structure (see Fig. 3.6), thus the fluctuations did not exist.

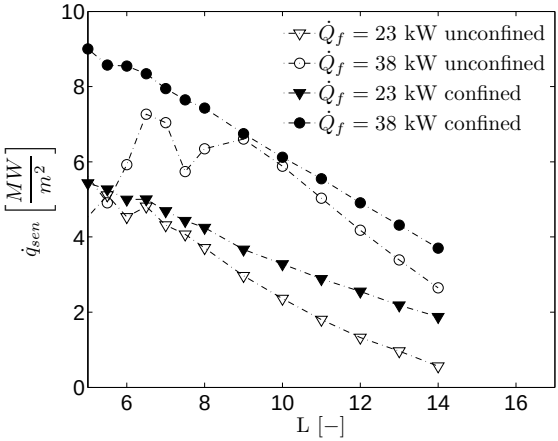

Figure 4.19: Heat flux obtained with the PSI calorimeter with and without confinement. The nozzle diameter was $D_1 = 7.9$ mm at different firing rats.

A different phenomenon was registered with the confined jet: the heat flux did not oscillate with dimensionless distance. The confinement seems to prevent the entrainment of ambient air such that no strong expansion and compression of the exhaust gases occurs. Therefore, no heat flux fluctuations were visible (see Fig. 4.19). When comparing the confined and unconfined heat transfer results, the confined flame jet transferred more heat to the impinged surface. This could be due to reduced entrainment of ambient air in the flame jet, which reduces the drop of gas temperature and velocity downstream.

In this work the heat transfer study with the confinement on the calorimeters is only presented, because this was the configuration of the impinging flame jet in the drilling experiments.

Influence of calorimeter design

The performance of both calorimeters was compared according to the results from the heat transfer study performed with the confined impinging flame jet. The PSI calorimeter has higher thermal resistance than the CU calorimeter, which influences the surface temperatures of the sensing element. The surface temperatures were between 250 and 600 °C and between 70 and 200 °C for PSI and CU calorimeter, respectively (see Fig. 4.20). As a consequence, more heat was transferred to the calorimetric water through the CU calorimeter (up to 10.5 $\mathrm{MW/m^2}$) than through the PSI calorimeter (up to 9.0 $\mathrm{MW/m^2}$) for the same process conditions (see Fig. 4.21).

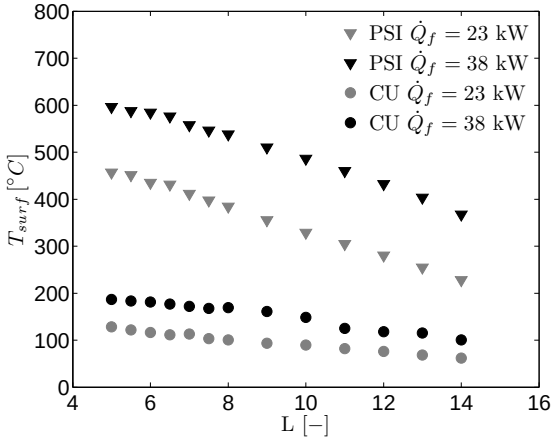

Figure 4.20: Surface temperatures of the PSI and CU calorimeter with dimensionless distance and firing rate. Nozzle diameter was $D_1 = 7.9\,\mathrm{mm}$ and the flame jet was confined.

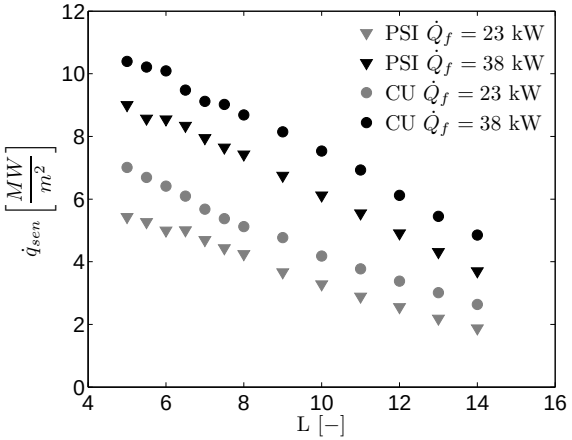

Figure 4.21: Heat flux measured with the PSI and CU calorimeter at different dimensionless distances and firing rates. Nozzle diameter was $D_1 = 7.9\,\text{mm}$ and the flame jet was confined.

The calculated heat transfer coefficient is presented in Fig. 4.22. For most operating conditions there was no apparent influence of the surface temperature on the heat transfer coefficient. Still, at dimensionless distances smaller than 9 and firing rate of 38 kW, the calculated heat transfer coefficient was smaller for the CU calorimeter than for the PSI calorimeter. In those cases the amount of transferred heat is very high, thus it is assumed that one part of the heat transferred through the sensing surface of CU calorimeter is lost: either it had not entered the cooling water in the control volume, or it might have been transferred to the guard ($\delta\dot{q} < 0$). Both could occur because the CU calorimeter had larger overall thermal conductivity than the PSI calorimeter. This difference is less noticeable for the firing rate of 23 kW. For dimensionless distances larger than 9 the surface temperature does not influences the heat transfer coefficient even though the surface temperature changed from 180 to 550 °C. Because drilling experiments are performed at dimensionless distance larger than 8, it is concluded that in experiments the heat transfer coefficient is not

surface temperature dependant. Therefore, it was assumed that the calculated values are also valid for the case the rock is impinged by the same flame jet.

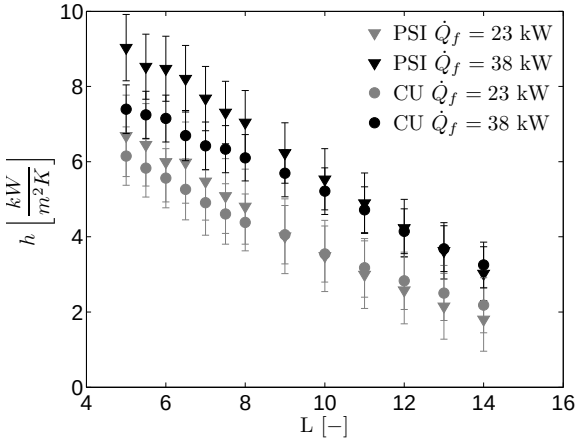

Figure 4.22: Heat transfer coefficient calculated with Eq. 4.16 from the measurement performed with both calorimeters. The nozzle diameter was $D_1 = 7.9\,\text{mm}$ and the flame jet was confined.

With the modelling results the heat distribution within the calorimeters and probable heat losses $\delta\dot{q}$ (Eq. 4.17) is estimated. The measured heat flux ($\dot{q}_{sen}$) is the heat through the impinged surface and ($\dot{q}_{real}$) is the modelling result. The heat transfer coefficient calculated from $\dot{q}_{real}$ and from $\dot{q}_{sen}$ are presented in Fig. 4.23 and 4.24 for selected operating points. Simulated and experimental results are in a good agreement and they differed in the range of measuring uncertainty (heat transfer coefficient calculated from the simulated heat flux is higher than the one calculated from the measured heat flux). In the model the calculated surface temperatures of the guard next to the air gap were higher than their respective surface temperatures on the sensing element, thus this temperature difference acted as driving potential for heat flux gain $\delta\dot{q}$ from the guard to the sensing element which was maximally 11.9% of $\dot{q}_{sen}$ for the tested operating points.

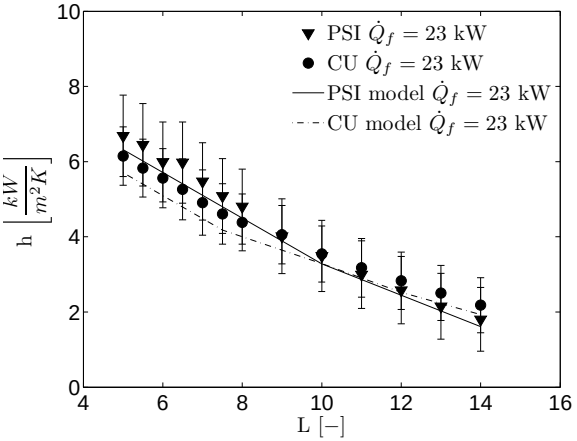

Figure 4.23: Heat transfer coefficient obtained with the PSI and CU calorimeter and with the model at different dimensionless distances and at $\dot{Q}_f = 23\,\mathrm{kW}$. Nozzle diameter was $D_1 = 7.9\,\mathrm{mm}$

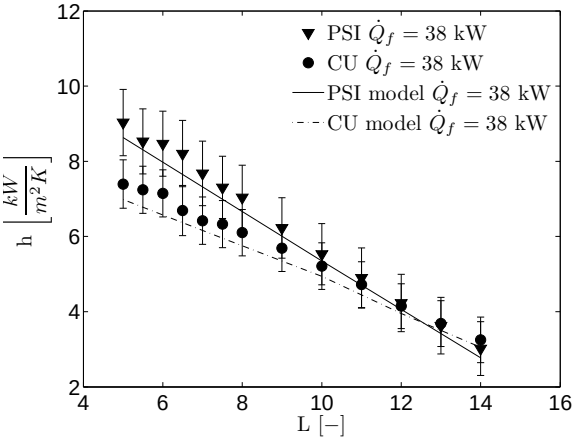

Figure 4.24: Heat transfer coefficient obtained with the PSI and CU calorimeter and with the model at different dimensionless distances and at $\dot{Q}_f = 38\,\mathrm{kW}$. Nozzle diameter was $D_1 = 7.9\,\mathrm{mm}$

Influence of firing rate and dimensionless distance

From Fig. 4.25 it can be seen that by increasing the firing rate the amount of transferred heat to the calorimeter was increasing. The maximum heat flux was around 9.0 $\mathrm{MW/m^2}$, which is larger than the maximum heat flux obtained by Rauenzahn [24] (7.4 $\mathrm{MW/m^2}$). By increasing the firing rate more reactants are combusted, more heat is released, the temperature at the nozzle and the mass flow of the exhaust gases increased, thus, more heat is transferred to the sensor. Moving the calorimeter further away from the exit nozzle, the heat flux decreased because more ambient air is entrained into the jet before it impinged the sensor surface. This entrainment is more intense at higher firing rate due to increased turbulence in the shear layer, which results in higher heat flux gradient with dimensionless distance. The corresponding heat transfer coefficient is presented in Fig. 4.26.

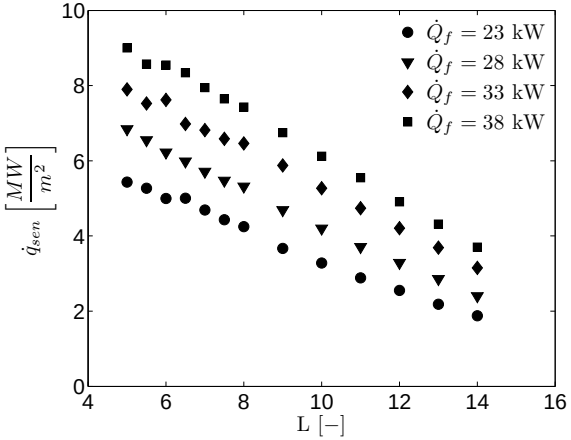

Figure 4.25: Heat flux measured with the PSI calorimeter at different dimensionless distances and firing rates. Nozzle diameter was $D_1 = 7.9\,\mathrm{mm}$ and the flame jet was confined.

The heat flux increased linearly with the firing rate having a higher gradient at smaller distances (see Fig. 4.27). The change in gradient can again be

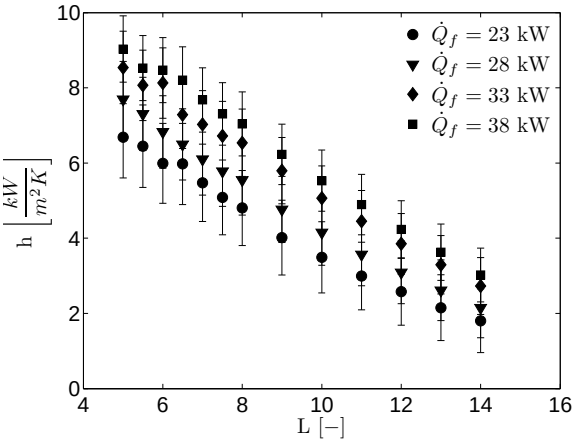

Figure 4.26: Heat transfer coefficient calculated from heat flux measurement obtained with the PSI calorimeter at different dimensionless distances and firing rates. Nozzle diameter was $D_1 = 7.9\,\text{mm}$ and the flame jet was confined.

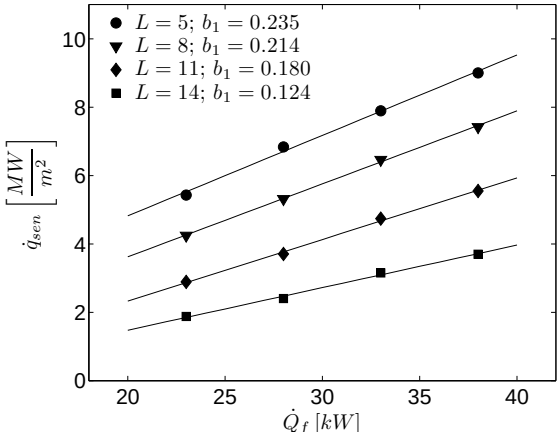

Figure 4.27: Heat flux measured with the PSI calorimeter at different firing rates keeping the dimensionless distance constant. Data was fitted with a linear dependency between the firing rate ($\dot{Q}_f$) and the heat flux ($\dot{q} = b_0 + b_1 \cdot \dot{Q}_f$). Nozzle diameter was $D_1 = 7.9\,\text{mm}$ and the flame jet was confined.

attributed to the air entrainment.

Even though the flame jet was confined, the obtained results confirm some conclusions conducted from the heat transfer study of free impinging flame jets presented in the work of Baukal et al. [58]. The conclusions are: 1) The heat flux increased as the dimensionless distance decreased and 2) keeping the distance and the oxidizer composition constant the heat transfer improved with the firing rate.

Influence of nozzle diameter

For the same firing rate lower heat fluxes were recorded with the larger nozzle diameter mounted at the burner. The reason for the reduced amount of transferred heat is the reduced gas velocity at the nozzle resulting into reduced turbulence of the impinging jet and lower the convection to the calorimeter. This is shown in Fig. 4.28 by comparing the heat fluxes obtained at 38 kW with nozzles of diameter D_1 and D_2 ($D_1 < D_2$). At $L = 5$, where the flame jet was completely confined, the heat flux with D_1 was 2.35 times larger than the heat flux with D_2, which is also the ratio of both nozzle areas. Fig. 4.28 also denotes the nozzle heat flux $\dot{q}_f$, which is the ratio between the firing rate $\dot{Q}_f$ and nozzle area A_{nozzle} as calculated in Eq. 4.24

$$\dot{q}_f = \frac{\dot{Q}_f}{A_{nozzle}} \tag{4.24}$$

With a firing rate of 23 kW and the small nozzle diameter D_1 the nozzle heat flux was the same as with the larger nozzle (D_2) at 54 kW ($\dot{q}_f = 474\,\mathrm{MW/m^2}$). At equal nozzle heat fluxes the recorded heat fluxes at the sensors were also equal at smaller dimensionless distances due to flame jets that are completely confined and less heat losses could result.

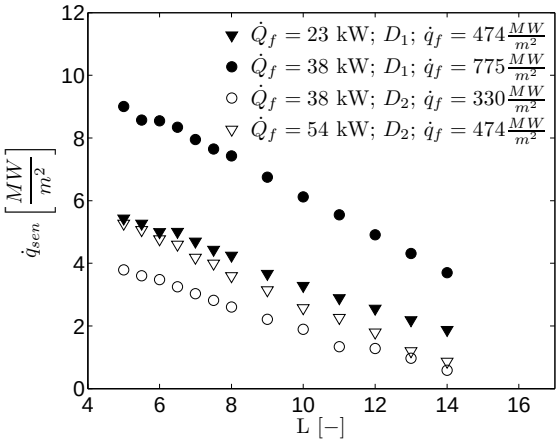

Figure 4.28: Heat flux measured with the PSI calorimeter at different dimensionless distances and firing rates. Nozzle diameters were $D_1 = 7.9\,\text{mm}$, $D_2 = 12.1\,\text{mm}$ and the impinging jet was confined. Nozzle heat flux ($\dot{q}_f$) was calculated as in Eq. 4.24

Comparison of firing rate, dimensionless distance and nozzle diameter influence

The influence of the tested parameters on heat transfer from the impinging flame jet is summarized in Table 4.6. Those parameters are the firing rate ($\dot{Q}_f$), dimensionless distance (L) and nozzle diameter (D_{nozzle}). The maximum heat flux was measured at $\dot{Q}_f(\dot{q}_{max}) = 38\,\text{kW}$, $L(\dot{q}_{max}) = 5$ and $D_{nozzle}(\dot{q}_{max}) = 7.9\,\text{mm}$. The heat flux decreased by increasing the dimensionless distance and the nozzle diameter and by decreasing the firing rate. In order to decrease the heat flux up to 39 %, L had to be increased by 120 % for constant $\dot{Q}_f$ and D_{nozzle}, $\dot{Q}_f$ decreased by 38 % for constant L and D_{nozzle} and D_{nozzle} increased also by 38 % for constant L and $\dot{Q}_f$. As a conclusion, the properties related to the impinging flame jet (the firing rate and the nozzle diameter) have larger influence on the heat transfer than the properties related to the target (dimensionless distance). The same comparison was performed starting from another dimensionless distance: starting point two (at $\dot{Q}_f(\dot{q}_{max}, L = 8) = 38$ kW and $D_{nozzle}(\dot{q}_{max}, L = 8) = 7.9\,\text{mm}$) and three (at $\dot{Q}_f(\dot{q}_{max}, L = 12) = 38\,\text{kW}$ and $D_{nozzle}(\dot{q}_{max}, L = 12) = 7.9\,\text{mm}$). It was noticed that at larger dimensionless distances all process parameters have similar influence on the heat transfer. This was due to more air that was entrained in the jet and a change in any process parameter effects the heat transfer in a same way.

Table 4.6: Influence of process parameters of impinging flame jet on heat transfer.

Starting oper. conditions			$\Delta\dot{Q}_f[\%]$	$\Delta D_n[\%]$	$\Delta L[\%]$	$\Delta\dot{q}_{sen}[\%]$
$\dot{Q}_f[kW]$	$L_n[-]$	$D_n[mm]$				
38	5	7.9	-38	38	120	-39
38	8	7.9	-39	33	55	-36
38	12	7.9	-17	17	17	-24

4.3.7 Conclusion

The heat transfer study performed with both subsonic and supersonic underexpanded impinging flame jets can be concluded with the following:

- The underexpanded free flame jet impinging on the calorimeter generated a heat flux that fluctuated with dimensionless distance. This is a consequence of the shocks present in the flow of the exhaust gases. By confining the flame jet no oscillations were noticed due to limited jet expansion (shock absence).
- Two calorimeters have been constructed for the reported heat transfer study: PSI and CU calorimeter. The PSI calorimeter has lower thermal conductivity, thus, less heat is transferred to the calorimetric water compared to CU calorimeter. This resulted in higher surface temperatures of the sensing element, which makes the heat transfer study more relevant to the flame-jet drilling experiments.
- The results showed that the surface temperature of the target (up to 600 °C) did not remarkably influence the heat transfer from an impinging flame jet.
- Heat losses can be minimized by increasing the length of the observed control volume in the CU calorimeter, or by increasing the sensing surface in order to reduce the possible heat losses.
- The sensitivity of the heat transfer to the target when changing the process parameters related to the flame jet (the firing rate and the nozzle diameter) is larger than when changing the dimensionless distance, if the target is placed very close to the exit nozzle. This sensitivity reduced when the target is located further away from the nozzle.

4.4 Force of the impinging jet

The exhaust gases that impinge the rock surface during the flame-jet drilling have the role not only to transfer the heat but also to remove completely or partially detached particles. Therefore, both the jet momentum and heat are required for successful thermal spallation. The mass flow, fluid temperature, velocity profile and physical properties of the exhaust gases have influence on both the heat transfer and the force of the hot impinging jet. Therefore, the force of the impinging jet was measured in order to understand its relevance to the flame-jet drilling experiments and its dependence on firing rate $\dot{Q}_f$ and dimensionless distance L. The force of the jet is direct proportional to gas velocity and mass flow of the impinging jet.

4.4.1 Experimental setup

A scale was applied for the force measurement of the impinging jet. The measuring system is presented in Fig. 4.29. A cylindrical metal plate (target plate) was placed on the scale and impinged by the flame jet. To protect the scale from the heat, a rock sample was used as isolation between the target plate and the scale. Also, the same cylindrical confinement used in the heat transfer study (chapter 4.3) was placed on the impinged plate in order to confine the flame jet as in the drilling experiments. The objective is to record only the force acting on the surface of the target plate that simulats the impinged rock surface part. Therefore, the free surface of the scale and the confining cylinder were protected with a fixed thin plate (fixed protection) that allows the impinging jet only to act on the free surface of the target plate. The resulting signal of the scale is recorded electronically.

The obtained mass (m_{scale}) was converted into the force of the jet (F_{jet}) as in Eq. 4.25.

$$F_{jet} = m_{scale} \cdot g \tag{4.25}$$

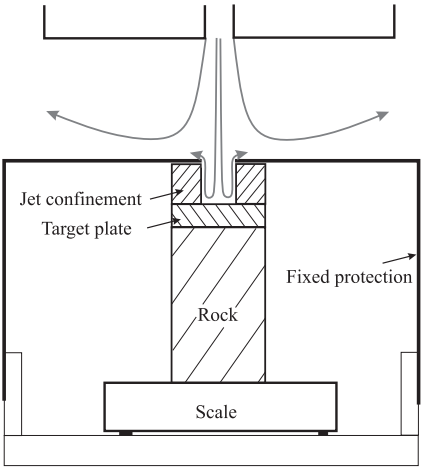

Figure 4.29: System for measuring the force generated by the impinging flame jet.

where g represents the standard gravity of Earth ($9.81\,\mathrm{m/s^2}$). The corresponding pressure on the free target plate surface was calculated as in Eq. 4.26

$$p_{jet} = \frac{F_{jet}}{A} \tag{4.26}$$

Both the firing rate and the dimensionless distance were varied, but mostly in the range where flame-jet drilling experiments could be performed. The installed nozzle had the diameter $D_1 = 7.9$ mm, because with that nozzle better heat transfer and larger jet velocities could be achieved.

4.4.2 Results

By increasing the firing rate, the pressure on the target p_{jet} was also increasing due to the larger mass flow rate of exhaust gases. This is graph-

ically presented in Fig. 4.30. Keeping the mass flow of the exhaust gases constant and moving closer to the source jet, also larger pressure on the target resulted. This is due to increase of the average jet velocity along the central jet axis. Still, the change in the dimensionless distance influences almost negligibly the pressure on target, probably due to small difference in the mass of the impinging gas. Mostly the firing rate has an effect on the measurement due to an influence on the mass flow of the jet. The maximal pressure on the impinged target was 13.4 kPa. Compared to the strength of the material that has to be exceeded in thermal spallation (compressive strength is 116.9 MPa and the tensile strength 9.54 MPa [79]) the applied pressure of the jet should have negligible influence on the stress field in the rock sample.

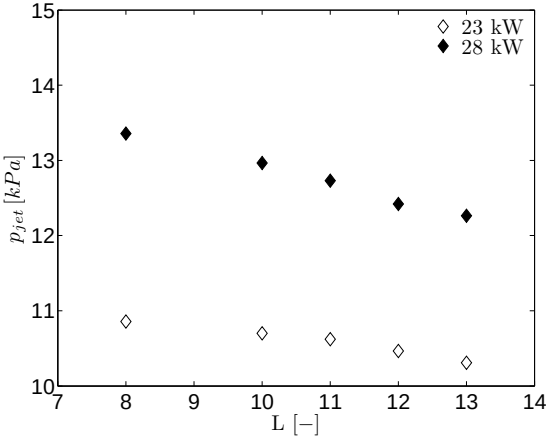

Figure 4.30: Pressure of the impinging jet on target for various firing rates and dimensionless distances. Exit nozzle diameter was $D_1 = 7.9$ mm.

4.4.3 Conclusion

The impinging flame jet made for flame-jet drilling exerted the pressure on the target surface up to 13.4 kPa. This is negligible compared to the

tensile or compressive strength of the Grimsel granite. Therefore, it is assumed that it does not influence the stress field in the rock sample and the crack propagation in the surface layers. It is assumed that the jet momentum mostly influences the removal of completely or partially ejected spalls from the rock surface and enables the new rock layers to be heated more efficiently.

Chapter 5

Flame-jet drilling at ambient pressure

Flame-jet drilling can be better understood and its efficiency be improved by following two research paths: 1) investigating the influence of rock properties on the thermal spallation, 2) investigating the influence of process parameters on the flame-jet drilling. Both paths are equally important, but in the field, the rock type that will be drilled cannot be altered. Therefore, engineers are only able to optimise the drilling by adjusting the process parameters (applied firing rate, fluid temperature, etc.). In the presented work it was decided to follow the second path with Grimsel granite being the rock type of choice.

As previously explained in chapter 2.1.1, thermal spallation is related to the developed thermal stresses within the rock surface. Thus, the best approach for understanding of the flame-jet drilling process would be directly measurement of thermal stresses in the rock sample while drilling. A convenient solution is to calculate the stress generation with a numerical model. The model used in this work describes only the stress development in a cylindrical shaped and confined rock sample with a flat surface part at the top where the heat source was applied on for less than a second (until the spallation onset). The obtained numerical results supported the interpretation of the experimentally obtained drilling results.

Several experiments are presented in literature showing that both heat transfer and jet momentum are important for spall creation and their removal [24, 86]. A rock sample cant be drilled by only increasing the rock temperature without presence of a gas stream (e.g. with a laser). The same experimental failure occurred when a rock sample was impinged by an air jet without preheating the air. Browning et al. [16] discovered that the spalls are not always entirely separated from the surface, thus adequate jet momentum is required to carry them and clear the impinged zone. Jets with larger momentum have also improved heat transfer efficiency (being reflected in the heat transfer coefficient) [109], thus, those two phenomena are difficult to separate when trying to understand the influence of each one individually. From the propulsion measurement, the resulting pressure on the impinged surface was maximal 13.4 kPa, which was assumed to have negligible influence on the stress field development in the surface rock layers. In the experimental study the focus was on the influence of uniaxial pressure and heat transfer on flame-jet drilling.

Influence of uniaxial pressure on drilling experiments

So far, it is unknown if stresses acting on a rock mass in nature would influence the generation of thermal stresses during flame-jet drilling. A stress tensor acting on a rock mass in nature is generally caused due to gravity and compressive stresses, and it varies from point to point [110]. Gravity causes axial stresses, thus it is expected that flame-jet drilling would be influenced when drilling deeper in the field. In order to study the impact of axial stresses on flame-jet drilling in laboratory, a uniaxial load was applied on rock samples. It was generated by a simple confinement consisting of two flanges positioned on both flat surfaces of the cylindrical rock probe (essentially forming a sandwich structure) and tightened with bolts and nuts. Up to 47 MPa of axial confining pressure was applied, which would correspond to the conditions in a depth of 2-3 km, when drilled through crystalline rock layers only [111]. The influence of uniaxial load on flame-jet drilling has never been experimentally studied before.

An additional advantage of the axial confinement was blockage of the sample expansion which resulted in axial breakage (mode I crack) as shown in Fig. 5.1 (left-hand side). The breakage occurred due to thermal stresses acting in the core of the sample in less than 10 s after exposing the whole top surface to the flame jet. The thermal stresses required for spallation were released through the created fracture and thermal spallation stopped. Similar problems are also described in the literature related to the flame-jet drilling technology [112]. With axial confinement, the applied load counteracted the generated stresses and mode I crack did not occur. As a result, the drilling lasted much longer creating large cavities and the drilling conditions in laboratory got closer to field conditions (Fig. 5.1, right-hand side).

Figure 5.1: Results of flame-jet drilling without (lefthand side) and with confinement (righthand side)

Influence of heat transfer on drilling experiments

Thermal spallation is a process induced by a steep temperature gradient in a rock sample. The way the rock sample is heated will define the temperature distribution within it. Exposing one part of the rock sample suddenly to a large amount of heat can cause the desired steep temperature gradient and the thermal spallation of the rock surface. Otherwise, when the heat transfer is low, the generated thermal stresses will not reach the values

required to initiate the thermal spallation. This phenomenon was analyzed according to the experimental and numerical results when changing the process parameters and heat transfer to the rock sample.

The results from the heat transfer study obtained with the PSI sensor as described in chapter 4.3 were used to quantify the heat transfer to the rock surface. The measured rock surface temperatures at the onset of spallation were around 500 °C (see chapter 4.2). Similar temperatures were recorded at the surface of the PSI calorimeter thus the obtained heat transfer coefficient can represent the heat transfer to the rock surface from the same flame jet.

5.1 Numerical model for thermal stress analysis

Theoretical analysis of the developed stresses in the rock sample was done by Patru [113] with the Matlab model developed by Meier et al. [114]. It helps to understand the thermal stress development in an observed object when applying a heat source and a load on it. The model is based on uncoupled, quasi-static, thermoelastic equations applied on 2D axisymmetric domain. It is defined with conservation equations for energy and momentum Eq. 5.1 and Eq. 5.2, respectively. The rock was modelled as a linearly elastic, isotropic, homogeneous, nonporous and brittle medium with the dimensions of the tested rock samples.

$$\rho C_p \dot{T} + \frac{E}{1-2\nu} \alpha T_0 \dot{\epsilon} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \tag{5.1}$$

$$\frac{\partial \sigma}{\partial x} + f = 0 \tag{5.2}$$

ρ presents the density, C_p is the heat capacity of the rock sample, $\dot{T}$ is

the time dependent temperature, E is Young's modulus, ν is Poissons ratio, α is a coefficient of thermal expansion, T_0 is the temperature at which no stresses occur, $\dot{\epsilon}$ is the time dependent strain tensor, k is the thermal conductivity, T is temperature, σ is the stress tensor and f_i is an external body force per volume. The properties of the rock from the Grimsel test site were applied (Table 3.1, [79, 80]) with assumption that they are independent of temperature. This assumption does not substantially affect the main conclusions of this work. The equations are valid for the cylindrical coordinate system presented in Fig. 5.2. To complete the system of equations the constitutive equation, or so called generalized Hook's law for the isotropic, linear elastic body under thermal and mechanical load was also considered. It presents the stresses-strains relations inside the body [114].

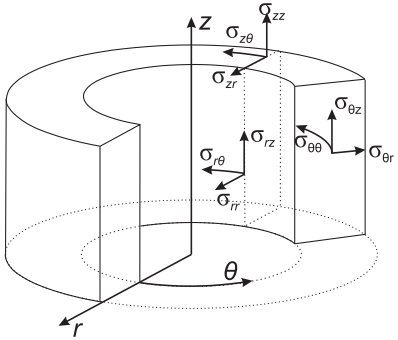

Figure 5.2: Normal and shear stresses in cylindrical coordinates

Initial and boundary conditions of the model are listed below and are graphically illustrated in Fig. 5.3:

- Energy conservation
  1. The initial temperature of the body is 298.15 K.
  2. At the top surface of the rock sample exposed to the flame jet ($0 < r < 0.02$ m, $z = 0.150$ m) convective heat transfer is applied.

It is defined with Eq. 5.3:

$$\dot{q} = h(T_{ref} - T) \tag{5.3}$$

where h is the heat transfer coefficient and the T_{ref} is the reference temperature, i.e. flame temperature (1866 K) and T is the surface temperature of the rock. The heat transfer coefficient was obtained from heat transfer study described in chapter 4.3. The heat transfer study was performed at steady-state, but the heat transfer to the rock sample is a dynamic process and the surface temperature changes with time. However, the heat transfer study showed that even though the surface temperature was varied between 180 and 550 °C (for all firing rates and dimensionless distance $L > 8$) the heat transfer coefficient remained unchanged, especially at the dimensionless distances at which the drilling experiments were performed ($L = 8 - 13$). Therefore, the results from the heat transfer study are applied in the stress analysis.

3. All other boundaries of the domain are defined as adiabatic ($\dot{q} = 0$).

- Momentum conservation

1. No external stress (traction free) is applied on the boundaries that are not confined (the surface part exposed to the flame jet and curved surface of the rock sample).

2. At the bottom of the rock a "$no - slip$" condition is applied (the body velocity at the boundary is equal to velocity of the boundary (zero)).

3. On the upper boundary of the rock ($r > 0.02$ m and $z = 0.150$ m), which is pressed with the flange, normal stress (from 0 to 47 MPa) and friction force are imposed.

4. The symmetry axis is defined as a free slip thus the displacement normal to the wall is zero.

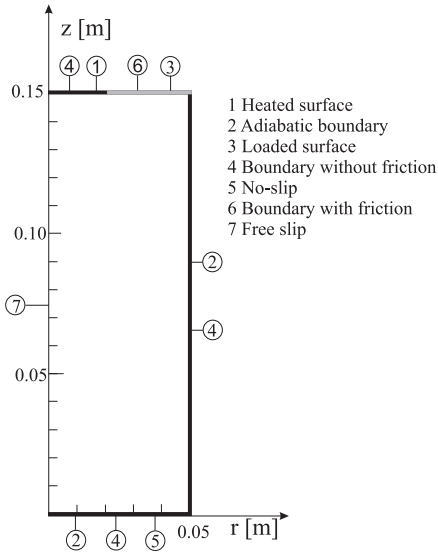

Figure 5.3: Schematic overview of the boundary conditions applied in the stress field model.

In the experiments at the applied process conditions, the onset of thermal spallation occurred less than 0.5 s after the beginning of the flame jet impingement. The model simulates the development of thermal stresses before the onset of spallation.

5.2 Experimental procedure

The rock samples were pressed uniaxially with two flanges and tightened with four bolts as shown in Fig. 5.4. In order to provide a more uniform distribution of the pressure along the surface, an additional disc made of

steel was inserted between the upper flange and the rock sample. Without the disc, mostly the edge of the rock sample would have been pressed due to possible elastic deformation of the flange while tightening the bolts. In order to be able to use the flanges many times they were constructed of Toolox33, which does not deform plastically after applying the confining pressure.

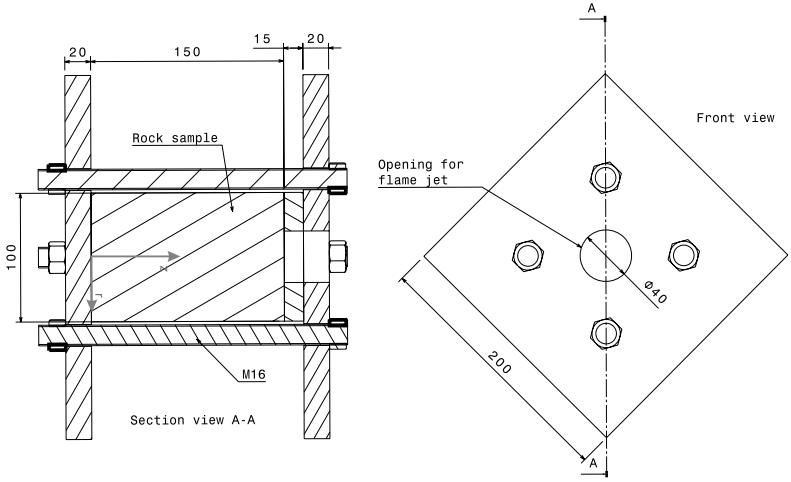

Figure 5.4: Schematic overview of the confined rock sample.

The maximal applied torque on the nut of each bolt, as suggested from the bolt manufacturer, was 180 N m. A load cell (Induk GmbH, HKMD43LT-G1) was used to measure the resulting force on one bolt as presented in Fig. 5.5. The bolt was placed in the centre of the cylindrical load cell, which was in between two nuts. The applied force was transferred from the nut to a disc and from the disc to the load cell. The cylindrical discs were installed in order to distribute the pressure of the nut to the surface of the load cell evenly.

In order to convert the measured force into pressure acting on the rock sample the following equation was used:

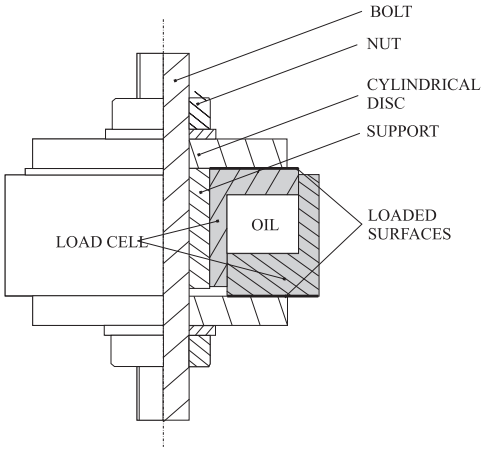

Figure 5.5: Schematic overview of force measurement coming from a nut with a load cell.

$$P_{conf} = \frac{4F_{bolt}}{A_{rock}} \tag{5.4}$$

where P_{conf} presents applied uniaxial load in Pa, F_{bolt} is the force applied with one bolt in N and A_{rock} is the loaded surface of the rock in m^2.

The influence of the uniaxial pressure on the drilling was studied for a torque equal to 0, 60, 100, 140 and 180 N m. A torque equal to 0 Nm means that no force was applied on the bolts. In this case, the only force acting on the rock is the weight of the flange.

The measured normal force increased linearly with increasing torque on the bolt. The corresponding pressure (Eq. 5.4) on the rock surface is presented in Fig. 5.6. The results are in agreement with the theory stating that the bolt clamping force is linearly dependent on the tightening torque. In literature, it was shown that the slope of the linear function is dependent on the friction coefficient (nut factor) and the bolt diameter which are both constant with applied torque [115]. The measurement was reproducible and

the same results were obtained with a different bolt. Without applying torque on the bolts only the flange was pressing the rock surface with its weight, which resulted in a pressure of around 9 kPa. This is negligible compared to the acting pressure when a torque was applied.

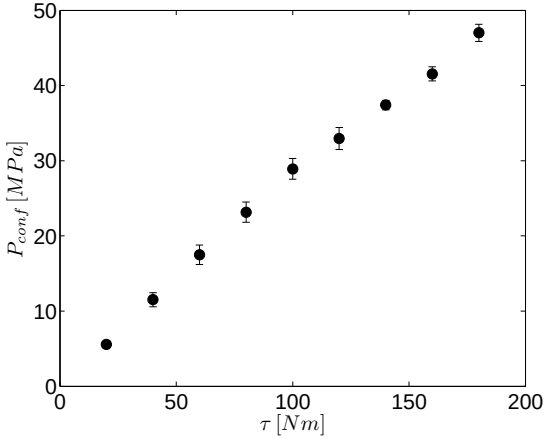

Figure 5.6: Uniaxial pressure on the rock sample depending on the torque applied per one bolt.

The heat transfer coefficient (h) is calculated from heat flux recorded with calorimeter ($\dot{q}_{sen}$) and the separately measured reference temperature (T_f). As a result, the uncertainty in h is a bit larger than in $\dot{q}_{sen}$. It was assumed that the heat flux obtained with the PSI calorimeter better represents the trend of heat transfer when varying process parameters, compared to the values obtained with the CU calorimeter. It must be kept in mind that the values are not equal to the heat fluxes transferred to the rock surface while spalling because of transient temperature change. Still, it is assumed that the trend of $\dot{q}_{sen}$ with the process conditions is the same as the trend of the heat flux transferred through the rock surface. Therefore, in order to investigate the influence of heat transfer on flame-jet drilling, the following series of experiments with smaller nozzle (D_1) were performed:

1. Constant firing rate, various dimensionless distances, various heat fluxes (empty circles in Fig.5.7, 1. in Tab. 5.1)
2. Constant dimensionless distance, various firing rates, various heat fluxes (empty squares in Fig.5.7, 2. in Tab. 5.1)

The selected operating points are graphically presented in Fig. 5.7 where the heat flux obtained with the PSI calorimeter (described in chapter 4.3) are presented. They are also listed in Table 5.1. Smaller nozzle was selected for the drilling experiments due to better het transfer results compared to the nozzle D_2.

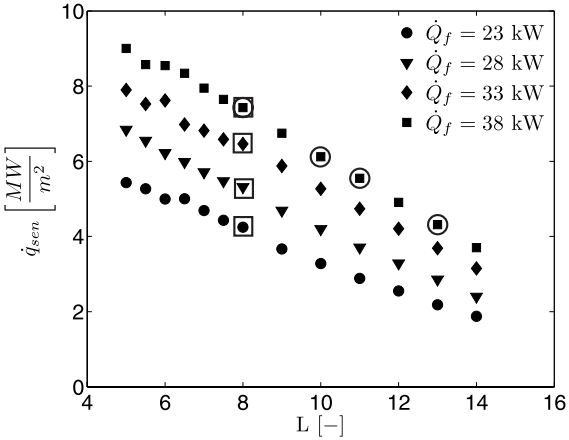

Figure 5.7: Heat flux obtained with the PSI calorimeter (dark points, see chapter 4.3) with selected operating conditions (empty squares and circles) for flame-jet drilling tests.

It is assumed that the nozzle diameter has also an influence on the flame-jet drilling. Therefore, both nozzles (D_1 and D_2) were tested in the flame-jet drilling experiments. The drilling results for the conditions at which the same heat flux was obtained with the heat flux sensor at two different nozzle diameter were compared. Therefore, in order to define the operating

Table 5.1: Two series of operating conditions for drilling experiments with nozzle D_1.

1.		2.	
$\dot{Q}_f$ $[kW]$	L $[-]$	$\dot{Q}_f$ $[kW]$	L $[-]$
38	8	23	8
38	10	28	8
38	11	33	8
38	13	38	8

conditions for drilling experiments, it is assumed that the heat flux transferred to the rock surface has the same trend with process conditions as the heat flux measured with the PSI calorimeter presented in 4.28. The operating points are shown in Fig. 5.8 and listed in Tab. 5.2.

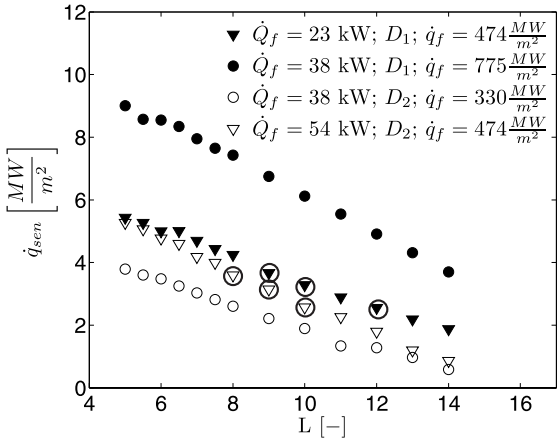

Figure 5.8: Circled points represent the operating conditions for drilling experiments in order to test the influence of the burner exit nozzle diameter

When the desired process conditions were reached, the confined rock sample was first repositioned upward in axial direction toward the flame jet

Table 5.2: Operating conditions of drilling experiments when testing the influence of different nozzle diameters.

D_{nozzle} $[mm]$	$\dot{Q}_f$ $[kW]$	L $[-]$	$\dot{q}_{sen}$ $[\frac{MW}{m^2}]$
7.9	23	9	3.9
12.1	54	8	4.6
7.9	23	10	3.4
12.1	54	9	3.4
7.9	23	12	2.5
12.1	54	10	2.6

with the linear actuator and then laterally directly below the flame jet. The time recording started at the moment when the flame jet impinged the rock surface. The position of the rock sample was not changed while drilling. The total drilling time was 30 s for all experiments except if explicitly state otherwise (chapter 5.3.1). When the drilling time elapsed, the rock sample was removed laterally from the flame jet. The removed rock volume ($\dot{V}_{removed}$) was calculated as in Eq. 5.5

$$\dot{V}_{removed} = \frac{\dot{m}_{removed}}{\rho} \tag{5.5}$$

where ρ is density of the rock (see Table 3.1) and $m_{removed}$ is the removed rock mass, which is a mass difference of the rock sample before and after the drilling. The measurement was repeated three times for each selected set of process parameters.

5.3 Results

5.3.1 Flame-jet drilling evolving with time

During the flame-jet drilling the created cavity in the rock sample was extending in length and diameter with time. The drilling could last until the diameter of the cavity was almost equal to the diameter of the rock sample. At that moment the thin walls broke because of the applied uniaxial load and the drilling could not proceed. For 47 MPa of applied axial load this happened after approximately 120 s of drilling. The drilling results were evaluated by removal rate ($\dot{V}_{removed}$, removed rock volume per time in cm^3/min), which is shown in Fig. 5.9. It was increasing with time while drilling (see also Fig. 5.10), because the rock surface that was exposed to the impinging flame jet was becoming larger, thus more surface was spalling.

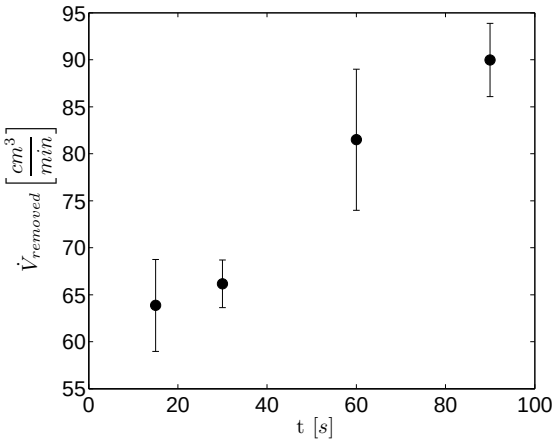

Figure 5.9: Influence of time on removed volume while drilling rock samples confined with 47 MPa. Applied firing rate was 38 kW.

The cavity evolution with time is presented in Fig. 5.10. Each rock sample

that was drilled for a certain period of time was cut afterward to obtain the cross section of created cavity. It can be seen, that the cavity grew more in depth than in width with time. Diameter (D_{cav}) and depth (L_{cav}) of the cavity are linear dependent with time in the observed period (up to 120 s) where $\frac{dL_{cav}(t)}{dt} = 1.8\frac{dD_{cav}(t)}{dt}$). The thermal spallation occurred at the whole rock surface exposed to the flame jet, but the removal was more intense at the stagnation point of the impinging jet. This occurred probably because of better heat transfer from the fluid to the rock surface in that region inducing larger compressive stresses. When the rock sample was drilled for longer than 30 s, the cavity was becoming wider than initial impinged free surface. For a period of 30 s, the rock sample confined with 0 MPa could successfully be drilled without causing its breakage. Therefore, this period was selected for examining the influence of uniaxial load and heat on the flame-jet drilling. After 30 s the standard deviation of the removal rate was 5 %, which indicated that the results are reproducible even though not much rock volume was removed.

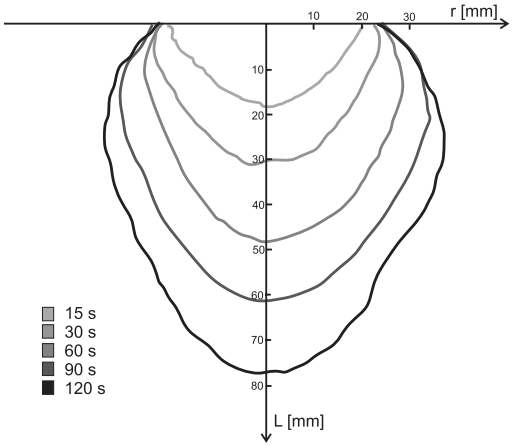

Figure 5.10: Hole propagation with time, uniaxial load 47MPa.

For a comparison of the flame-jet drilling results with those obtained with a conventional drilling technique applied in the field different performance

indicators can be used. One of them is the so-called rate of penetration or average drilling velocity when drilling a deep geothermal well. It indicates the moving velocity of the cavity bottom while drilling. According to Fig. 5.10, when drilling with the flame jet the hole diameter increases with time when the burner is kept at the same position. Thus, the drilling velocity is dependent on the target hole diameter and on burner dimensions (e.g. exit nozzle diameter, outside diameter of the burner that has to enter the hole). In Fig. 5.11, the resulting drilling velocity is presented. It is largest at the beginning of the drilling process, but as the hole becomes deeper and the distance between rock surface and nozzle increases, the drilling velocity decreases due to worse heat transfer. If the distance could be kept constant at the initial distance, it is assumed that the average drilling velocity would be equal to the initial velocity (8 cm/min or 4.8 m/h). This is in the range of the rate of penetration for a 5 km deep geothermal well drilled in Basel (4 m/h [116]).

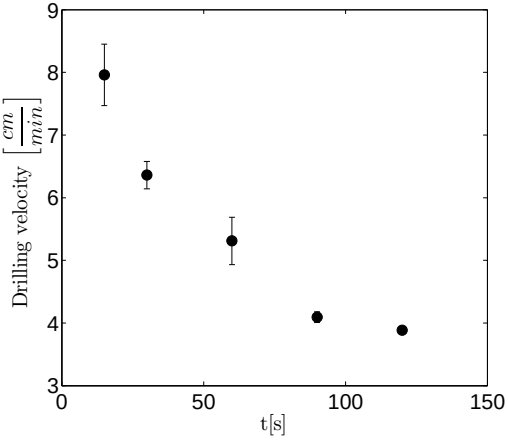

Figure 5.11: Evolution of the drilling velocity with time. Applied uniaxial load was 47 MPa and firing rate 38 kW.

5.3.2 Flame-jet drilling evolving with pressure

Fig. 5.12 shows that the removal rate increases with the applied load for the tested loading range (0−47 MPa). This indicates that the efficiency of the flame-jet drilling would increase with depth while creating a borehole, which presents an advantage of this technology over the conventional ones. The tensile strength of the rock is reached faster with larger uniaxial pressure at the same firing rate.

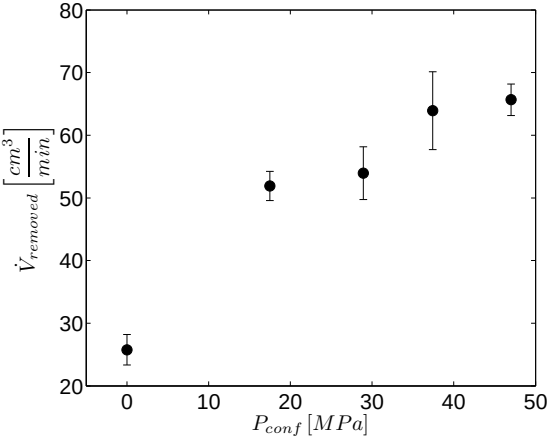

Figure 5.12: Influence of uniaxial pressure on removed volume while drilling for 30 s. Applied firing rate was 38 kW.

Ejection of spalls is a very fast process (faster than 10 000 Hz as observed with a high speed camera) thus the steps of thermal spallation mechanism (crack extension and buckling) are difficult to detect experimentally. Therefore, the increase of the drilling efficiency with uniaxial load was investigated with the numerical model. It shows that the uniaxial load and the heat source induce stresses in the rock probe creating regions of compression and tension. Fig. 5.13 presents the normal stresses in radial (r) and axial (z) directions (σ_{rr}, σ_{zz}, respectively) with 47 MPa of applied uniaxial load and after 0.01 s of heat exposure. Negative values represent

compression, and positive ones tension. On the left-hand side of Fig. 4.25 a narrow region below the surface (including the surface itself) is subject to compression with a very steep gradient along z - axis. Below the compressive region there is a larger region of tension with lower gradient of stresses. The obtained stress distribution is in agreement with the numerical results from Germanovich [25], who observed only the influence of heat on stress development. The right-hand side of Fig. 4.25 shows the stresses in axial direction. The region below the loaded surface ($0.02 < r < 0.05$ m) is exposed to axial compression and due to the elastic properties of the body the compression is transferred to the rock core. Below the free surface (5 mm thickness) the axial stresses are equal to zero because no uniaxial loading is applied.

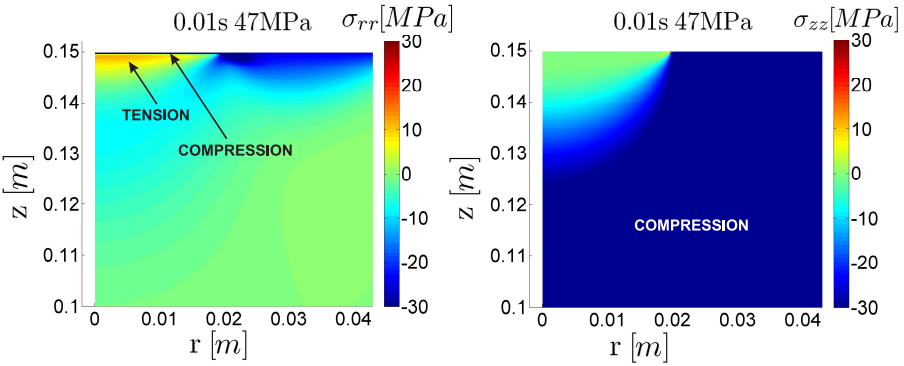

Figure 5.13: Stress field in radial and axial direction at 0.01 s and with 47 MPa uniaxial load.

The influence of time at constant load (47 MPa) and the influence of uniaxial load at the same moment (0.01 s) on development of normal stresses in axial (z), radial (r) and tangential (θ) directions (σ_{zz}, σ_{rr}, $\sigma_{\theta\theta}$, respectively) along the centerline of the modelled rock sample are presented in Fig. 5.14 and Fig. 5.15, respectively.

The intensity of the stresses only close to the impinged surface (free sur-

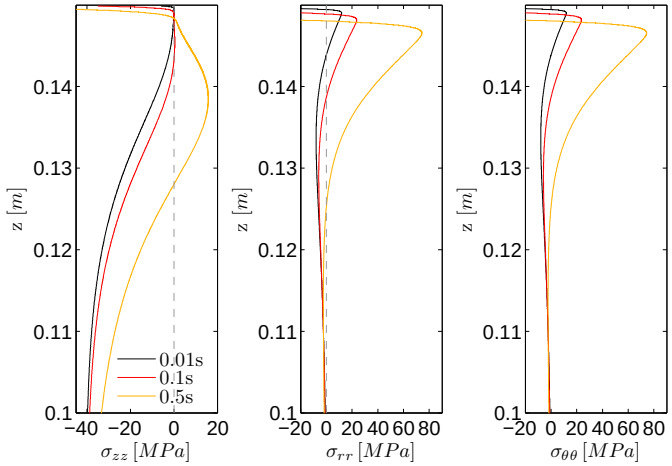

Figure 5.14: Stress development with time along the centerline of the modelled rock sample in cylindrical coordinates for the load of 47 MPa.

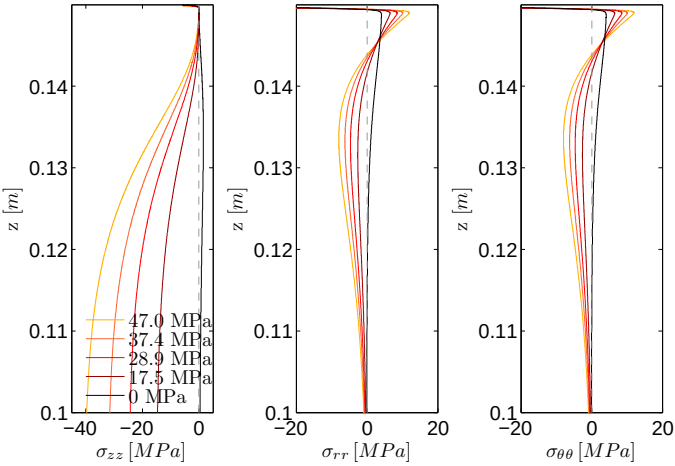

Figure 5.15: Stress development with uniaxial load along the centerline of the modelled rock sample in cylindrical coordinates at 0.01 s.

face) between 0.10 and 0.15 m in axial direction is shown. This is the region where the radial and the hoop stress (σ_{rr} and $\sigma_{\theta\theta}$, respectively) are mainly influenced by the uniaxial load and the applied heat. The radial compressive stresses (σ_{rr}) at the top layer decrease negligible compared to the stresses when no load is applied. Therefore, the improvement of experimental drilling with uniaxial pressure is related to the increase of tensile stresses in tensile region below the surface. The tensile stresses additionally disintegrate the material and weaken the connection between the grains. Reduced connection strength between grains simplifies the breaking process when entering the compressive region during thermal spallation, which increases the drilling efficiency. Previous theories of thermal spallation mostly highlighted the influence of compressive stresses, but the importance of the tensile region was only discussed in the theoretical work done by Meier et al. [114] and is confirmed with the experimental results presented in this work. As presented in experiments, there was no mode I cracking of the confined rock sample. The reason is, as Meier et al. [114] showed with a numerical analysis after 10 s, that the tension that causes mode I cracking in the core of the rock sample is reduced due to uniaxial load.

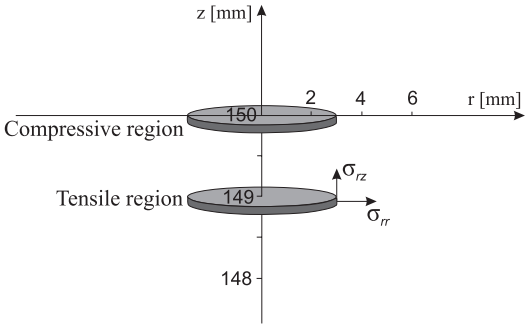

Figure 5.16: Graphical interpretation of the pad that represents a future spall when placed in compressive and tensile region. The average stresses were calculated from the stresses acting on the surface of the pad.

The buckling phenomenon of a spall before ejection can be better presented

by stresses that act on the incipient spall when it is not yet detached from the rock. This spall is located at the surface in the compressive region. It was approximated with a circular pad (see Fig. 5.16) that simulates the future spall (250 µm thickness and 3 mm in diameter). Stresses causing crack formation and buckling act along the curved surface of the pad, thus they were analyzed. The same stresses were evaluated when the pad is placed in the tensile region (as shown in Fig. 5.16), actually at the location where the tension is the largest. The acting stresses are interpreted with average stresses ($\bar{\sigma}_{rz}$ and $\bar{\sigma}_{rr}$) calculated as in Eq. 5.6 and Eq. 5.7, respectively. Indices ij and kk represent the coordinates rz and rr, respectively. Those were the stresses of the curved pad surface with remarkable large values compared to the stresses acting on other surfaces of the pad.

$$\bar{\sigma}_{ij} = \frac{\iint\limits_{Sn} r\sigma_{ij} dj d\theta}{\iint\limits_{Sn} r dj d\theta} \qquad (5.6) \qquad \bar{\sigma}_{kk} = \frac{\iint\limits_{Sn} r\sigma_{kk} dj d\theta}{\iint\limits_{Sn} r dj d\theta} \qquad (5.7)$$

The average stress along the curved surface showed the largest change with applied uniaxial load. It is presented in Fig. 5.17. In the compressive region considerably high stresses were only existent in radial direction, which decrease with the load. This can not be the cause for the improved drilling, because those stresses are required for the thermal spallation. Contrary to the compressive region, the average tensile stresses acting in radial and axial direction ($\bar{\sigma}_{rr}$ and $\bar{\sigma}_{rz}$, respectively) increase both with the load, which again indicate that the tension influences the drilling efficiency when increasing the uniaxial load as mentioned before.

5.3.3 Flame-jet drilling evolving with heat flux

The drilling results for various heat transfer conditions between the flame jet and the rock surface are presented in Fig. 5.18 and 5.19. Each figure

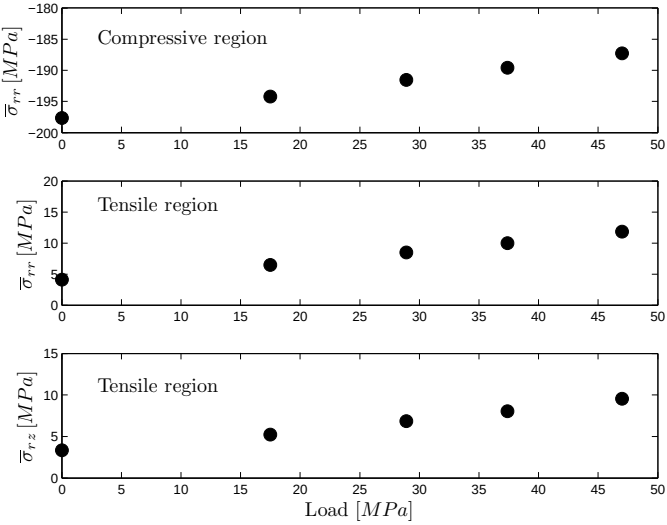

Figure 5.17: Average stress acting on a circular pad (250 µm thickness and 3 mm in diameter) presented in the region of the largest compression and tension.

has two vertical axes: the removal rate on the left-hand side and the heat flux obtained with the PSI sensor for the same operating conditions on the right-hand side (MW/m^2). The results indicate an increase of removal rate due to improved heat transfer. This occurred 1) when increasing the firing rate for constant dimensionless distance ($L = 8$) (presented in Fig. 5.18) and 2) when decreasing the dimensionless distance for constant firing rate (38 kW) (presented in Fig. 5.19).

At higher firing rate, more fuel was combusted releasing more heat and increasing the mass of exhaust gases that impinged the rock surface with higher velocity. As a consequence, both the momentum and the transferred heat flux were increased that improved the removal rate. Increasing the firing rate from 23 to 38 kW, the heat flux to the PSI calorimeter was increased up to 75 %, but the removal rate up to 95 %. When reducing the dimensionless distance, less ambient air entrains the jet [117]. Therefore, the exhaust gases reached the rock surface with higher mean temperature and higher mean velocity [118]. This increased mostly the transferred heat flux to the rock surface compared to larger dimensionless distances. Decreasing the dimensionless distance from $L = 13$ to 8 increased the heat flux up to 75 % and the removal rate up to 110 %. It is assumed that improving the heat transfer in both sets of operating conditions actually influences the thermal stresses in the surface rock layers, which is reflected on the removal rate.

When changing the firing rate and dimensionless distance the resulting heat transfer coefficient is also changing. For the modelling, the values of the heat transfer coefficient were those obtained in the heat transfer study (see Fig. 4.26). They were applied as the boundary condition of the model together with the corresponding reference temperature (as presented with Eq. 5.3).

The experimental drilling results showed that the removal rate was increasing with the firing rate (from 23 to 38 kW, Fig. 5.18) keeping the dimensionless distance constant ($L = 8$). The calculated thermal stresses at those conditions are presented in Fig. 5.20 with a closer look in Fig.

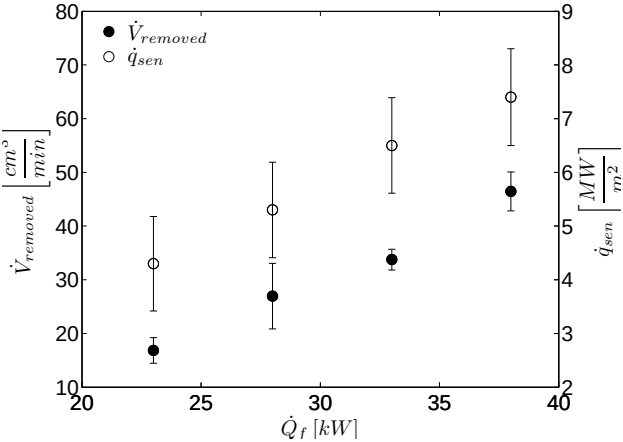

Figure 5.18: Removal rate when increasing the firing rate at a constant dimensionless distance ($L = 8$) and corresponding heat flux obtained with the PSI calorimeter for the tested operating conditions.

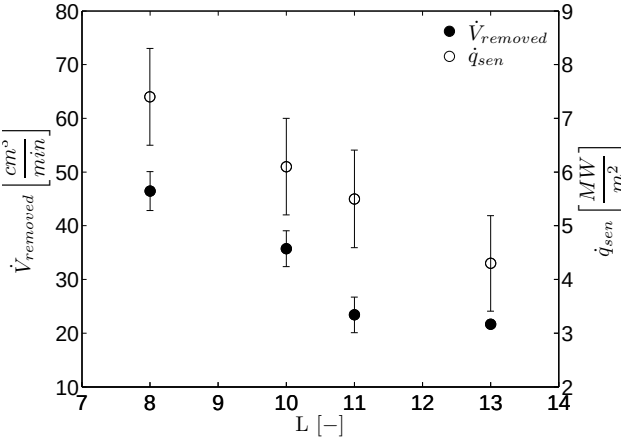

Figure 5.19: Removal rate when increasing the dimensionless distance at a constant firing rate ($\dot{Q}_f = 38$ kW) and corresponding heat flux obtained with the PSI calorimeter for the tested operating conditions.

5.21.

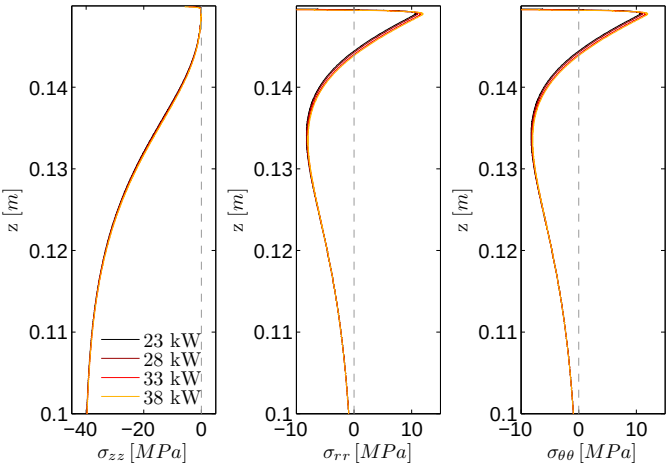

Figure 5.20: Stress development with firing rate at constant dimensionless distance $L = 8$. The uniaxial load is 47 MPa and the simulation moment is 0.01 s.

It is shown that both tensile and compressive stresses in radial direction close to the surface were increasing with the firing rate. The increase in compressive stresses is much larger than the increase in tensile stresses. The compressive stress at the largest firing rate is 50 % higher compared to the lowest firing rate. Obviously, the increase of the heat flux does not much influence the tensile stresses below the compressive region, thus the removal rate is mostly influenced by the compressive stresses at the surface. They are responsible for the spalls creation and their ejection from the surface in this case. The increase of the compressive stresses is also presented with average stresses acting on the surfaces of a circular pad placed in the compressive and tensile region (compressive region: $\bar{\sigma}_{rr}$ and $\bar{\sigma}_{zz}$, tensile region: $\bar{\sigma}_{rr}$ and $\bar{\sigma}_{rz}$ in Fig. 5.22). The other average stresses (compressive region: $\bar{\sigma}_{rz}$, tensile region: $\bar{\sigma}_{zz}$) change negligibly. The stresses that are influenced the most are those acting on the curved

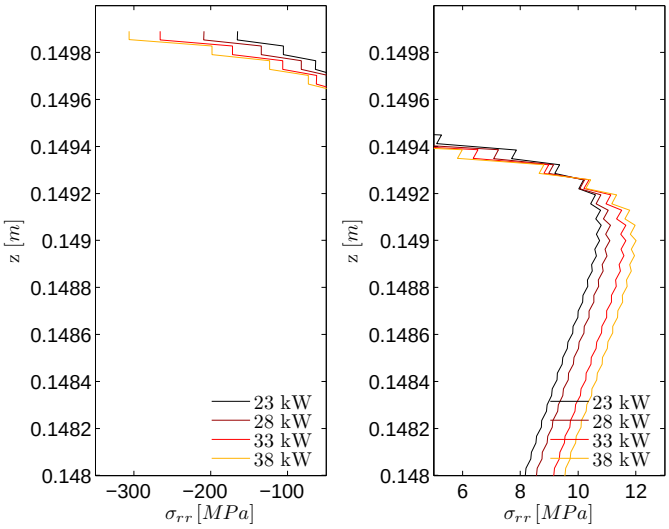

Figure 5.21: Compressive stresses (lefthand side) and tensile stresses (righthand side) acting close to the impinged surface along the centerline depending on the applied firing rate, keeping the dimensionless distance constant ($L = 8$). The uniaxial load is 47 MPa and the simulation moment is 0.01 s.

surface of the pad in the compressive region in radial direction ($\bar{\sigma}_{rr}$). The average stresses are calculated as in Eq. 5.6 and Eq. 5.7. Indices ij and kk represent the coordinate rz and rr, or zz, respectively.

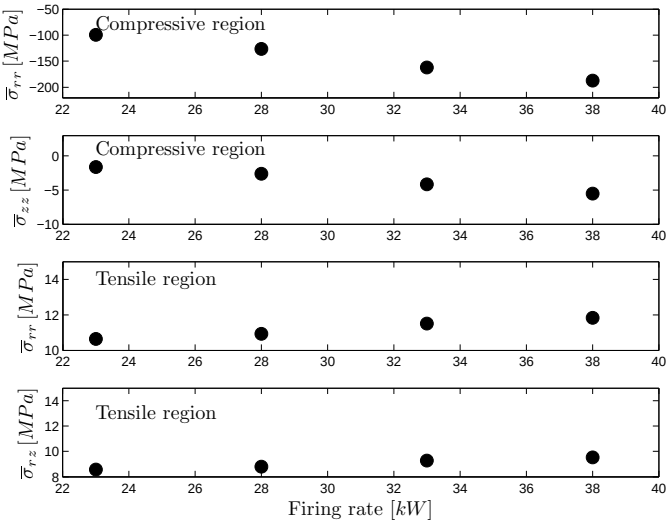

Figure 5.22: Average stress acting on a circular pad (250 µm thickness and 3 mm in diameter) presented in the region of the largest compression and tension for $L = 8$. The results are valid for 0.01 s.

When reducing the dimensionless distance at constant firing rate the heat transfer increases, thus the compressive stresses at the top layers also increase. The increase in compressive stresses is presented in Fig. 5.23. In parallel, the drilling experiments show the increase of the removal rate when reducing the dimensionless distance in Fig. 5.19. The increase of compressive stresses is actually the reason for the increase of drilling efficiency in the experiments. As in the previous case a change in tensile stresses is negligible compared to the change in compressive stresses, thus it is assumed that it does not contribute to the removal rate when changing the heat transfer conditions.

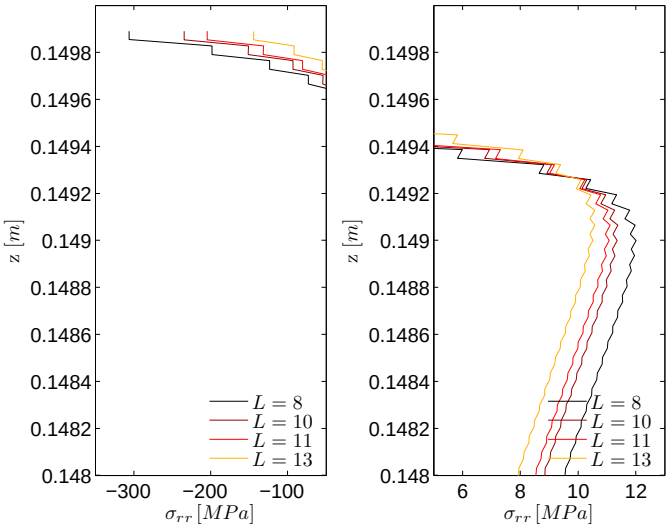

Figure 5.23: Compressive stresses (lefthand side) and tensile stresses (righthand side) acting close to the impinged surface along the centerline depending on the dimensionless distance, keeping the firing rate constant (38 kW). The uniaxial load is 47 MPa and the simulation moment is 0.01 s.

The average stresses over a circular pad when changing the dimensionless distance at constant firing rate are summarized in Fig. 5.24. They were calculated also with Eq. 5.6 and Eq. 5.7. As in the previous case, when changing the firing rate for the constant dimensionless distance, only the most influenced stresses acting on the surfaces of the circular pad are presented. The largest change is noticeable in the compressive region on the curved surface of the pad ($\bar{\sigma}_{rr}$).

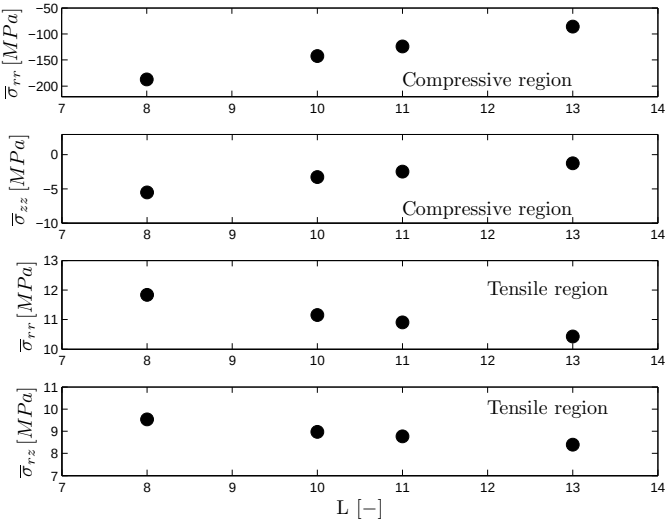

Figure 5.24: Average stress acting on a circular pad (250 µm thickness and 3 mm in diameter) present in the region of the largest compression and tension at 38 kW. The results are valid for 0.01 s.

5.3.4 Flame-jet drilling evolving with nozzle diameter

The obtained removal rate and the corresponding heat flux measured with PSI calorimeter for two nozzle diameters are presented in Fig. 5.25. With the larger nozzle the impinging flame jet was wider and a larger part of the

rock surface was heated. As a consequence, more rock mass was removed. The effect of nozzle diameter influencing the flame-jet drilling was better illustrated in Fig. 5.26. Those cavities were created with D_1 nozzle at 23 kW and $L = 9$ and D_2 nozzle at 54 kW and $L = 8$ after 90 s of drilling. At those conditions almost equal heat fluxes were recorded with the PSI calorimeter. The obtained cavities had the same depth but different diameter, which is related to the width of the impinging flame jet.

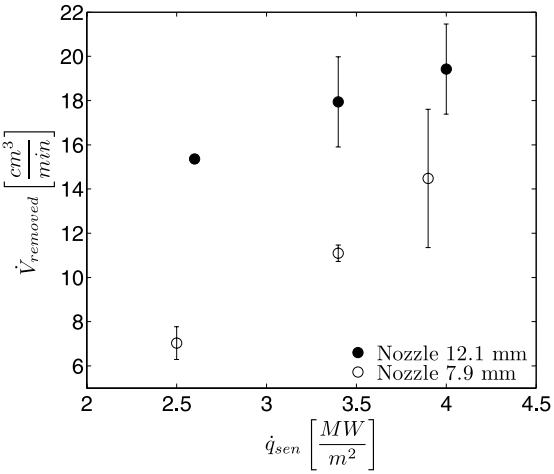

Figure 5.25: Removal rate depending on the nozzle diameter. The operating conditions were at the same relative firing rate ($\dot{q}_f = 474\,\mathrm{MW/m^2}$). The heat fluxes on the horizontal axis were obtained with the PSI calorimeter at different dimensionless distances listed in table 5.2.

5.3.5 Particle size distribution

As experimentally analysed in this work, the heat flux influences the removal rate in the flame-jet drilling experiments. Besides that, Browning [119] reported that it also influences the size of the spalls: higher heat fluxes generate smaller spalls. The proposed theory is that higher heat fluxes in-

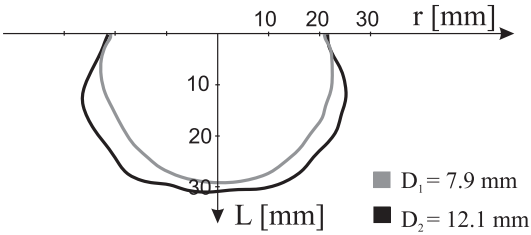

Figure 5.26: Cavity shape after 90 s of drilling with nozzle D_1 at 23 kW and $L = 9$ and D_2 at 54 kW and $L = 8$. At those conditions the heat fluxes obtained with the PSI calorimeter were equal.

duce a larger temperature gradient in the rock sample, thus the thermal stress required for fracturing can be achieved in a very small volume. If the heat flux is lower, the resulting temperature gradient in the rock is smaller, thus the required thermal stresses can be achieved only in a larger volume. In order to confirm the theory, the spalls were collected and analysed after flame-jet drilling experiments at 23 and 38 kW of firing rate and at dimensionless distance $L = 8$. The particles were collected from everywhere in the insulation box where it could be approached with a small broom used only for this purpose. Collected particles were separated by size with different sieves (2.8, 1.4, 1.0 and 0.5 mm). Afterward, the particles smaller than 0.5 mm were additionally analysed with a laser diffraction device (Sypma TEC).

The recorded macroscopic pictures of spalls are presented in Fig. 5.27. Some of the spalls had the size of a grain (around 2 mm). The smallest one could still contain more than one crystal. Those with the size around 2 mm and less are mostly round, but the larger spalls ($3 \div 4$ mm) had the shape of a flake that probably resulted due to larger preexisting flaws in the rock. The spallation mechanism was observed in addition with a high speed camera. The videos were recorded with a velocity of 10 000 frames\s which was at the limit to record the ejecting of the spall from the rock surface. It was noticed that initial spalls in the buckling phase broke into many small ones. Thus the initial size of a spall (while buckling) is in some

cases larger than the size of the resulting ejected spalls. On the other hand, it was also noticed that really small spalls having a size of only a couple of crystals were also ejected individually from the surface.

Fig. 5.28 presents the mass fraction of the particles depending on their size. According to the results smaller particles (less than 1 mm) were obtained at the higher firing rate, and larger particles at the lower firing rate. This confirms the theory that the transferred heat influences the compressive stresses in the surface layers and the size of the spalls, too.

The particle size distribution was only analysed for particles smaller than 500 µm. The resulting frequency particle size distribution is shown in Fig. 5.29 where it is not very clear which firing rate generates smaller particles. It is only noticeable that the distribution of particles obtained when drilling with 23 kW is broader than when drilling with 38 kW flame jet, thus the particles have a wide range of size. Also, the peaks in Fig. 5.29 for both firing rates are between 200 and 300 µm, thus the frequency particle size distribution can not confirm the proposed theory on microscale. Furthermore, the cumulative distribution presented in Fig. 5.30) indicates that up to 500 µm more small particles were produced with the less heat, which is also in contradiction to the proposed theory. An explanation could be that the sample was not representative and very small particles could not be efficiently collected (less than 500 µm in diameter). Because the insulation box was ventilated during the experiments, it is possible that particles smaller than 500 µm were lost and taken away with the ventilation, thus their amount can vary in the analysed samples. Therefore, it was assumed that the uncertainty in the number of particles in a collected sample increases with smaller diameter.

5.4 Conclusion

Influence of various operating conditions was tested when drilling a Grimsel granite with the impinging flame jet. The influence of firing rate, di-

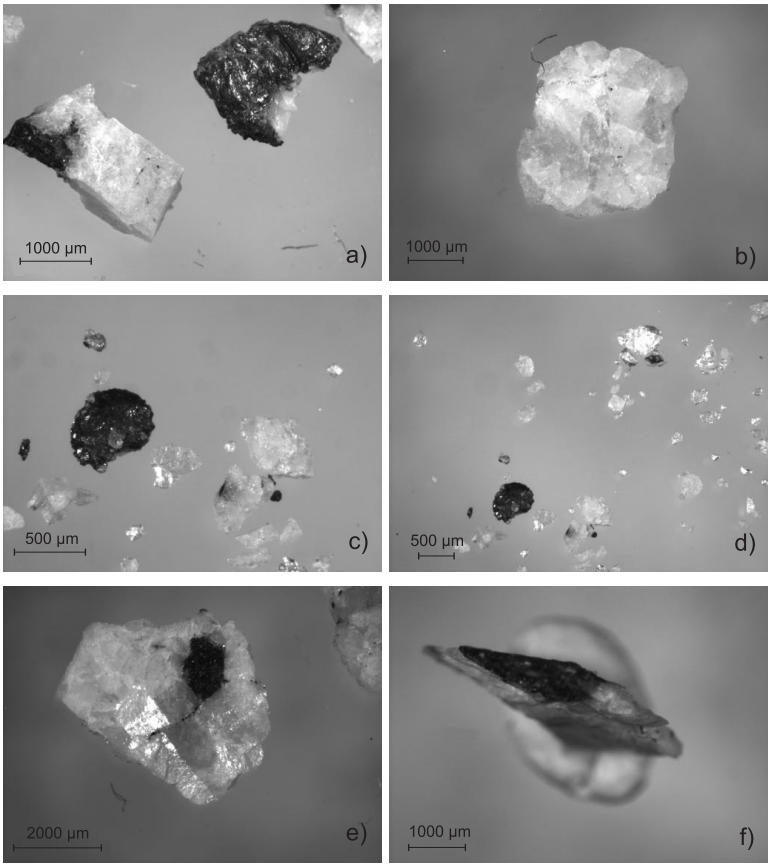

Figure 5.27: Obtained spalls observed under microscope.

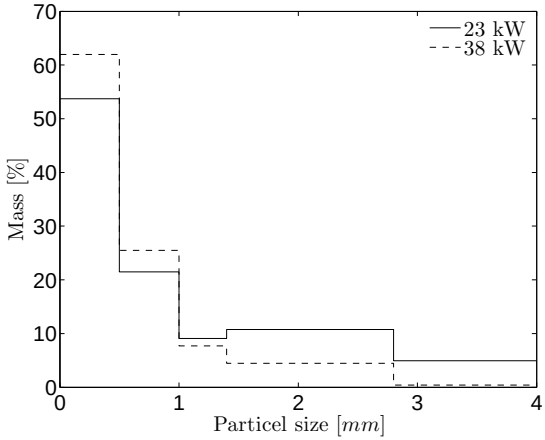

Figure 5.28: Mass fraction of the spalls depending on their size in a collected sample. The rock samples were drilled at $L = 8$, but at different firing rates.

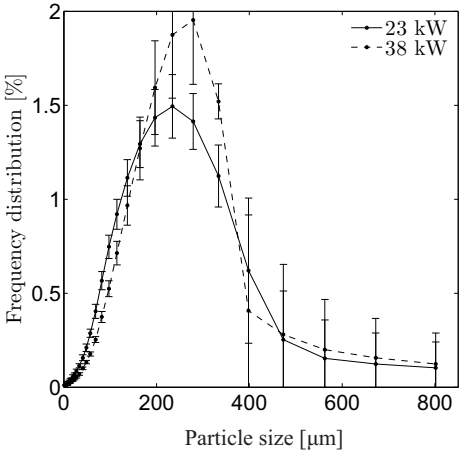

Figure 5.29: Frequency particle size distribution for the particles sieved with 500 µm sieve. The rock samples were drilled at $L = 8$, but at different firing rates.

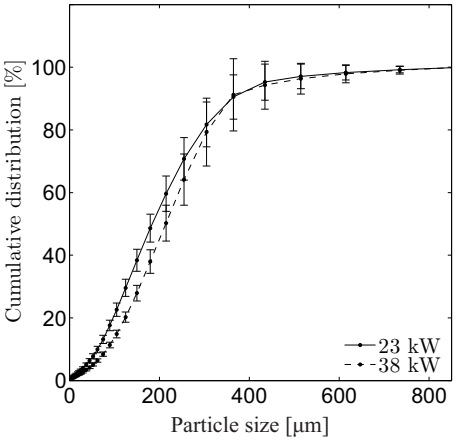

Figure 5.30: Cumulative particle size distribution for the particles sieved with 500 μm sieve. The rock samples were drilled at $L = 8$, but at different firing rates.

mensionless distance, nozzle diameter and uniaxial load applied on the rock sample were investigated. When increasing the uniaxial load (up to 47 MPa) and keeping the firing rate and the dimensionless distance constant the removed rock volume (removal rate) was increasing. According to the stress analysis, an increase of the uniaxial load increases the tensile stresses beneath the impinged surface, which is thus related to the improved penetration. A possible explanation is that the generated tension helps weakening the connection between the grains in the rock sample, which simplifies the drilling. As soon as the disintegrated region becomes compressed due to applied heat, the buckling of the surface and spalls creation occurred easier. In summary, the new contribution to the thermal spallation mechanism theory is that tensile stresses below the compressive region help weakening the rock layers. It is also concluded that drilling at larger depths, where larger uniaxial load is present, with an impinging flame jet the drilling efficiency would be improved.

Flame-jet drilling experiments were performed when changing the firing

rate and the dimensionless distance between the rock surface and the exit nozzle in a way that the heat transfer is improved (reducing dimensionless distance or increasing the firing rate). As a results the temperature gradient below the surface is steeper, which increases the stresses in the compressive region. Larger compressive stresses are beneficial for flame-jet drilling, thus higher removal rates are reached. The tensile stresses do not play a significant role when influencing the heat transfer to the rock surface.

Surface temperature measurement showed the temperatures up to 600 °C cause thermal spallation. Larger temperatures are not required. Therefore, the improvement of the technology can be done by improving the amount of transferred heat to the rock surface. This can be achieved 1) by combusting more fuel and producing more heat (being careful not to cause melting), or 2) by improving the efficiency of the heat transfer between the impinging jet and the rock surface and reducing the fluid temperature. The first solution would require changes in the supply equipment and the second one requires changes in the drilling tool and, maybe, applied reactants.

Chapter 6

Flame-jet drilling tool for the field tests

When drilling in the laboratory many issues that can occur in the field can not be addressed. They can be divided in two groups: drilling tool issues (design of the burner in order to drill larger depths, high temperatures sustainability, insulation of electrical connections for igniter and their robustness, hose pipe design around the water and gas lines) and rock nature issues (presence of large cracks, varying mineral composition, varying spallation behaviour of the drilled material and inhomogeneous stress distribution). Experience from drilling in the field can be very helpful to address those issues, thus a new burner was designed at IPE in order to perform a test in laboratory as a preparation for the field experiments. The new burner was designed by Thierry Meier because the burner presented in chapter 3.1 was not made for drilling deep holes. Still, the experience collected during flame-jet experiments with the FJDS is applied when designing the new burner, which performance was tested with the existing setup in the laboratory on a 0.5 m long Grimsel granite block. The experiment and the results are presented in this chapter.

6.1 Experimental setup

All equipment presented in chapter 3 (see Fig. 3.2) was used for the drilling test of the new large rock probe. Only the burner was replaced with the new one. The new burner is made of a stainless steel double-walled tube with an outside diameter of 80 mm, length of 0.8 m and it is cooled with water (see Fig. 6.2). The gas mixture (methane and air) is ignited with a hot surface igniter in a form of a coil made of NiCr alloy [120]. All pipes and electrical cables are routed through one hose pipe mounted on the top of the burner (see Fig. 6.1). The products of the combustion are passing through the exit nozzle of 6 mm in diameter with de Laval shape. To provide a distance between the nozzle and the rock surface 58 mm long feet is mounted at the bottom of the burner. The experimental test was performed with a firing rate of 38 kW. In order to provide a perpendicular position of the burner toward the rock surface a holder is mounted at the top of the rock sample, which is also designed in a way to collect the created spalls and to direct them to a separation cyclone. The tested rock sample was a rectangular cuboid with dimensions of 0.5 x 0.7 x 0.7 m (where 0.5 m is the height). It is assumed that those dimensions are able to prevent mode I cracking of the rock sample.

6.2 Results

The rock sample was successfully drilled through (see Fig. 6.3). The hole resulted in a straight shape with a slightly varying diameter of around 95 mm. At the very beginning of the drilling the burner could not smoothly be placed in the hole and remaind at one position for some time (around 10 min), thus the cavity below the burner was becoming wider than required. After entering the burner in the hole, the flame-jet drilling could successfully be continued. The bottom of the rock was reached after 22 min of drilling. The drilling velocity (rate of penetration) was approximated to be around 1.5 m/h. Particles could be successfully separated from the exhaust

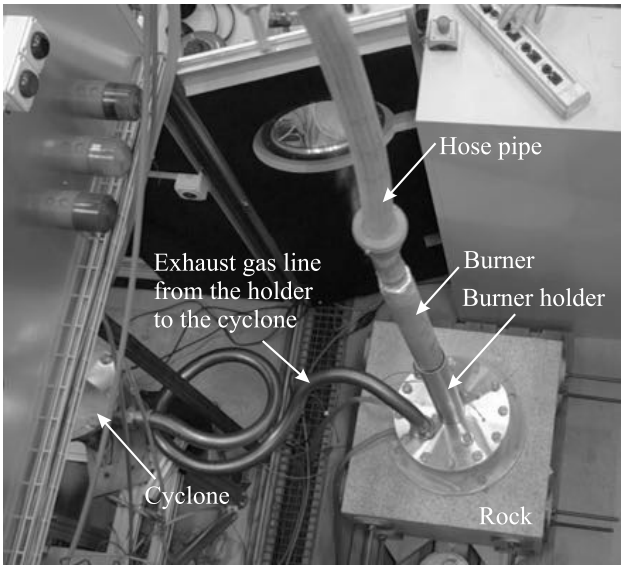

Figure 6.1: Flame-jet drilling technology applied on a larger rock probe. The methane, air and water lines, together with control system used in the previous drilling experiments was also applied in this test. Differences to the old setup are in the burner, burner holder and separator of particles (cyclone).

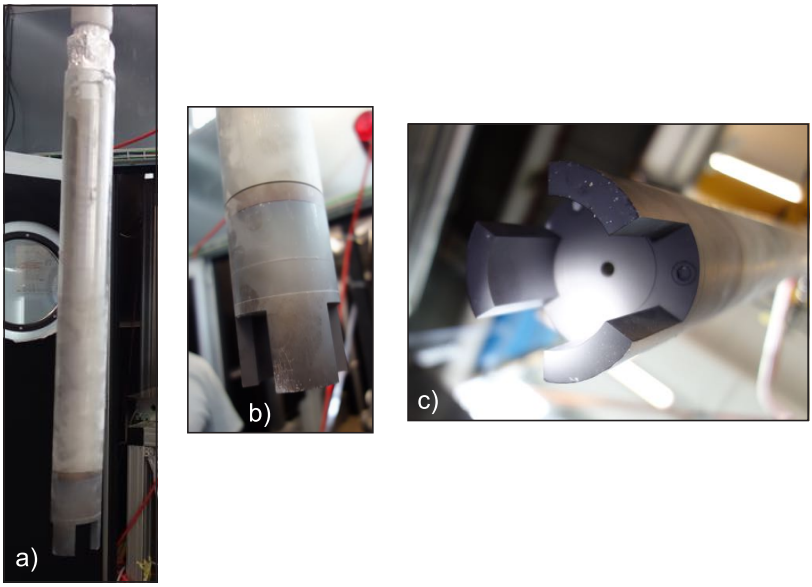

Figure 6.2: The burner applied for the drill test of the large rock sample. Drilled depth 0.5 m in 22 min, a) the burner assembly, b) distance feats, c) exit nozzle

gas line and collected. The burner was relatively robust and it sustained high temperatures and the motion through the cavity.

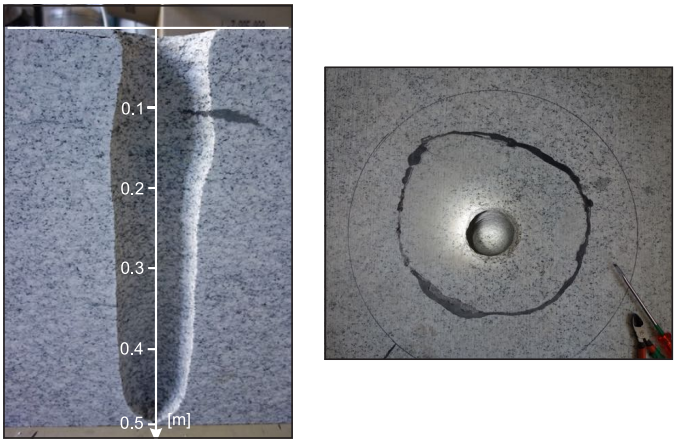

Figure 6.3: Drilled rock sample with flame-jet drilling technology.

6.3 Conclusion

The flame-jet drilling technology could be used for drilling up to 0.5 m. This was the largest drilled depth at IPE. The designed burner performed very well and it can be applied for drilling in larger depths. Any problems while drilling related to the rock nature could not be addressed.

Chapter 7

Conclusion

The presented work was focused on gaining knowledge required to develop the flame-jet drilling technology as an applicable method for geothermal well formation. It is a contact-less drilling technology where a flame jet impinges the rock surface, generates a high temperature gradient below the rock surface and causes localized fracturing of the brittle material due to the developed thermal stresses. As a result, small flakes (so-called spalls) are ejected from the surface. The fracturing mechanism is known as thermal spallation. Flame-jet drilling is very dependent on the developed thermal stresses when applying a heat source (flame jet) on the rock surface. Therefore, this work was aimed at improved understanding of the interaction between the stress field and process parameters that influence the impinging flame jet.

The thermal spallation mechanism presented in literature emphasizes the importance of compressive stresses at the impinged surface and in the near-surface regions acting parallel to it. They indirectly influence the enlargement of already existing flaws and help crack propagation due to the resulting tensile stresses at the tips of the cracks. The thermal stresses obtained with a numerical model in this work were in agreement with literature showing that there is always a thin region below the impinged surface of compressive stresses acting in radial direction with a steep drop along the surface-normal axis. Below the compressive region, there is a region of

tensile stresses acting in radial direction with a lower stress gradient than in the compressive region. However, previous theories did not consider the tensile region as relevant for spallation. With the experimental and theoretical results of this work, it was proven that together with the compression the tensile region has also an important role in flame-jet drilling. It additionally disintegrates the material and weakens the connection between the grains before they enter the compressive region. This can simplify the fracturing process and increase the thermal spallation efficiency, which is a novel contribution to the theory of thermal spallation mechanism.

Consequently, there are two ways to improve the flame-jet drilling technology: (1) By increasing the stresses in the compressive region or (2) by increasing the stresses in the tensile region. (1) The compressive region can be influenced mostly by the temperature gradient below the surface, which is in relation to the transferred heat flux from the impinging flame-jet. A larger heat flux transferred through the rock induces a steeper temperature gradient. This directly increases the compressive stresses at and below the surface, which increases formation frequency of cracks, thus it reduces the time until the surface buckling and ejection of the spalls occurs. This was proven with flame-jet drilling experiments by evaluating the removal rate of the rock that was increasing with the heat flux to the rock surface. Also, by increasing the heat flux, the particle size distribution shifted towards smaller dimensions, because the required compressive stresses for buckling and spalls ejection are even reached in a smaller rock volume. Therefore, the particle size distribution could be a useful indication of the flame-jet drilling quality due to the missing optical access to the created hole while drilling in the field. (2) The stresses in the tensile region can be increased by increasing uniaxial pressure. In laboratory, this was achieved by applying uniaxial load on the tested rock samples (up to 47 MPa). As a result, the removal rate increased. However, uniaxial pressure is no process parameter adjustable by the drilling engineers when drilling in the field. But, this discovery is beneficial for flame-jet drilling technology, because the axial stress increases with depth due to gravity, thus the flame-jet drilling efficiency should improve when drilling deeper underground, contrary to

conventional drilling techniques.

A mathematical model was applied to simulate the development of thermal stresses in the rock when being impinged by the flame jet used in this study. The obtained results are in agreement with literature and could be successfully used to describe the cause of obtained experimental results. However, the model was not designed to predict drilling and hole formation.

Flame-jet drilling was improved by increasing the heat flux transferred to the rock surface. According to the heat transfer study, the most influencing parameter on the heat transfer was the firing rate. It influenced the amount of released heat and the heat transfer coefficient. Therefore, in order to increase the removal rate and the rate of penetration when drilling Grimsel granite (or a similar granite) with a constructed flame-jet system, the firing rate has to be increased. This requires, for example, larger flow rates of the fuel and oxidizer and larger supply pressures. On the other hand, flames with temperatures higher than rock melting temperatures are not necessarily required for successful flame-jet drilling. The surface temperature of the rock at the onset of spallation was smaller than 600 °C. Therefore, if the heat transfer between the impinging jet and the rock surface is very high, the fluid temperature can be much lower than flame temperatures applied in this study (between 1200 and 1800 °C) in order to avoid eventual fusion.

Chapter 8

Outlook

Within this work it was proven that flame-jet drilling is a feasible technology for drilling of Grimsel granite. A square rock sample with $0.245\,m^3$ in volume was drilled successfully in laboratory without any serious complications concerning the drilling tool. The influence of some process parameters on flame-jet drilling technology was also described, but still there are some remaining parameters that have not been investigated.

Concerning the process parameters it was concluded that larger heat fluxes transferred through the rock surface increase the removal rate. Therefore, the heat transfer to the rock surface can be improved either by increasing the firing rate, or by replacing the impinging fluid with more dense fluid (e.g. steam, having a possibility to reduce the fluid temperature and prevent rock fusion). Due to limitation of the setup firing rate could not be improved, but this can be solved having a larger supply pressure in the air line. Also, water was injected in the combustion chamber to reduce the fluid temperature while keeping the heat transfer coefficient high. Due to inadequate water injection system and low residence time of water in the burner its evaporation could not be reached. This solution can be further improved and tested.

It is known which heat fluxes are required to perform flame jet drilling. This knowledge can be transferred to the high pressure plant constructed for drilling in aqueous environment. That means to perform a heat trans-

fer study in hydrothermal and high pressure conditions and compare the results with the heat transfer study at ambient pressure. In case of a large differences the heat transfer under high pressure can be improved by adjusting the nozzle design and reduction of entrainment phenomenon.

One parameter that is changing in nature is the uniaxial pressure, which increases with a well depth. Its influence on drilling was tested in laboratory and described in presented work. Still, there are many other natural factors that could influence the drilling result, e.g. presence of large cracks and inhomogeneity of stresses in horizontal direction. Their influence on drilling could be tested directly in field but due to usually complex nature it is difficult to relate drilling results to only one phenomenon. Therefore, the mentioned effects should be simulated on rock samples and investigated also in laboratory. Still, experiments in the field are very valuable to obtain a knowledge about the performance of the drilling equipment (e.g. durability of a drill bit and other elements, motion of the drill bit, ignition in larger depths, etc.). Therefore, experiments in the field and laboratory should be performed in parallel and the obtained knowledge in both areas could be combined in order to improve the understanding of the flame-jet drilling technology and its development.

Different rock types were tested in order to decide which rock type should be applied in the study. Without any scientific explanation it was noticed that drilling of some rock types is faster than of the others. Therefore, further research should be performed to understand which physical property and mineral structure is important for thermal spallation. If the phenomenon is understood, flame-jet drilling can be accordingly adjusted in order become a universal drilling technology.

Currently, the numerical model used in this work simulates the development of thermal stresses in a rock sample when being exposed to the flame jet and uniaxial load. Removal of elements (e.g. cavity formation) and drilling process is not included. In order to do so, the flaws existence, their enlargement into a crack, buckling and spalls ejection has to be taken into account. The flaws could be modelled as small voids in the material filled

with air that become larger according to the theory of crack propagation existing in literature [25]. Afterward, the model can be further improved considering the mineral composition, grained structure, etc. The final goal of the model could be prediction of the flame-jet drilling results depending on rock structure and applied drilling tool. In that case required properties of the drilling tool could be estimated depending on the field conditions and desired well depth.

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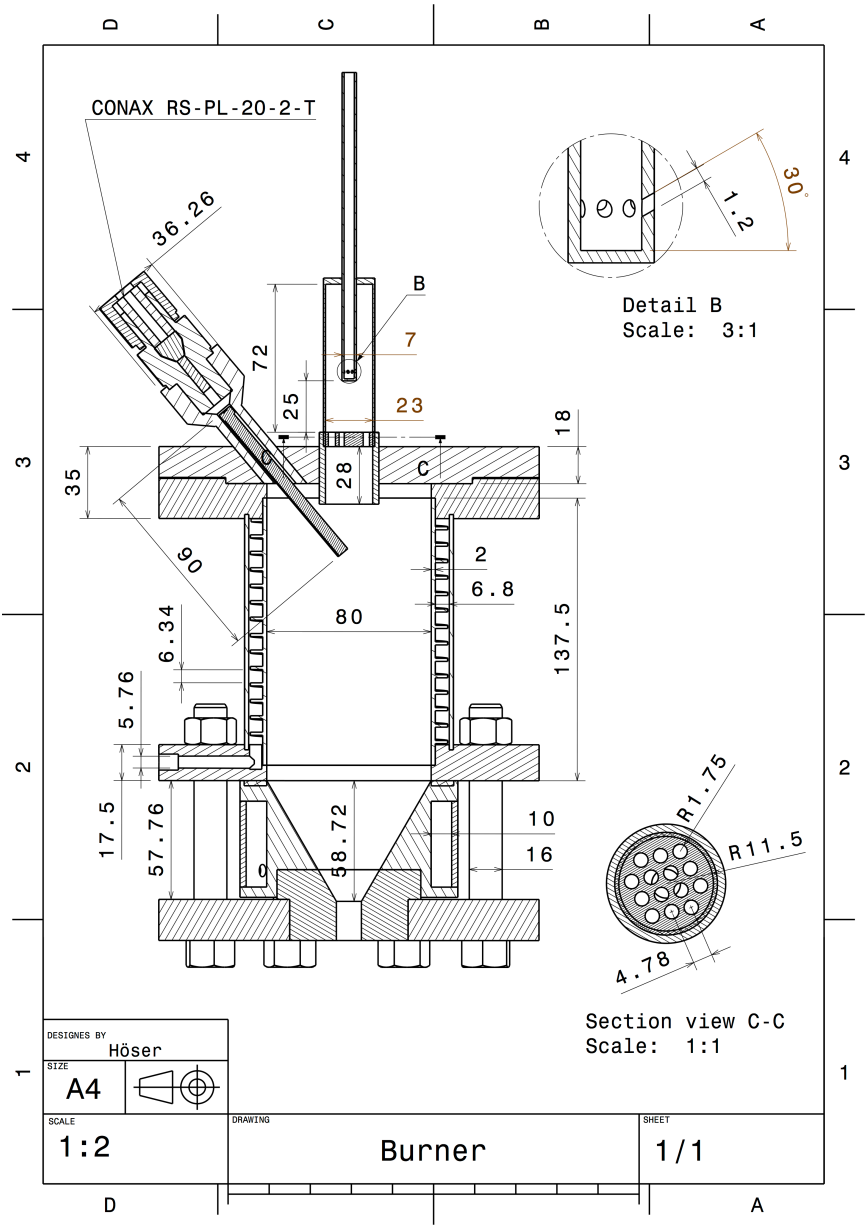
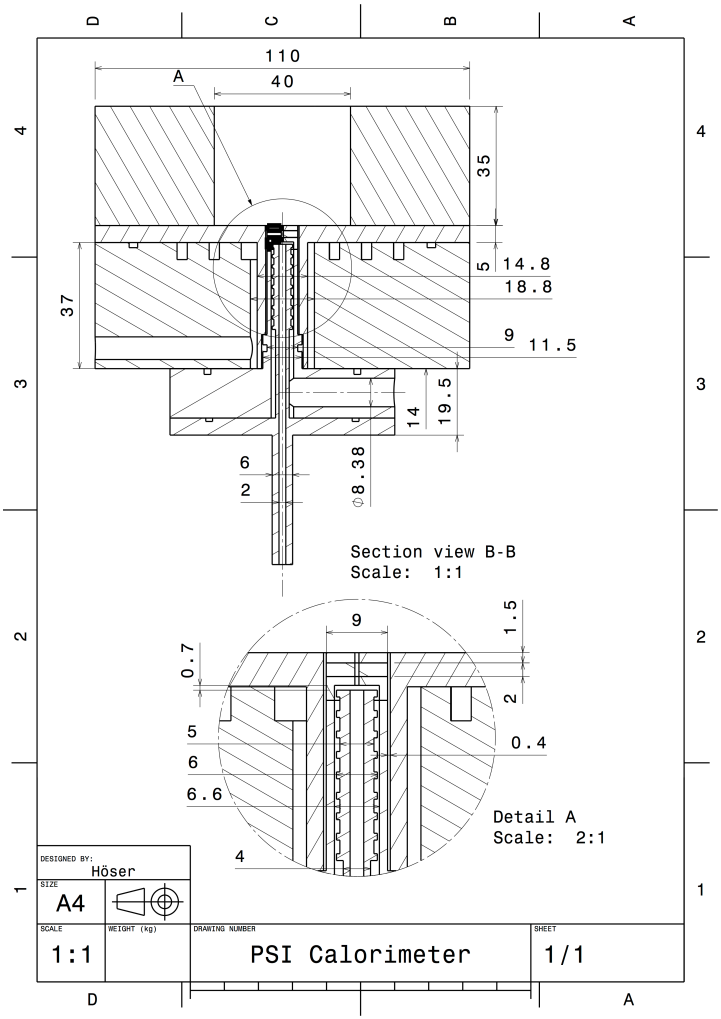
Appendix A

The design of the burner


CONAX RS-PL-20-2-T
36.26
72
25
23
7
B
C
28
35
18
90
2
6.8
80
137.5
6.34
5.76
17.5
57.76
58.72
10
16
Detail B
Scale: 3:1
30°
1.2
R1.75
R11.5
4.78
Section view C-C
Scale: 1:1
DESIGNES BY
Höser
SIZE
A4
SCALE
1:2
DRAWING
Burner
SHEET
1/1

Appendix B

The design of the PSI calorimeter


D
C
B
A
110
40
A
35
5
14.8
18.8
37
9
11.5
14
19.5
6
2
8.38
Section view B-B
Scale: 1:1
9
1.5
0.7
2
5
0.4
6
6.6
4
Detail A
Scale: 2:1
DESIGNED BY:
Höser
SIZE
A4
SCALE
1:1
WEIGHT (kg)
DRAWING NUMBER
PSI Calorimeter
SHEET
1/1

List of publications

Journal publications

1. D. Höser, R. Wallimann and P. Rudolf von Rohr, Uncertainty analysis for emissivity measurement at elevated temperatures with an Infrared Camera. *International Journal of Thermophysics 37:14*, 1-17, 2016.
2. D. Höser and P. Rudolf von Rohr, Experimental heat transfer study of confined flame jet impinging on a flat surface. *Experimental Thermal and Fluid Science*, under review.
3. D. Höser, T. Meier, A. Patru, M. Kant and P. Rudolf von Rohr, Influence of uniaxial load on flame jet drilling - an experimental study. *International Journal of Rock Mechanics and Mining Sciences* - under review.
4. M. Schuler, T. Rothenfluh, P. Stathopoulos, D. Brkic and P. Rudolf von Rohr, Simulating supercritical water jets with a variable turbulent Prandtl number. *Chemical Engineering and Technology* - 18961902, 2014.

Conference contributions

1. D. Brkic, M. Kant, T. Meier, M. Schuler and P. Rudolf von Rohr, Influence of process parameters on thermal rock fracturing under ambient conditions. *Proceedings World Geothermal Congress*, Melbourne, Australia, 2015.
2. T. Meier, D. Brkic, M. J. Schuler, M. Kant and P. Rudolf von Rohr, The Potential of Hydrothermal Flames to Induce Spallation in Gneiss. *Proceedings World Geothermal Congress*, Melbourne, Australia, 2015.

3. M. Kant, D. Becker, D. Brkic, T. Meier and P. Rudolf von Rohr, Investigation of a novel drilling technology - influence of the surface temperature for hydrothermal spallation drilling. *Proceedings World Geothermal Congress*, Melbourne, Australia, 2015.

4. D. Brkic, R. Wallimann and P. Rudolf von Rohr, Calibration of an infrared camera for rock surface temperature measurement. 4th European Geothermal PhD Day, Szeged, Hungary, 2013

5. M. Kant, D.Höser, M.Schuler, E.Rossi and P. Rudolf von Rohr, Thermal spallation drilling - an alternative drilling technology for hard rock drilling, International Conference and Exhibition for Advanced Drilling Technology, Celle, German, 2016.

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